Quantitative Research

Comparative Study on Blue Light Exposure in Grade 11 Students from Livingstone International School, National University Clark, and Angeles University Foundation

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Abstract

This quantitative research will focus on the blue light exposure emitted from digital devices on the sleep quality and mental health of grade 11 students. This study aims to compare adolescents' awareness, behaviors, and coping strategies regarding blue light exposure through comparative structured surveys, hypothesizing that there is no significant difference between the schools. This comparison is to be conducted between three different schools: Livingstone International School, Angeles University Foundation, and the National University located in Clark. Grade 11 participants from the strands GAS, ABM, HUMSS, and STEM/STEAM will be assessed using approved tools such as the General Anxiety Disorder scale and the Pittsburg Sleep Quality Index.

Grade 11 students are among those most affected by extended screen time for both academic and recreational purposes. Blue light disrupts circadian rhythms by suppressing melatonin production, which is critical for initiating and maintaining sleep. This disruption leads to delayed sleep onset, reduced sleep duration, and poor sleep quality. Poor sleep, in turn, exacerbates mental health challenges, including heightened risks of anxiety, depression, and impaired cognitive performance.

With a total of 66 students at the end of the survey period February 16 to March 3, 2025, we reject the null hypothesis, finding there a significant difference between Livingstone International school and Angeles University Foundation to National University of CLark, where Livingstone International School has the highest scoring in severe anxiety and Angeles University Foundation for poor sleep. In contrast, National University was scored to be the lowest, finding their scores to be normal and have quality sleep. This shows through the graphs that the lifestyles of the students and their academic environment matter in affecting how they use their devices and how they choose to respond to issues they face. Having them lose their sleep using their devices creates a difficulty for them to rest, increasing anxiety levels and melatonin. This affects their circadian rhythms and has them behave differently during the daytime, repeating a cycle of difficulty operating in classes and outside.

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CHAPTER I
INTRODUCTION

Background of the study

The twenty-first century has a high exposure to blue light from electronic devices such as smartphones, tablets, and computers. This exposure towards adolescents has links to disruption in the Circadian rhythms and sleep quality, with effects on mental health. Adolescents in grade 11 experience elevated screen time due to demands of education and leisure, increasing their vulnerability to adverse Blue Light effects. Adolescents are susceptible to development shifts in circadian timing, making them prone to delayed sleep phases. Further research indicates that evening blue light exposure significantly suppresses melatonin production in a study by Behar and Radrizzani (2022) and Prather and Hurst(2022), highlighting how disrupted sleep patterns can affect mental health issues such as anxiety.

Statement of the Problem

Despite large evidence and knowledge spread online on the adverse effects of blue light, adolescents lack a comprehensive understanding of how their screen habits impact their health. Unaware of these physiological and psychological consequences, prolonged evening exposure to blue light despite existing interventions such as blue light filters and recommendations for screen reduction desired to be applied consistently to lower reduction, Addressing these gaps is critical to mitigate the risks associated with blue light exposure and improve adolescents' sleep hygiene and mental health. To know how to respond and help the students, this research seeks to answer the following:

1. Do the grade 11 students of three different schools perceive the impact of blue light exposure on their sleep quality and emotional well-being?
   This question is directed to help provide a foundational background towards the main question's results. Helping to better understand the differences between the three schools and evaluate the reasons for the difference.
   1.1. What are the participants' experiences regarding the effects of screen time before bed?

1.2. How do participants describe changes in their emotional state related to their screen time habits?

1.3. What coping mechanisms do participants use to manage blue light exposure and its impacts on sleep?

2. Is there a significant gap between the three schools' students in Awareness, behavioral Patterns, and Coping Strategies?

   This is the main question aimed at getting an idea of the subjective interpretations and generally agreed-upon experiences related to the issue of blue light exposure and its impact on sleep quality and emotional well-being during adolescence. Strive to identify whether adolescents are aware that their exposure to blue light influences their circadian rhythm, sleep quality and duration, and other secondary outcomes on mental health by impacting mood, stress levels, and emotional regulations listed within this study.

   2.1. (Awareness) How do the grade 11s perceive the existence of blue light and its impact? What do they believe or think about it and its role in affecting sleep and mental health?

   2.2 (Behavioral Patterns) Why do the grade 11s use blue light-emitting devices such as smartphones, tablets, or computers in the evening? What is their habit or routine that would increase the exposure?

   2.3. (Coping Strategies) What coping strategies do the grade 11s use to reduce their exposure to blue light, including blue light filters, limiting screen time at night, or participating in other sleep hygiene behaviors?

This investigation will gain an understanding of how youth conceptualize and cope with the advancements in technology use, sleep health, and emotional well-being, which in turn could help highlight the need for educational or technological interventions to address these.

Hypothesis/ses of the Study

1. Null hypothesis (H_0)

   There is no significant difference in awareness, behavioral patterns, and coping strategies between students from the three schools.

2. Alternative hypothesis ($H_{a)}$

   There is a significant difference in awareness, behavioral patterns, and coping strategies between students from the three schools.

These hypotheses are to be tested on each portion individually, not all in general.

Theoretical Framework

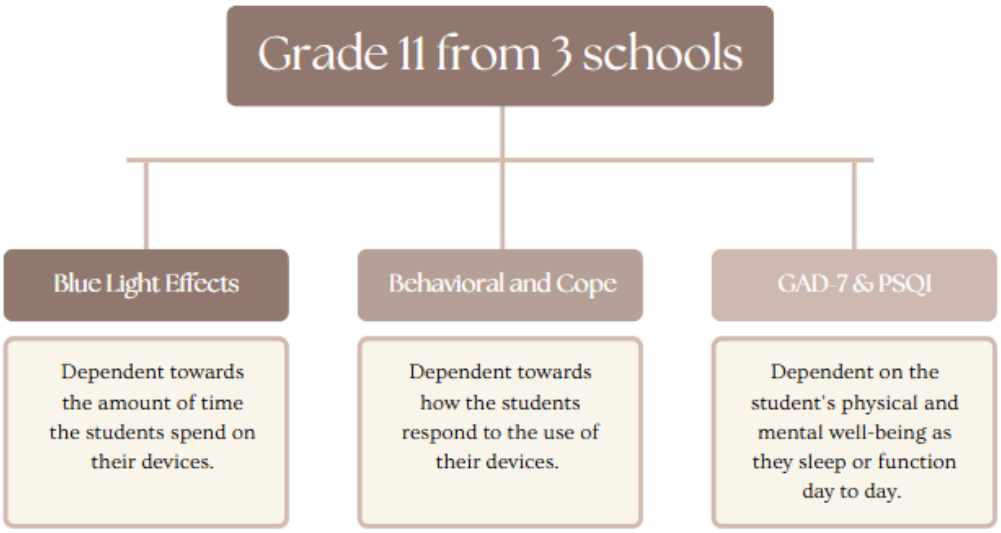


Table 1. Theoretical Framework Mapping Chart

Conceptual Framework

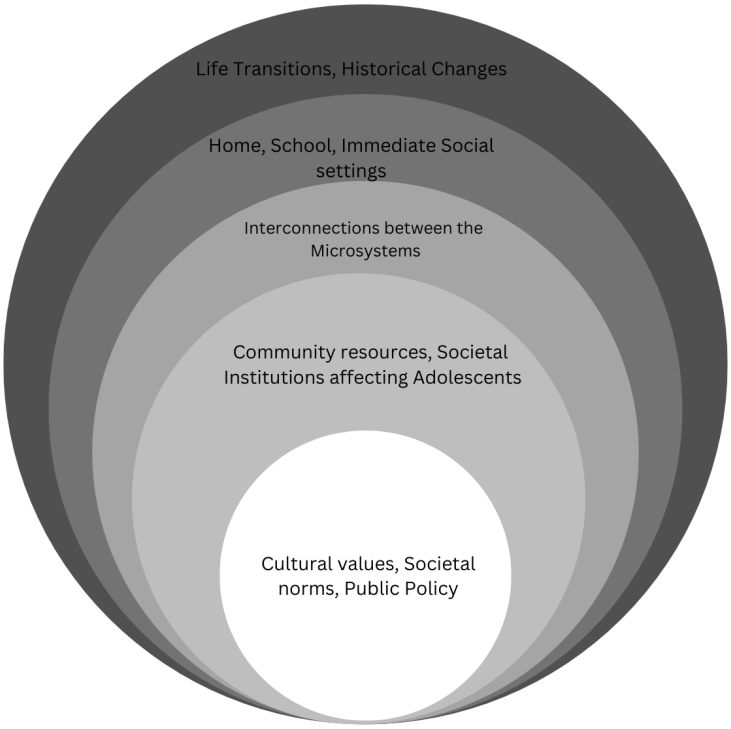


Figure 1. Conceptual Framwork

Conceptual Paradigm

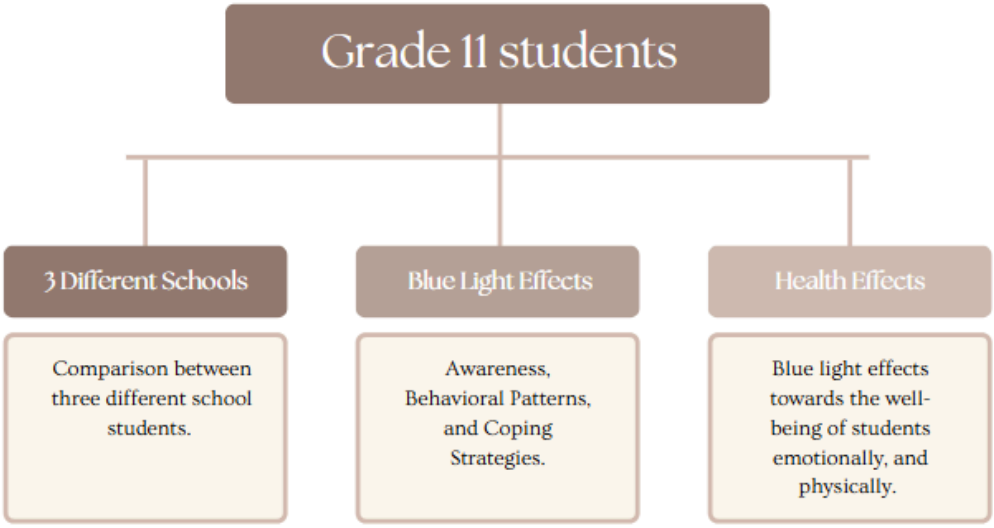


Table 2. Conceptual Paradigm of the Study

Significance of the Study

Understanding how teenagers perceive the impact of exposure to blue light is a critical step toward understanding larger concerns about their health and well-being. This study could provide information for developing interventions in the form of educational programs, better sleep hygiene practices, and the adoption of blue-light-filtering technologies. The findings could also serve as a starting point in mitigating mental health risks, enhancing academic performance, and promoting emotional stability during the critical developmental phase.

Scope and Delimitations of the Study

The study focuses on 66 Grade 11 students from three educational institutions, broadening the scope for a moderate-depth exploration of individual experiences through a comparative study. The limitation begs the restriction to have further research for broader populations. The study excludes younger students and adults. This study was also only conducted within a month and lacks bias due to the participants' personal experiences, implying all students agreed to the survey concerning Blue Light and its effects without in-depth interviews for personal experiences.

Definition of Terms

1. *Blue Light:* A type of visible light with a wavelength between 380 to 500 nanometers(nm). Containing the shortest wavelength and highest energy of all visible lights. This affects humans' CIrcadian and Melatonin levels and has twice the amount of damage done by green light to the human retina. Some sources may be sunlight or artificial sources such as fluorescent lights, LED TVs, computer monitors, smartphones, and tablet screens.
2. *Circadian Rhythm: A* Human's natural 24-hour cycle is called, at times, the "Body's clock" as it controls sleep, body temperature, hormones, and appetite. One disorder one can attain is the sleep-wake cycle disorder when the body's internal clock is out of sync with the environment.
3. *STEM/STEAM students:* Students pursuing in the field of Science, Technology, Engineering, and Mathematics + Arts.
4. *ABM students*: Students pursuing in the field of Accounting, Business, and Management.
5. *GAS students*: Students pursuing in the field of General Academic Strand.
6. *HUMSS students*: Students pursuing in the field of Humanities and Social Sciences Strand.

7. *Sleep Quality:* A measurement of how well a person sleeps determined by how long, how they sleep, and how they feel after waking up.
8. *Pittsburg Sleep Quality Index:* The PSQI is a self-report questionnaire that assesses sleep quality over a 1-month time interval. The measure consists of 19 individual items, creating 7 components that produce one global score, and takes 5–10 minutes to complete.
9. *General Anxiety Disorder:* The GAD-7 is the most frequently used diagnostic self-report scale for screening, diagnosis, and severity assessment of anxiety disorder.
10. *Livingstone International School:* LSIS shortened term.
11. *Angeles University Foundation*: AUF shortened term.
12. *National University of Clark*: NU shortened term.
13. *Significant Difference*: A noticeable change between two groups in a study, showing that the difference is unlikely due to chance.
14. *Mean*: The average value of a set of numbers, found by adding them up and dividing by the total count.
15. *P-Value:* A number that shows how likely a result happened by chance; a smaller value (usually below 0.05) means the result is statistically significant.

CHAPTER II
LITERATURE REVIEW

The Sleep Quality of each generation becomes more apparent with the evolution of technology to have been greatly affected disrupting their Circadian rhythm or sleep cycles, disrupting their physical and mental health. For many students, it is common to sleep less than 7 hours, the minimum required for young adolescents as their bodies grow. With the new century, there has been enlightenment towards the mental health of society. More people are concerned for their and their peers being more aware day by day. These are 15 studies or articles that will help provide insights or deeper pieces of evidence to prove how important it is to be aware of the effects of Blue Light Exposure on sleep quality and mental health in adolescents and how this may improve future generations and current ones.

An Article by Marcia Ines Silvani and the Faculty of Medicine in Switzerland, and Robert Werder and Claudio Perret both Sports medicine institution professors in Switzerland about The Influence of Blue light on sleep, performance, and well-being in young adults as a systematic review discusses how the increasing use of blue light-emitting electronic devices effects of blue light exposure on sleep and mental health are scrutinized. This systematic review outlines how blue light impacts physiological processes and how the melatonin secretion process for the regulation of sleep and mental health in general is deeply crucial, especially for adolescents who are in their developmental stage with increased device use and are highly susceptible to these effects. The main findings are that evening exposure to blue light delays sleep onset and reduces overall sleep duration, resulting in increased fatigue and impaired cognitive performance the preceding day. The review further discusses how disordered sleep can worsen mental health problems, such as anxiety and depression in adolescents, creating a vicious cycle of poor sleep and psychological distress. The body of the research contained increased screen time, especially before bedtime, correlating with poor sleep quality. The review cites evidence that shows adolescents who report high levels of screen time are more

likely to suffer from sleep disturbances and mental health challenges. In summary, the systematic review argues that the physiological disturbances from blue light not only affect sleep but also contribute to a decline in mental health, making it a significant concern for adolescents who are still developing their coping mechanisms. Poor sleep quality, as mentioned, can lead to lower academic performance.

And hinder social relationships, perpetuating feelings of isolation and exacerbating mental health challenges. The connection to my study is the theme in regards to the systematic review that comes with an important conclusion since adolescents expose themselves more than ever before to the emission of devices. By acknowledging the negative effects of blue light, the possibility of developing measures like blue light filtering apps or limiting screen time at bedtime can be realized. Investigating this relationship could inform interventions aimed at helping young adults develop better sleep hygiene and mental well-being, equipping them to navigate their increasingly digital world more effectively.

The second Article by Prather and Hurst established the foundation of the complex relationship between sleep and circadian rhythms, This showed how these two aspects work together to influence overall health, especially in adolescents who are often subjected to irregular sleep patterns due to lifestyle factors and digital media use. The article focuses on the mechanisms of how sleep and circadian cycles work, exploring how interruptions to one trigger the disturbances of the other. In terms of breadth, authors would sum up some existing literature about what was known by physiological and psychological effects attributed to the alignment or misalignment of sleep and circadian rhythms, then address related factors that impact the overall quality of adolescent sleep such as light exposure-environmental factors that, in their effects, make a real difference. To support this hypothesis, documented evidence from various studies proves the negative impact of low-quality sleep on mental health, especially in adolescents. For instance, they mention findings that associate more screen time, especially before bed, with reduced sleep duration and quality, thus heightening the risk of anxiety and depression. The authors discuss how the blue light emitted from screens disrupts the production of melatonin, a hormone that regulates sleep-wake cycles. Such disruption can cause delay in falling asleep and reduced sleep duration, which can worsen mood disorders common among adolescents. Additionally, the authors discuss potential interventions, such as reducing screen time and using blue light filters, to minimize the effects. In conclusion, findings from the article underscore a relationship between sleep quality, blue light exposure, and mental health, which underlines the need for raising

awareness and developing intervention strategies that would improve sleep hygiene among adolescents.

The third article, by Wahl, Engelhardt, and three other professors, goes into the discussion of the circadian rhythm, the inner clock, and how blue sets the human rhythm. Exposing the body to blue light, which is emitted by all modern light-emitting devices, has become an increasingly important concern because it influences circadian rhythms, mainly in adolescents. This rising use of smartphones, tablets, and computers during the evening hours can disturb sleep and mental health due to the suppression of melatonin levels, delay in falling asleep, and increase in alertness at night. These factors are associated with adverse effects on emotional regulation and cognitive performance in this vulnerable group of adolescents, who are already vulnerable due to changes in their circadian rhythms and increased device use. The use of LE devices increases alertness and suppresses melatonin, which further leads to low-quality sleep and high mental health risks. Sleep disturbance in adolescents may contribute to worsening stress, anxiety, and depression. Targeted interventions are therefore required. The findings are a direct reflection of how exposure to blue light directly relates to disrupted sleep patterns, which then cascade into broader mental health challenges. Adolescence, being the period of heightened neurobiological and social change, amplifies these risks, which makes preventive strategies like regulating screen use and promoting healthier habits all the more important.

Andre Comiran Tonon and 18 other researchers in Brazil, Washington, and the United Kingdom study bring out the significant association between sleep disturbances, disruptions in circadian rhythm, and the prevalence of major depressive disorder among adolescents. It calls attention to the need to understand how such factors as nocturnal light exposure, particularly blue light, contribute to these mental health challenges among this age group. Introduce the need for there to be at least risk factors that could be used to inform a preventive strategy because most adolescents have limited access to effective treatments. Artificial light at night, ALAN exposure, is associated with great effects on the sleep of adolescents, as well as their mental and cognitive performance. Higher ALAN exposures are related to late bedtime and higher incidence of mood disorders because suppressing melatonin affects the quality of sleep onset. These problems will be compounded among individuals with major depressive disorder since their sleeping difficulties include insomnia. Poor sleep hygiene is already a common problem among teenagers, and irregular sleep patterns and nighttime light exposure only make it worse. Blue-enriched light, emitted from devices, delays circadian alignment and reduces

restorative sleep. This results in decreased sleep efficiency, heightened risks of depression, anxiety, and stress, and negative impacts on academic and emotional well-being. Furthermore, although blue light can promote alertness and reduce sleepiness during the day, it can impair accuracy and cognitive control during tasks, indicating a trade-off between wakefulness and accuracy. The complex relationship between ALAN, sleep disturbances, and mental health points to the need for targeted interventions that could promote better sleep hygiene and minimize these effects on adolescents' health and performance. The effects of blue light are mediated by its capacity to disrupt the natural sleep-wake cycle. Adolescents, who are already predisposed to delayed circadian phases, face compounded risks from blue light, leading to shorter sleep durations and heightened emotional instability. These disruptions not only undermine mental health but also exacerbate the challenges of academic and social demands during adolescence. The complex connection between blue light exposure and the degradation of sleep quality and, consequently, mental health highlights the growing need for effective interventions as well as educational programs educating adolescents about the risks of nocturnal light exposure, particularly in today's technological settings.

The fifth Article about the impacts of blue light exposure from electronic devices on circadian rhythm and sleep disruption in adolescent and young adult students by Mudassir Alam and three other biologists in India begins with the role of blue light in disrupting circadian rhythms and negatively impacting sleep quality. It focuses on how adolescents are uniquely vulnerable to these effects due to their developmental sleep needs. Scientific findings show that exposure to blue light, particularly in the evening hours, suppresses the secretion of melatonin, an important hormone involved in sleep regulation. This impact would lead to poor outcomes, not only in the duration and quality of sleep but also in mental health. An important reason to study this deeper. The supporting evidence identified in the article suggests that the core argument posited by the article—blue light being harmful to sleep quality and mental stability is highly plausible. As such, the physiological grounding of melatonin suppression accounts for the anecdotal reason that exposure to blue light consistently, especially in the evening, causes adolescents' circadian rhythms to be dysfunctional. This is alarming in that sleep plays a large role in the cognitive processes and emotional balance that characterize adolescents. Analysis reveals that even if blue light can boost alertness and cognitive performance in the short term, these advantages are overridden by the negative effects of low-quality sleep, considering that adolescents usually operate in stressful environments. Moreover, the mental effects, such as increased anxiety and depression symptoms, that accompany sleep deprivation make interventions reducing exposure to blue light, especially during

late evenings, critical. In conclusion, the relationship between blue light exposure, sleep quality, and mental health is a concern for adolescents. It may be possible to improve sleep quality by mitigating blue light exposure, thereby enhancing the mental health outcomes of young people.

The sixth article about the 24-hour lighting scheme to promote alertness and circadian entertainment in railroad dispatchers on rotating shifts a field study conducted by PhD Sahin and PhD Figuerio, both light researchers in New York, United States of America abstract outlines a study focused on improving the sleep quality and circadian rhythms of shift-working railway personnel who are prone to fatigue as a result of disruptions to their light-dark exposure. The main objective of the research was to assess the effectiveness of a lighting intervention designed to realign circadian rhythms, improve sleep quality, and reduce sleepiness, which ultimately impacts performance in safety-critical roles. The study was specifically focused on the effects of various lighting interventions (a combination of red, blue, and white light) to correct circadian misalignment resulting from shift work. The study aimed to achieve four goals: (1) improve the alignment of the light-dark cycle with natural biological rhythms, (2) advance the onset timing of melatonin production in low-light conditions, (3) enhance both objective and subjective sleep quality, and (4) decrease subjective sleepiness, especially during off-duty hours. This successfully introduced greater synchrony between day-shift workers' light exposure and their rest-activity patterns. Although this did not advance the timing of the onset of melatonin, it was associated with lower sleep disturbances and had a positive, albeit nonsignificant, effect on subjective sleep quality. Of critical importance, subjective sleepiness was reduced while off-duty during the day shift. Although none of the desired outcome measurements advancement of the onset of dim light melatonin had been fully met. While the intervention appeared promising in improving sleep quality and off-duty sleepiness, all of the overall effects promoted circadian alignment rather than improving sleep onset timing and quality during duty hours directly.

The seventh Article relates to the effects of evening smartphone use on sleep and declarative memory consolidation in male adolescents and young adults conducted by Christopher Höhn and four other peers, providing how the exposure to blue light from digital screens has increased significantly with the widespread use of smartphones and other LED-based devices. Teenagers are more vulnerable as they are in their development stage and often use such devices near bedtime. Such disruption of natural circadian rhythms can have effects on not only sleep quality but also broader aspects of mental health, including emotional regulation and cognitive functions. Supporting evidence from

the article showed how melatonin effects are suppressed due to blue light and how adolescents recover faster than young adults. How at bedtime adolescents had normal melatonin levels but adults were still suppressed, and how neither group exhibited significant effects on sleep-dependent memory performance in their research. The analysis based on this is how age-related differences in physiological resilience in adolescents may be overcome by the quicker recovery of melatonin. How adult sensitivity suggests longer recovery times and a potential for greater impacts on deep sleep stages. Finally, how is memory consolidation still maintained despite melatonin or sleep-wake cycle changes? These results are consistent with guidelines for sleep hygiene that limit screen time before bed. The data indicates age-specific differences that may be used to create age-specific recommendations for teens and young adults.

The eighth article focuses on different sleep patterns during the COVID-19 pandemic and its association with mood, exercise, and light exposure, written by Mirian David, Gilberto Vieira, and nine other researchers. The study aimed to assess the sleep-wake patterns, mood, perceived stress, and behaviors of physical exercise and exposure to natural light among college students during the COVID-19 pandemic. The study took place between June and August 2020 and addressed how these factors affected the mental health and sleep quality of students. The importance of sunlight exposure and physical activity in good sleep hygiene and mental well-being during a global health crisis is pointed out by the findings. Students with higher sunlight exposure and exercise levels showed better quality sleep. Students who suffered from high daytime sleepiness and low sleep quality also had increased levels of anxiety, depression, and perceived stress; therefore, there was a relationship between disturbed sleep and mental health. These aspects were worse in women who had bad-quality sleep and more problems in mental health. This study gives a comprehensive overview of how environmental and behavioral factors affect sleep and mental health during a stressful time such as a pandemic. It indicates that not all students have the same sleep disruption but rather that individual habits like sunlight exposure and exercise could buffer the negative impacts of poor sleep. The association between poor sleep and higher levels of anxiety and depression reinforces the importance of addressing sleep quality as part of mental health care, at least during such crises as the COVID-19 pandemic.

The ninth article, by Hohman and Wong, on sleep quality and cognitive performances' role to whether quantity matters discusses the critical relationship between sleep quality, quantity, and cognitive functioning, especially among adolescents. It stresses that adequate sleep is crucial during this developmental stage due to significant physical and

emotional changes that occur. It underscores the importance of both the quality and duration of sleep to achieve optimal cognitive performance and overall well-being. Results show that poor sleep quality leads to undesirable outcomes concerning attention, memory, and emotional stability. Some studies documented positive correlations with improved sleep quality and higher cognitive scores, especially among adolescents, that enable them to achieve more in school. The article provides interventions for better sleep habits, such as a need for educational systems and parents to encourage sleep hygiene, which will improve the cognitive performance of adolescents as well as their health in general. The article significantly throws light on the adverse outcomes of sleep deprivation, framing those outcomes within the larger scope of adolescent cognitive and emotional development. It effectively merges quantitative data with qualitative assessment, focusing on the need for more targeted interventions toward improving the quality of sleep in adolescents.

The tenth article, by Gibbons and Rieger, is a literature review of the impact of circadian rhythms on sleep patterns. It explains how Circadian rhythms are intrinsic 24-hour cycles that play a very important role in sleep and many other physiological processes. Provides extensive insight into how disruptions of these rhythms impact sleep quality and health in general. It is thus essential to understand the association between circadian rhythms and sleep, particularly when factoring in extrinsic factors such as blue light exposure, which has become rampant in modern life, especially among adolescents and young adults. Research shows that exposure to blue light at night seriously compromises sleep quality by delaying the secretion of melatonin and altering its circadian rhythms. It is also established that adolescents are more susceptible to such impacts because their natural circadian rhythms already experience shifts in the process of development. Systematic reviews indicate that a greater percentage of adolescents suffer from sleep disturbances caused by exposure to blue light than adults. Research shows that blue light not only reduces the amount of sleep but also affects the quality of sleep because it increases sleep latency and reduces sleep efficiency. These same disruptions in this group of individuals, given blue light exposure often associated with delayed sleep timing and lowered melatonin levels, might exacerbate mental health disorders in adolescents, who already face high-stress academic pressures and the dynamics of socialization. The study emphasizes that efficient interventions should target light exposure, specifically among populations that are also at risk for both disorders of sleep and related psychological issues. Stress management, education in sleep hygiene, and reduced evening exposure to blue light could create significant benefits both in sleep quality and in mental health outcomes in adolescents.

The eleventh article published about the sleep characteristics across the lifespan in 1.1 million people from the Netherlands, United Kingdom, and United States is a systematic review and meta-analysis written by 83 authors that explains how blue light emitted by electronics has recently received attention concerning its potential adverse effects on our health. It possesses a wavelength that affects the circadian rhythm, thereby affecting sleep patterns and, consequently, general well-being. This systematic review seeks to summarize the available evidence related to the impact of exposure to blue light on sleep quality, performance, and mental health, especially among adolescents who are most exposed to screens. Supporting evidence for the sentence includes various studies that have documented altered sleep patterns and increased reports of insomnia linked to prolific digital screen use among adolescents. For example, a study revealed that increased screen time correlates with delayed bedtimes and insufficient sleep durations among children, raising concerns about the long-term mental health ramifications stemming from disrupted sleep cycles due to blue light. The analysis reveals a dual perspective on blue light exposure: while it can enhance alertness during the day, its evening effects are predominantly negative. This dichotomy suggests the nuances in how blue light interacts with human biology, particularly in a demographic that increasingly relies on technology for education and leisure. The findings imply a need for strategies that balance the benefits of daytime blue light exposure with the necessity of safeguarding sleep hygiene in adolescents. This critical discourse about how widespread electronic use affects not only sleep duration but also mental states, such as anxiety and depression, which are rife in this age group, is underlined by the reviewed literature. An urgent need to address these issues through educational programs and parental guidance on technology use and its implications for sleep and health among adolescents is accentuated.

The twelfth article published by Harvard Health 2020 is on the effects of blue light on sleep quality and solutions, blue light glasses, and whether they can improve one’s sleep. The possible impacts of blue light on sleep and human health in general make the topic quite important. Light from electronic devices has been known to disrupt sleep cycles and cause interference with one's circadian rhythm among teenagers. Knowing how the effects of blue light will influence the quality of their sleep and, consequently, their mental well-being has become imperative in this growing digital age. The accumulated research demonstrates that though exposure to blue light increases alertness during the day, night exposure to it disrupts sleep. Teenagers, being the most vulnerable section to this, are at higher risks of being affected cumulatively due to the basic need for them to

gain enough sleep to adequately facilitate cognitive and emotional growth in their brains. This interference with sleep due to the growing use of digital equipment compromises their academic performance as well as their mental well-being.

The thirteenth article from Harvard Health in 2021 is about the dark side of blue light. Exposure to blue light, especially from electronic devices, has gained much attention because of its significant effect on sleep quality and general health. Harvard Health Publishing points out how artificial lighting, particularly blue light, interferes with the biological clock of the body and can pose risks to health, such as diabetes and heart disease. The increased reliance on artificial light changes our natural circadian rhythm, thereby making sleep-related problems and mental health issues worse, especially in teenagers. Blue light exposure has been demonstrated to affect sleep quality by suppressing melatonin production, which in turn delays the onset of sleep and changes the circadian rhythms. This effect has been associated with an increased risk of mental illness, such as anxiety and depression. However, it has also been found that blue light enhances cognitive performance, including mood, learning, and decision-making, during the day. Systematic reviews indicate a balance is required, especially among vulnerable groups such as adolescents, as blue light is associated with impairing sleep but also enhancing alertness. Analysis of the evidence yields a complex relationship between light exposure to blue and adolescent health. While daytime exposures to blue light increase alertness and cognitive performance, blue light at night is particularly detrimental to sleep quality; it even worsens existing mental health conditions. Among adolescents, circadian rhythms of wakefulness during the day and at night naturally predispose people to be awake at certain times, leading to sleep deficiency, which significantly impacts student performance and emotional stability. Reduce exposure at night to maximize sleep and mental health with appropriate exposure during the day, which shall enhance cognitive abilities and academic performance. Thus, an effective strategy should manage blue light exposure to maximize the beneficial effects of blue light at any time of the day by minimizing the adverse effects through exposure at night. A nuanced approach is necessary to promote healthy sleep patterns and support the mental well-being of adolescents.

The fourteenth article, Harvard Health 2019 on whether blue light from electronic devices increases one's risk of macular degeneration or blindness, explains how blue light emitted by various electronic devices has generated issues about its possible contribution to eye health problems, more specifically, macular degeneration and blindness. Specialists often have questions about whether exposure might increase the risk for this

kind of disease, primarily among smartphone, tablet, or computer users. While blue light does possess certain characteristics that could impact health, this is not directly relevant to macular diseases. Emphasizes that the emission of blue light through digital devices does not pose an imperative threat to the retina or an elevated predisposition to macular degeneration. Blue light is emitted more vigorously in natural sources, specifically sunshine, than in everyday gadgets. It is indeed proven that long exposure to display screens in the late afternoon affects sleep as there is a hindrance in the release of the hormone melatonin, which is responsible for the sleep-wake cycles.

The fifteenth article about how blue light affects sleep was written by Mikulska and Skoczylas, both researchers in Poland. Their article explains that the circadian rhythm, which controls sleep patterns, is significantly affected by light exposure, especially blue light. Although blue light is useful for alertness and cognitive performance during the day, exposure to it at night inhibits the secretion of melatonin, causing poor sleep quality. Adolescents are particularly vulnerable to circadian disruptions, and they often have longer exposure to blue light through screens in the evening, which can affect their sleep and psychological well-being. This suggests the need for controlling blue light exposure, particularly at night, for adolescents to improve sleep health. Supporting evidence explains how Blue light entrains the body clock by acting on the SCN, however, night exposure desynchronizes the rhythm. How long prolonged evening exposure to LED screens delays the onset of sleep and impairs the quality of sleep. It also mentions there is inconsistent evidence where some studies indicate decreased sleep duration and efficiency among adolescents who are exposed to blue light before bedtime. Increased latency or wait to sleep and decreased deep sleep(N3 stage) have also been reported in some studies. The psychological effects of the study mention how chronic sleep deprivation due to circadian rhythms may aggravate anxiety and depression in adolescents. The analysis is how the physiological susceptibility of adolescents to blue light is based on their more rapid melatonin production and, therefore, a rather sensitive sleep-wake cycle. The widespread use of LED devices makes the problem acute, as youth frequently employ such devices just before bedtime. These findings outline the need for interventions such as blue-light filtering and behavioral modifications to protect adolescents from the adverse effects of nighttime blue-light exposure on sleep and mental health.

In summary, the articles collectively examine the effects of blue light exposure on sleep quality and mental health in adolescents, emphasizing the biological and behavioral disturbances that occur due to the overuse of electronic devices during the evening. Some key findings include the suppression of melatonin production, delayed sleep onset, and a general decrease in overall sleep quality, resulting in increased sleep latency and decreased restorative sleep. This disruption exacerbates mental health issues, such as anxiety and depression, that are already common in the adolescent developmental stages. During adolescence, natural circadian rhythms tend to be phase-delayed, making adolescents vulnerable to the adverse effects brought about by late-night screen exposure. Moreover, other literature indicates potential interventions through blue-light filtering technology or setting curfews for devices to minimize these negative impacts.

Studies emphasize the two-way relationship between poor quality of sleep and mental health, revealing that poor quality of sleep not only degrades cognitive and emotional functions but also sets a cycle of chronic sleep deprivation with psychological stress. The above findings urge for targeted intervention in exposure to blue light, as this may further enhance sleep hygiene and adolescent mental health in an increasingly digitalized world.

CHAPTER III
RESEARCH METHODOLOGY

Seeks to explore the influence of blue light exposure on sleep quality and mental health in adolescents and understand their perceptions, behaviors, and coping strategies. Recapping to know how to respond and help the student's research seeks to answer the following:

1. Do the grade 11 students of three different schools perceive the impact of blue light exposure on their sleep quality and emotional well-being?
   This question is directed to help provide a foundational background towards the main question's results. Helping to better understand the differences between the three schools and evaluate the reasons for the difference.
   - 1.1. What are the participants' experiences regarding the effects of screen time before bed?
   - 1.2. How do participants describe changes in their emotional state related to their screen time habits?
   - 1.3. What coping mechanisms do participants use to manage blue light exposure and its impacts on sleep?
2. Is there a significant gap between the three schools' students in Awareness, behavioral Patterns, and Coping Strategies?
   This is the main question aimed at getting an idea of the subjective interpretations and generally agreed-upon experiences related to the issue of blue light exposure and its impact on sleep quality and emotional well-being during adolescence. Strive to identify whether adolescents are aware that their exposure to blue light influences their circadian rhythm, sleep quality and duration, and other secondary outcomes on mental health by impacting mood, stress levels, and emotional regulations listed within this study.
   - 2.1. (Awareness) How do the grade 11s perceive the existence of blue light and its impact? What do they believe or think about it and its role in affecting sleep and mental health?
   - 2.2 (Behavioral Patterns) Why do the grade 11s use blue light-emitting devices such as smartphones, tablets, or computers in the evening? What is their habit or routine that would increase the exposure?

2.3. (Coping Strategies) What coping strategies do the grade 11s use to reduce their exposure to blue light, including blue light filters, limiting screen time at night, or participating in other sleep hygiene behaviors?

This investigation will gain an understanding of how youth conceptualize and cope with the advancements in technology use, sleep health, and emotional well-being, which in turn could help highlight the need for educational or technological interventions to address these.

Design of the Study

This qualitative study employs semi-structured interviews and open-ended surveys to explore adolescents' experiences with blue light exposure. "Research is formalized curiosity. It is poking and prying with a purpose," by Zora Neale Hurston(1891-1960). Qualitative research is a method focused on understanding human behavior, experiences, and social phenomena through non-numerical data, involving more collecting and data such as interviews, focus groups, observations, and textual materials to explore the "why" and "how" of a topic.

Population and Sampling

The study involves 66 grade 11 participants from Livingstone International School, Angeles University Foundation, and the National University of Clark. 25 participants from Livingstone International School, 20 participants from Angeles University Foundation, and 21 from National Univesirty of Clark. A total of 21 male students and 45 female students. Ages from the years 2005 to 2008, critical development stage from middle to late adolescent transition. In this age group, there is more development in advanced reasoning and reflective capabilities. Having experienced significant cognitive, emotional, and social changes, this age group is insightful for studies on identity formation, decision-making, or peer influence. This is also an age for balance, wherein they are young enough to recall early adolescent experiences yet old enough to anticipate young adult challenges. This sample was chosen via purposive sampling as it was aimed to be a case-controlled study, having it be concerning their medical situations with the effects of blue light, and to ensure an accurate longitudinal study.

Research Instrument

The researcher will use a comparative survey based on validated instruments like the Pittsburgh Sleep Quality Index(PSQI) and the General Anxiety Disorder-7(GAD-7). The survey will contain three questions per sub-topic (Awareness, Behavioral Patterns, Perceived Effects, and Coping Strategies) to understand and explore the knowledge and insights of the current grade 11 student. This portion is to gather more overview knowledge about Blue light, whereas the PSQI is to evaluate how much sleep the students had during February in total, asking questions of how long they stay awake in bed to how long they get the total amount of hours asleep. The purpose is to compare their awareness and how they implement their knowledge. This follows the GAD-7, where the results will determine the effects of sleep deprivation and answer any other factors involving their emotional distress apart from using their devices in the late evening. Before the testing of the PSQI and GAD-7 scores of the grade 11 students, there will be questions regarding their awareness, behavioral patterns, perceived effects, and coping strategies to help build further a foundation on their differences in blue light exposure.

Validation of Instrument

Per identifying the perceived effects and awareness of Blue light effects on the Grade 11 students, the researcher can find two standard survey tests, the Pittsburgh Sleep Quality Index(PSQI) and the General Anxiety Disorder-7(GAD-7), both having been used many times in the field of psychology and for any civilian as it is published publicly with the scoring graph attached.

Method and Data Collection

The researcher will be using one instrument, the survey will follow in total, lasting approximately 6 to 12 minutes. The surveys would then be collected by the researcher, and the data of each participant would be compared and analyzed to evaluate the understanding of Grade 11 students on the effects of blue light. The survey is focused into a structured survey divided into 9 sections. To be collected from February 16 till March 3, 2025, a total of 15 days on 3 separate Google Forms.

Plan of Data Analysis

To provide simple summaries about the measures and samples of this study, descriptive statistics will be used in analyzing the data alongside One-way ANOVA, Post Hoc Tests, and the Kruskal-Wallis Test in consideration to comparing three different groups. The One-Way ANOVA is for three or more independent groups on a single variable, telling if there is a significant difference between at least one pair of groups. Post Hoc Tests if ANOVA finds a difference to identify which specific group is different, and the Kruskal-Wallis Test for if the data result isn't normally distributed.

1	***Making Survey***	
	Introduction (Gender, Strand)	1
	Consent	2
	Awareness, Perceived Effects, Behavioral Patterns	3
	GAD-7 & PSQI	4
2	***Sending Survey***	
	Branch out to Peers	1
	Branch in reuqest to teachers for assistance	2
3	***Data Analysis***	
	Google Excel - *Tallying the results of awareness(2 questions), GAD-7 (7 questions), and the PSQI (19 questions).*	
	Jamovi - *To find the significant differences in the research in both results and samples.*	
4	***Results***	
	Descriptive Analysis based on results	1
	Graphs and Charts from Survey	2
	Conclusion	3

Table 3. Plan of Data Analysis Table chart

CHAPTER IV
RESULTS

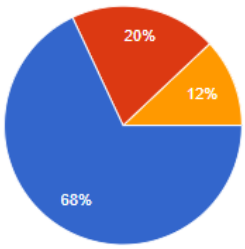
The results are divided into two portions, The first portion is a comparison between the graph and chart responses created from the Google forms from the participants' responses. This portion is to compare and see what the top responses from each group are and how they respond to Blue Light effects during February. The second portion is using Jamovi to use ANOVA to acquire the significant difference between all three samples.

1. Sample Space Identities

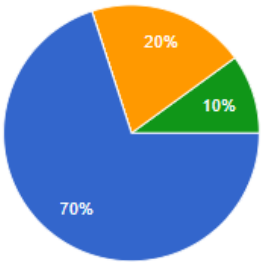
a. What strand are you pursuing?

Blue - STEM/STEAM
Red - ABM
Yellow - HUMSS
Green - GAS

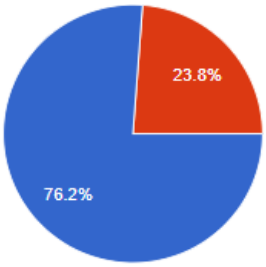
1.1. Livingstone International School



1.2. Angeles University Foundation



1.3. National University of Clark

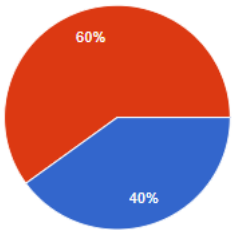


b. Gender

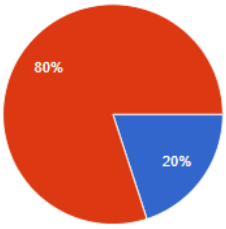
Red - Female

Blue - Male

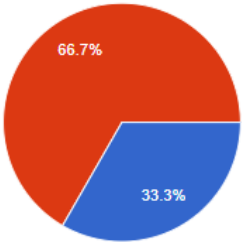
1.4. Livingstone International School



1.5. Angeles University Foundation

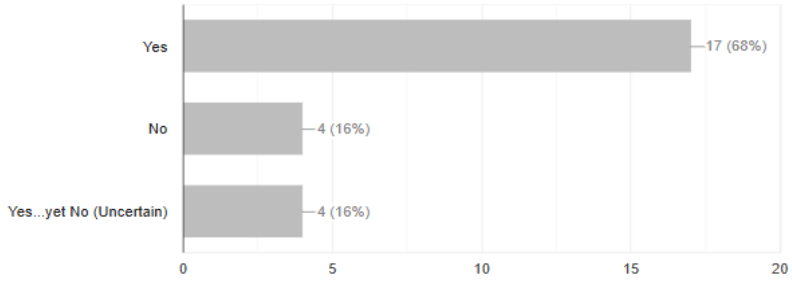


1.6. National University of Clark

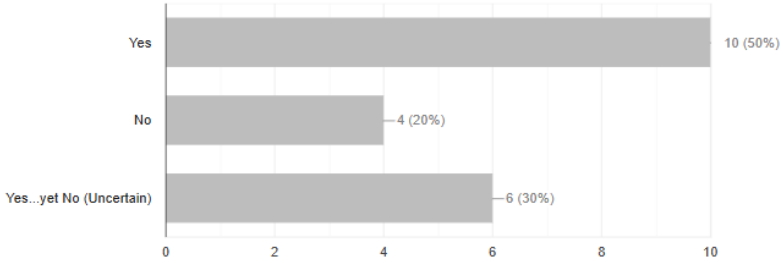


2. Blue Light (this section aimed to see how they perceived the term blue light)
Do you know what Blue Light means?

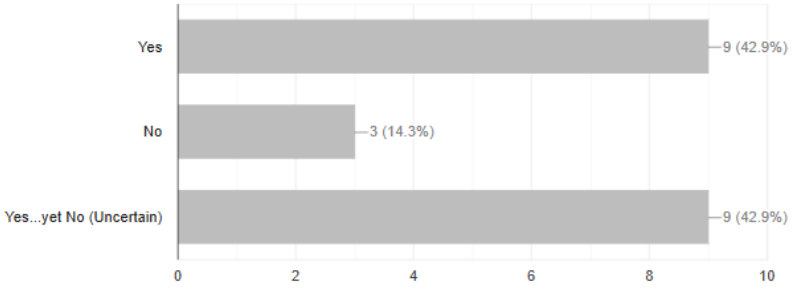
2.1. Livingstone International School



2.2. Angeles Univeristy Foundation

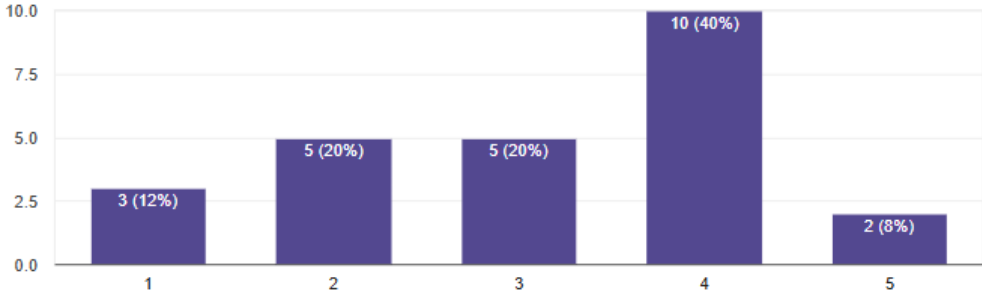


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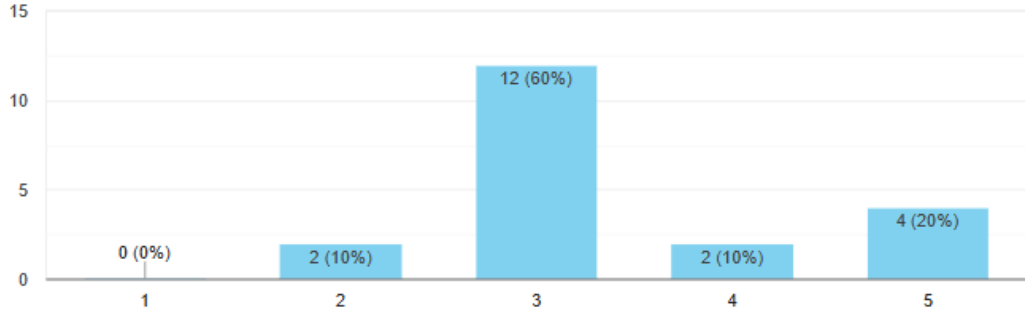


Confidence Level

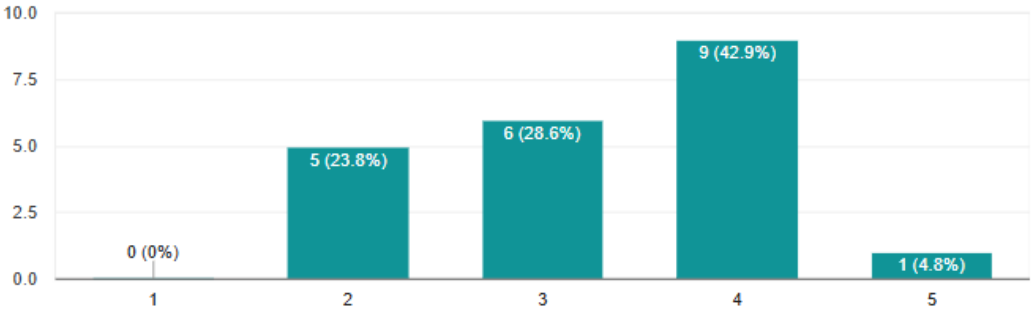
2.4. Livingstone International School



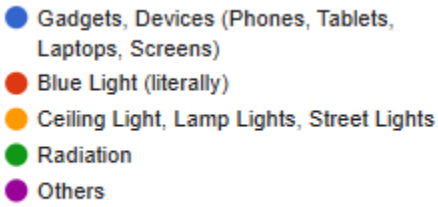
2.5. Angeles Univeristy Foundation



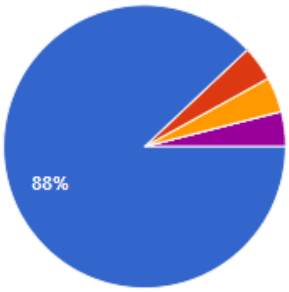
2.6. National University of Clark



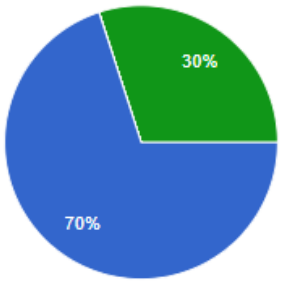
What do you think when you hear blue light?



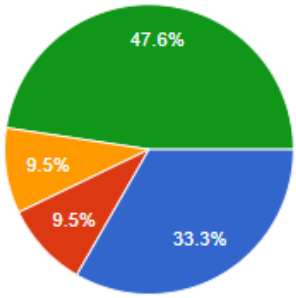
2.7. Livingstone International School



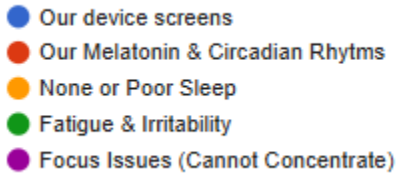
2.8. Angeles Univeristy Foundation



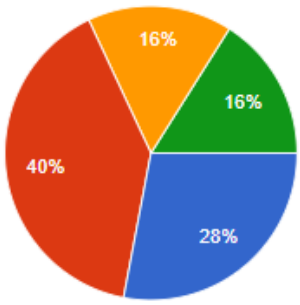
2.9. National University of Clark



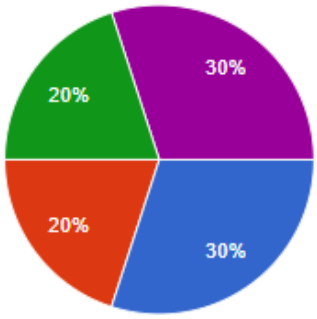
What do you think blue light affects mainly?



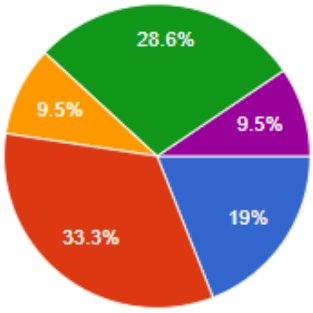
2.10. Livingstone International School



2.11. Angeles Univeristy Foundation



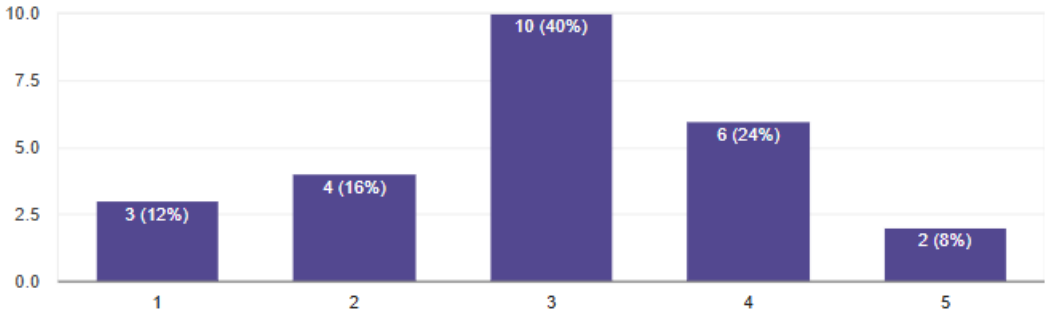
2.12. National University of Clark



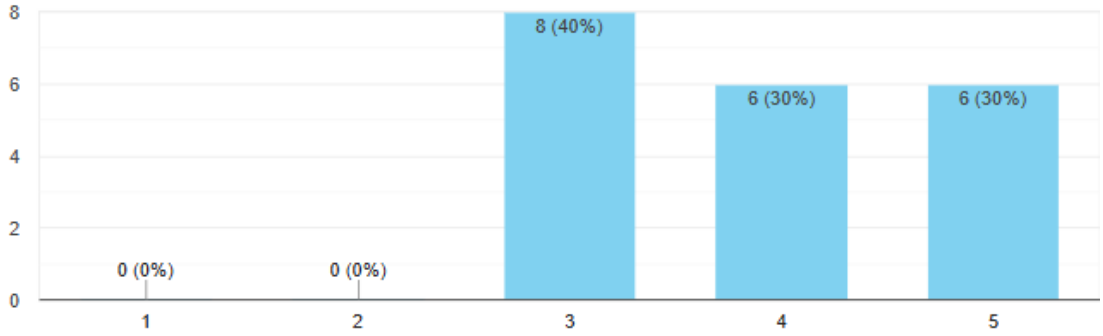
3. Awareness

What do you know about Blue Light and its effects on sleep and Mental Health?

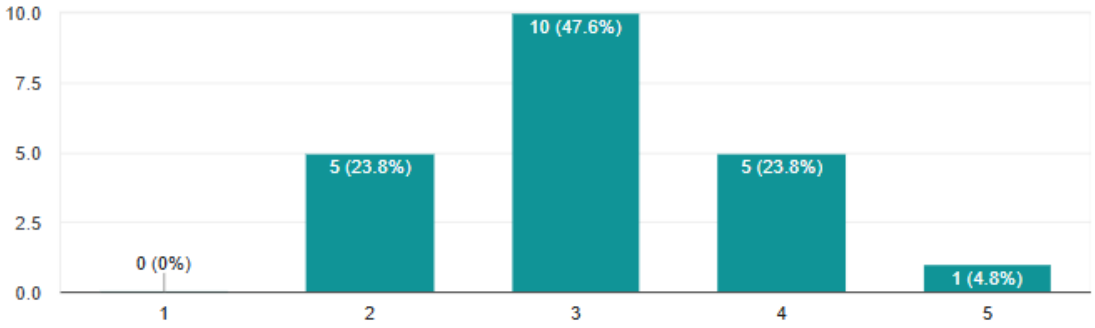
3.1. Livingstone International School



3.2. Angeles Univeristy Foundation

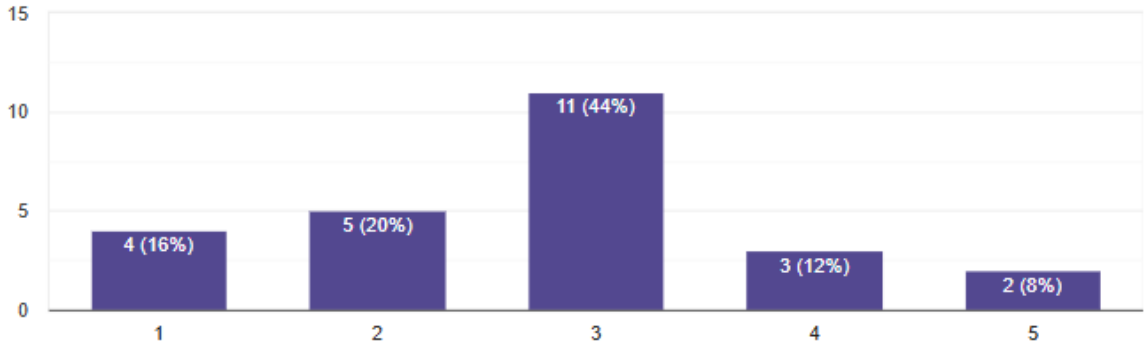


3.3. National University of Clark

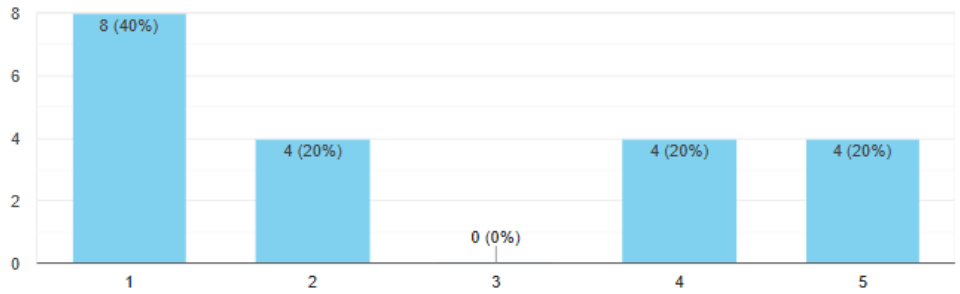


Have you ever heard about how blue light influences melatonin levels or circadian rhythms?

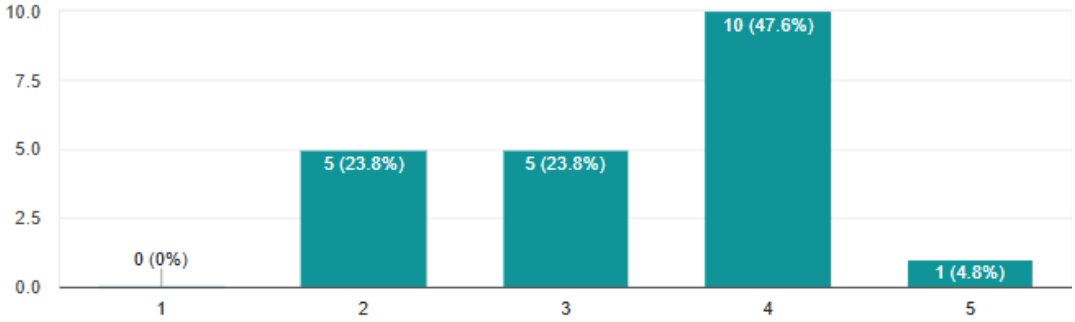
3.4. Livingstone International School



3.5. Angeles Univeristy Foundation

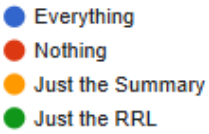


3.6. National University of Clark

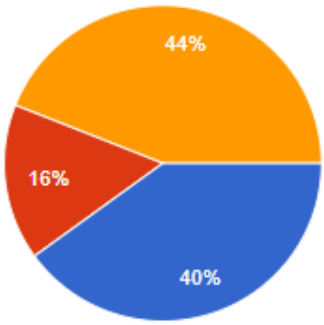


4. **Blue Light Definition** - This portion is to provide the students knowledge of the topic of the survey. (5th section of the form)**

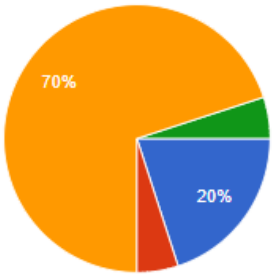
What part did you read?



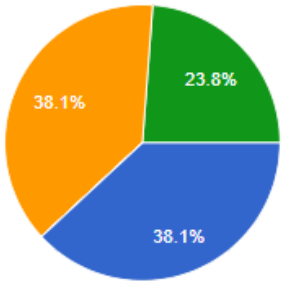
4.1. Livingstone International School



4.2. Angeles Univeristy Foundation



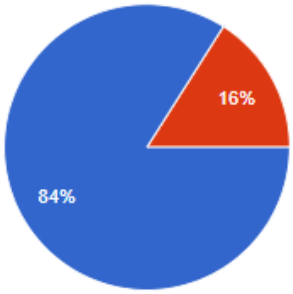
4.3. National University of Clark



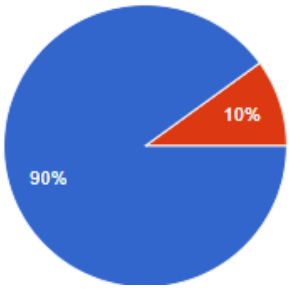
Did you Google or search online for further explanations?



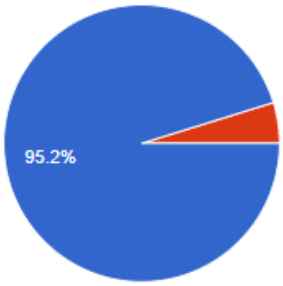
4.4. Livingstone International School



4.5. Angeles Univeristy Foundation

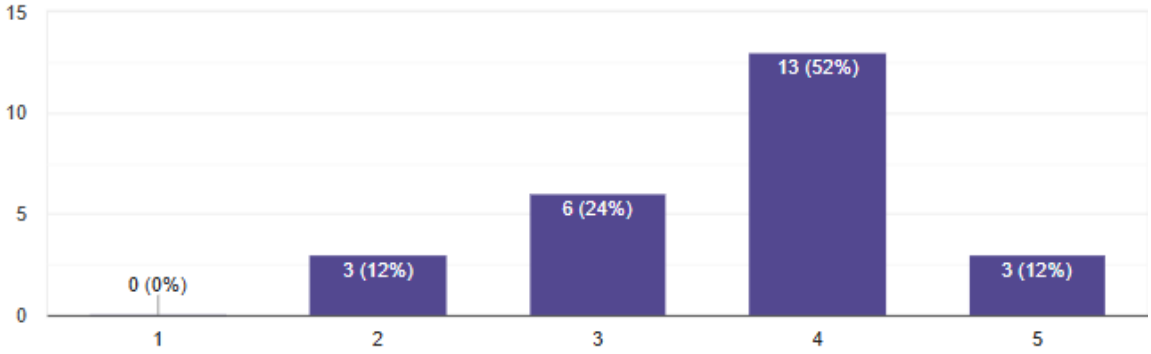


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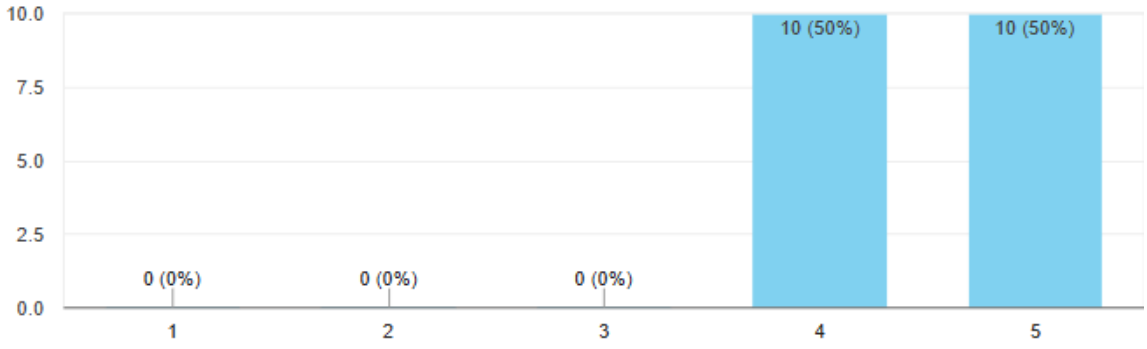


Re-asking, how much do you know about Blue light and its effects on Sleep and Mental Health?

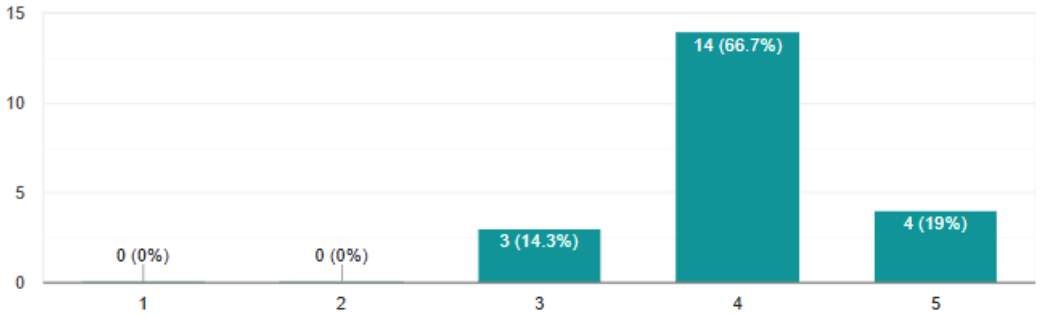
4.7. Livingstone International School



4.8. Angeles Univeristy Foundation



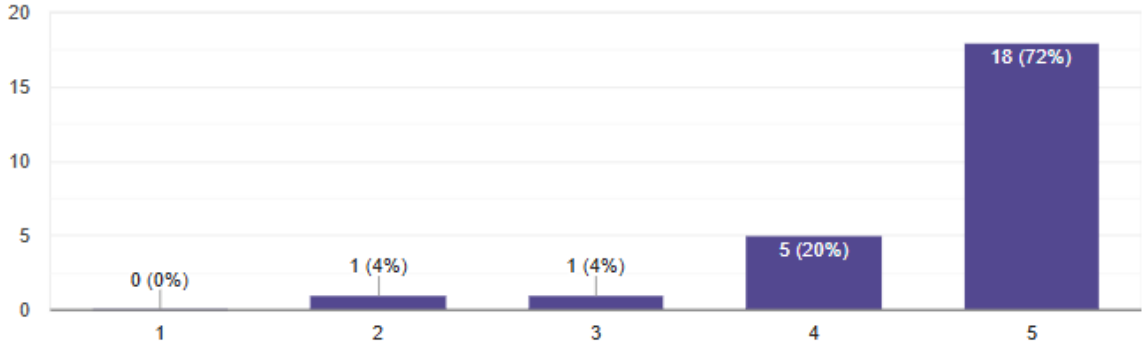
4.9. National University of Clark



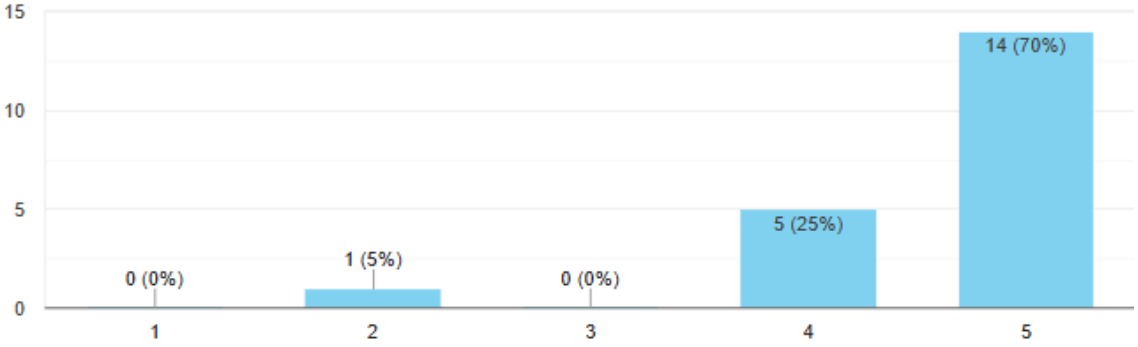
5. Behavioral Patterns, Perceived Effects, & Coping Strategies

How often do you use electronic devices at night?
1 = Not so Often 5 = Very Often

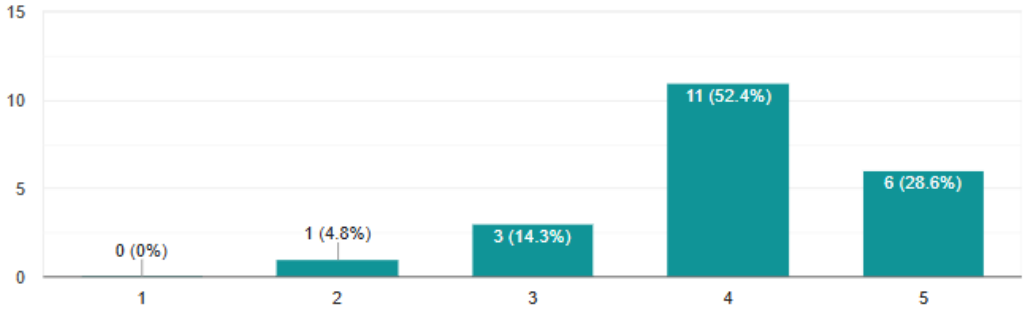
5.1. Livingstone International School



5.2. Angeles Univeristy Foundation



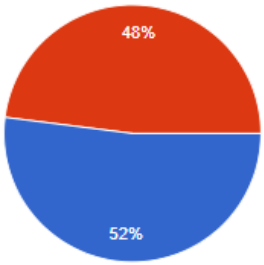
5.3. National University of Clark



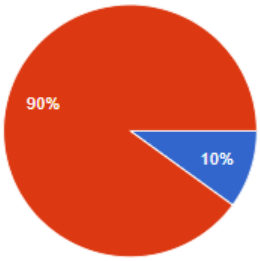
Do you take any breaks from screen use before going to bed?



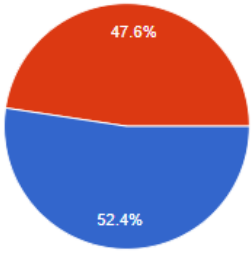
5.4. Livingstone International School



5.5. Angeles Univeristy Foundation



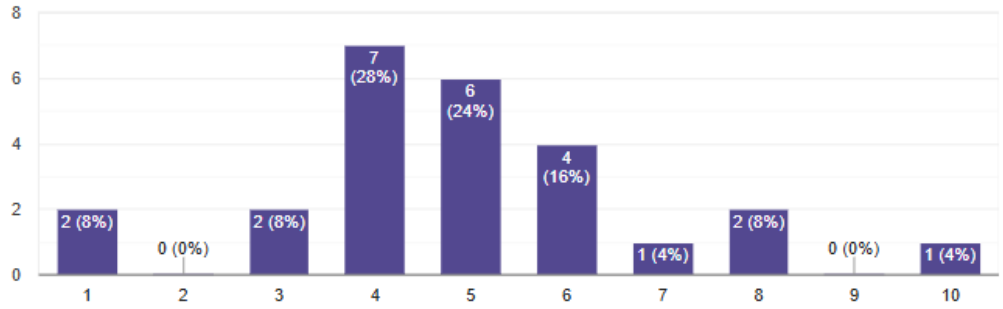
5.6. National University of Clark



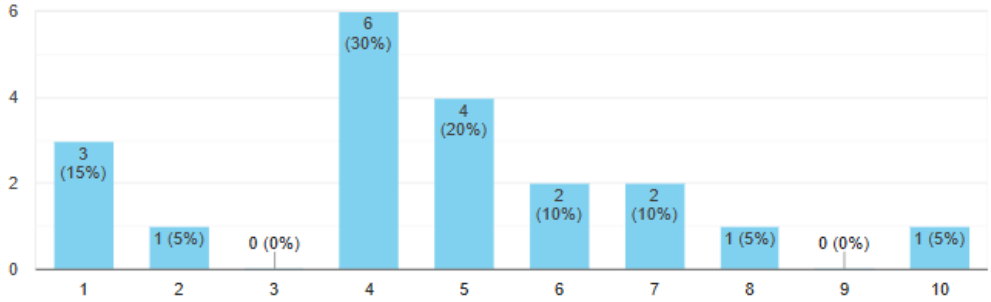
How long do you use your devices before you sleep?

1 = 8:00 pm 10 = 6:00am

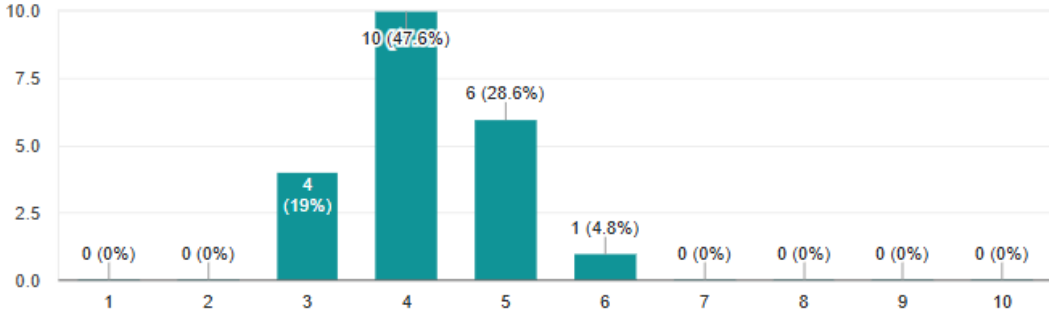
5.7. Livingstone International School



5.8. Angeles Univeristy Foundation



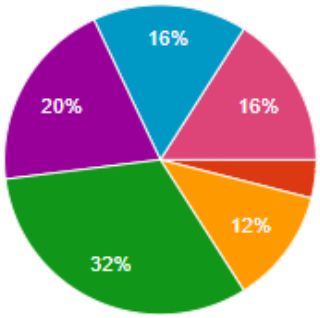
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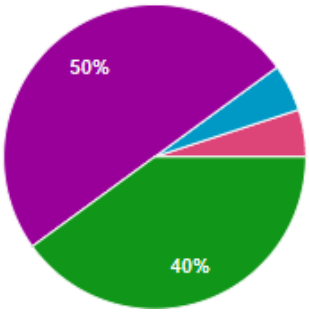
What is your average screen time? (all devices or phone)



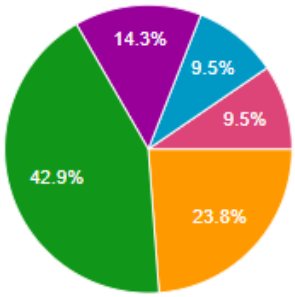
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5.11. Angeles Univeristy Foundation

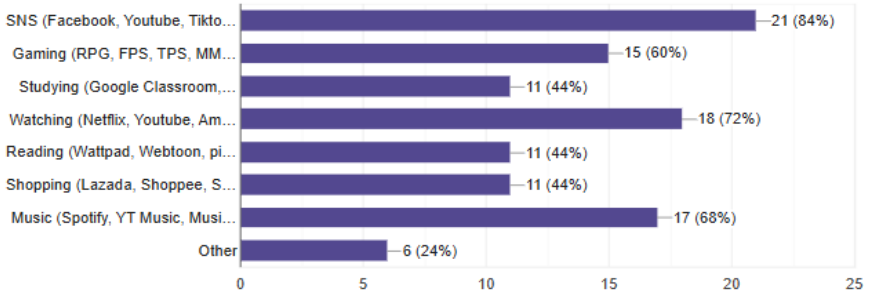


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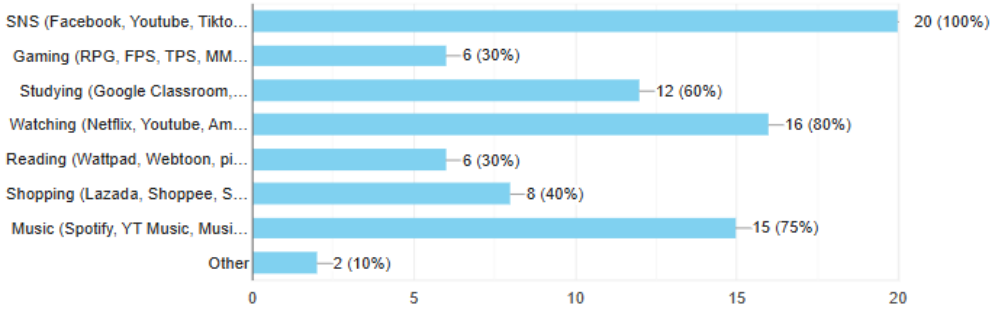


What types of activities (e.g., Social Media, Gaming, Studying) do you typically engage in on these devices?

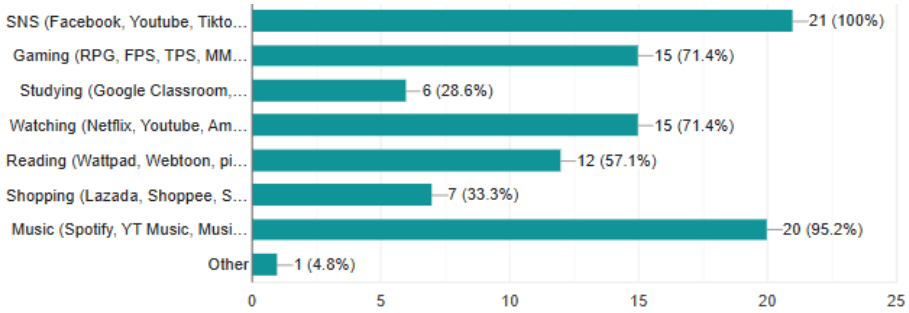
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5.14. Angeles Univeristy Foundation



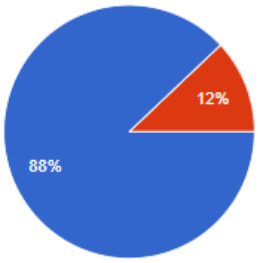
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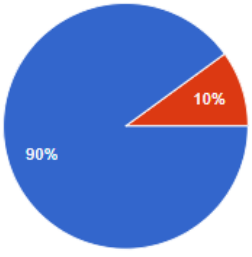
Do you feel that using devices at night affects your ability to fall asleep or stay asleep?



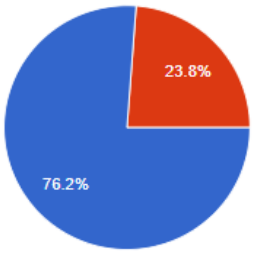
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5.17. Angeles Univeristy Foundation



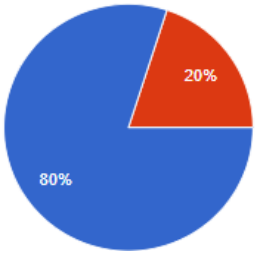
5.18. National University of Clark



Have you noticed any changes in your mood, stress levels, or energy after a night of disrupted sleep?



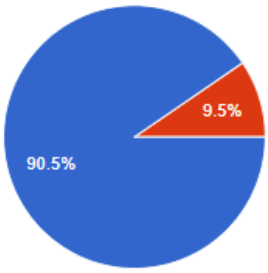
5.19. Livingstone International School



5.20. Angeles Univeristy Foundation

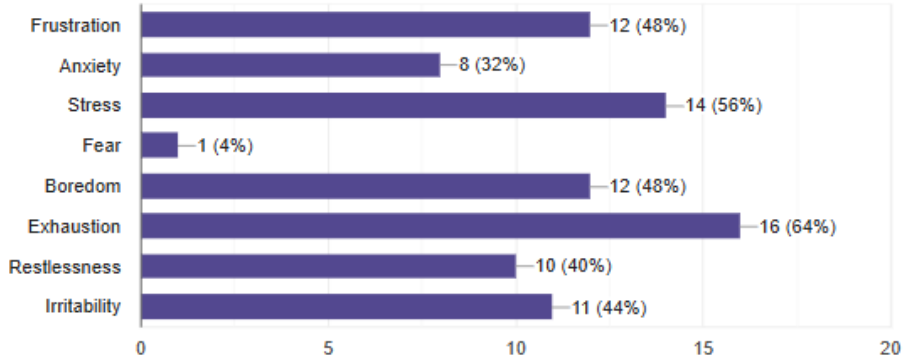


5.21. National University of Clark

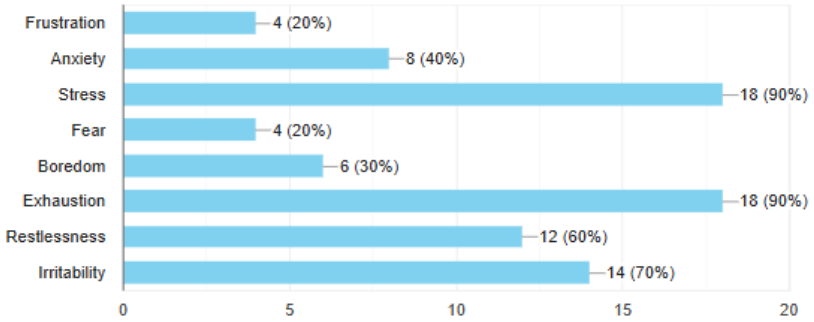


What emotions do you associate with sleep difficulties? (Pick at most 3)

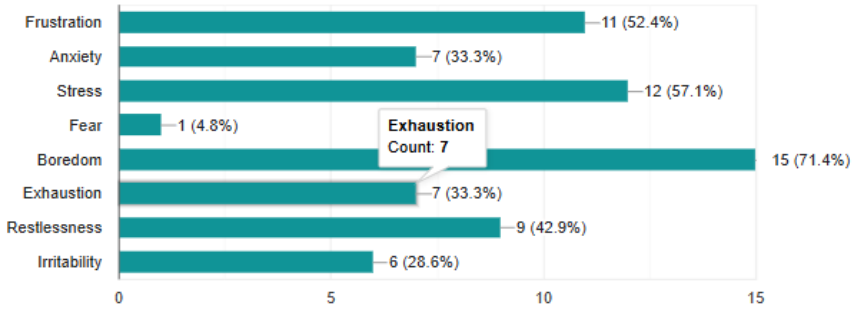
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5.23. Angeles Univeristy Foundation



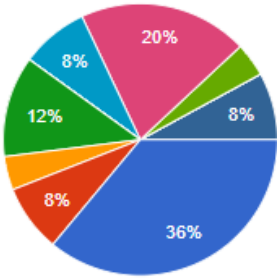
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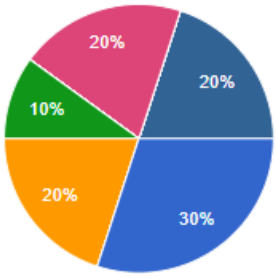
Have you tried any methods to reduce your screen at night? (Max 3)



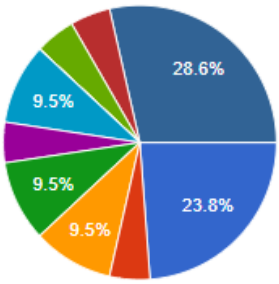
5.25. Livingstone International School



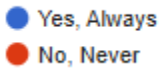
5.26. Angeles Univeristy Foundation



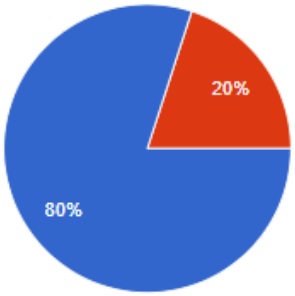
5.27. National University of Clark



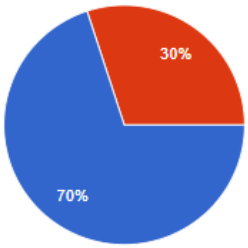
Do you use blue light filters or apps to manage screen exposure in the evening? (Warm light, Eye protection, or Dark mode)



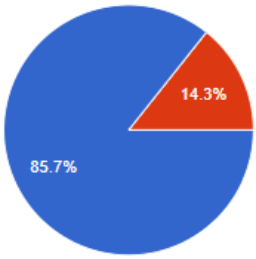
5.28. Livingstone International School



5.29. Angeles Univeristy Foundation



5.30. National University of Clark



This is the second portion containing ANOVA results for the Awareness section, the General Anxiety Disorder scale, and the Pittsburgh Sleep Quality Index. As mentioned previously, the One-Way ANOVA is for three or more independent groups on a single variable, telling if there is a significant difference between at least one pair of groups. Post Hoc Tests if ANOVA finds a difference to identify which specific group is different, and the Kruskal-Wallis Test for if the data result isn't normally distributed.

1. Awareness

1.1. The mean awareness scores for the three schools are quite similar, with AUF students averaging 6.50, NU students 6.43, and LSIS students 5.76.

1.2. The standard deviation (SD) indicates moderate variation, particularly for AUF(SD = 2.33), and LSIS(SD = 1.98) suggests some students scored significantly higher or lower than the mean.

1.3. The p-value (0.385) indicates that the difference between schools is not statistically significant.

1.4. The Games-Howell test showed no significant difference between any pair of schools, wherein all p-values > 0.005.

1.5. ***The students across the three institutions have similar levels of awareness.***

2. General Anxiety Disorder scale GAD-7

2.1. LSIS students reported the highest average anxiety levels (Mean = 12.56), followed by AUF students (Mean = 10.10), while NU students had the lowest anxiety levels (Mean = 5.67).

2.2. The standard deviations(SD) suggest high variability in anxiety levels, particularly for LSIS students (SD = 4.47) and AUF (SD = 4.18), meaning some students have extremely high or very low anxiety.

2.3. The p-value (< 0.001) confirms that the difference between schools is statistically significant.

2.4. Post Hoc Test Comparison

2.4.1. AUF vs NU: AUF students have significantly higher anxiety than NU students (Mean Difference = 4.43; p = 0.002).

2.4.2. NU vs LSIS: LSIS students have significantly higher anxiety than NU students (Mean Difference = 6.89; p < 0.001).

2.4.3. AUF vs LSIS: No significant difference (p = 0.151), but LSIS still shows a higher mean.

2.5. Students from ***LSIS exhibit the highest levels of anxiety, followed by AUF, while NU has the lowest anxiety levels***. The possible explanations provided by their responses lead to it leaning between more pressure or highly competitive environments leading to greater anxiety levels. NU students might have better stress management mechanisms, leading to lower anxiety levels.

3. Pittsburg Sleep Quality Index PSQI

3.1. Higher PSQI scores indicate poorer sleep quality.

3.2. AUF (Mean = 7.85) and LSIS (Mean = 7.80) students report significantly poorer sleep quality compared to NU students (Mean = 4.86).

3.3. The p-value (< 0.001) confirms a significant difference between the schools.

3.4. Post Hoc Test Comparison:

3.4.1. AUF vs NU: AUF students have significantly poorer sleep quality than NU students (Mean Difference = 2.99; p = 0.002).

3.4.2. NU vs LSIS: LSIS students have significantly poorer sleep quality than NU students (Mean Difference = -2.94; p <0.001).

3.4.3. AUF vs LSIS: No significant difference (p = 0.998), meaning their sleep quality is similarly poor.

3.5. NU students have significantly better sleep quality than both AUF and LSIS students. Poorer sleep quality in AUF and LSIS students could be

linked to higher academic pressure, greater screen time before bed, more extracurricular demands, less effective coping mechanisms for stress, and balancing school-home life.

4. Final Conclusion

The null hypothesis (H_0) is rejected for behavioral patterns(Anxiety) and coping strategies(Sleep Quality), confirming significant differences between schools. It is not rejected for awareness, meaning students from all schools have similar awareness levels. Further inventions should focus on anxiety and sleep quality improvements in LSIS and AUF.

4.1. ANOVA results show a difference between the three schools ($p<.001$) for both GAD-7 and PSQI

4.2. The Post Hoc Test shows that NU significantly differs from the other two schools in anxiety and sleep quality.

4.3. Kruskal-Wallis test confirms these differences in case normality was a concern.

One-Way ANOVA

One-Way ANOVA (Welch's)

	F	df1	df2	p
Awareness	0.978	2	40.2	0.385
GAD-7	18.663	2	41.2	< .001
PSQI	10.521	2	40.7	< .001

Group Descriptives

	Schools	N	Mean	SD	SE
Awareness	AUF	20	6.50	2.33	0.521
	NU	21	6.43	1.60	0.349
	LSIS	25	5.76	1.98	0.397
GAD-7	AUF	20	10.10	4.18	0.934
	NU	21	5.67	3.35	0.731
	LSIS	25	12.56	4.47	0.895
PSQI	AUF	20	7.85	2.94	0.658
	NU	21	4.86	2.13	0.464
	LSIS	25	7.80	2.99	0.597

Assumption Checks

Homogeneity of Variances Test (Levene's)

	F	df1	df2	p
Awareness	2.732	2	63	0.073
GAD-7	0.777	2	63	0.464
PSQI	2.087	2	63	0.133

[3]

Post Hoc Tests

Games-Howell Post-Hoc Test – Awareness

		AUF	NU	LSIS
AUF	Mean difference	—	0.0714	0.740
	p-value	—	0.993	0.502
NU	Mean difference		—	0.669
	p-value		—	0.422
LSIS	Mean difference			—
	p-value			—

Games-Howell Post-Hoc Test – GAD-7

		AUF	NU	LSIS
AUF	Mean difference	—	4.43	−2.46
	p-value	—	0.002	0.151
NU	Mean difference		—	−6.89
	p-value		—	< .001
LSIS	Mean difference			—
	p-value			—

Post Hoc Tests

Games-Howell Post-Hoc Test – Awareness

		AUF	NU	LSIS
AUF	Mean difference	—	0.0714	0.740
	p-value	—	0.993	0.502
NU	Mean difference		—	0.669
	p-value		—	0.422
LSIS	Mean difference			—
	p-value			—

Games-Howell Post-Hoc Test – GAD-7

		AUF	NU	LSIS
AUF	Mean difference	—	4.43	−2.46
	p-value	—	0.002	0.151
NU	Mean difference		—	−6.89
	p-value		—	< .001
LSIS	Mean difference			—
	p-value			—

Games-Howell Post-Hoc Test – PSQI

		AUF	NU	LSIS
AUF	Mean difference	—	2.99	0.0500
	p-value	—	0.002	0.998
NU	Mean difference		—	−2.9429
	p-value		—	< .001
LSIS	Mean difference			—
	p-value			—

CHAPTER V
DISCUSSION

The findings of this study indicate that while students from AUF, NU, and LSIS exhibit similar levels of awareness, significant differences exist in anxiety levels and sleep quality. The results suggest that awareness alone does not necessarily translate into better mental health outcomes, highlighting the need for interventions beyond informational campaigns.

Regarding anxiety, LSIS students reported the highest levels, followed by AUF students, while NU students exhibited the lowest levels. The statistically significant differences between the groups suggest that institutional factors such as academic pressure, competition, and stress management strategies may play a role in anxiety variations. The high variability in anxiety levels, as indicated by the standard deviations, also implies that individual differences contribute to how students experience and cope with stress.

In terms of sleep quality, AUF and LSIS students reported significantly poorer sleep compared to NU students. Since poor sleep quality is often linked to heightened anxiety, these results suggest a reinforcing cycle where stress leads to sleep disturbances, which in turn exacerbate mental health challenges. Potential contributing factors include academic workload, screen time before sleep, and ineffective coping strategies. NU students' comparatively better sleep quality and lower anxiety levels indicate they may have access to better stress management mechanisms, such as healthier lifestyle habits, institutional support, or lower academic pressure.

Conclusion

This quantitative research focused on the impact of blue light exposure from digital devices on the sleep quality and mental health of Grade 11 students. By comparing awareness, behaviors, and coping strategies regarding blue light exposure across three schools—Livingstone International School, Angeles University Foundation, and National University Clark—this study aimed to determine if significant differences existed between the institutions. The null hypothesis (H0), which proposed no significant difference, was ultimately rejected for anxiety and sleep quality, confirming disparities between the schools. However, it was not rejected for awareness, indicating that students across all institutions had similar levels of knowledge regarding blue light exposure.

With a total of 66 students surveyed between February 16 and March 3, 2025, the results showed that LSIS students exhibited the highest levels of severe anxiety, while AUF students reported the poorest sleep quality. In contrast, NU students recorded the lowest anxiety levels and the best sleep quality. These findings suggest that lifestyle factors, academic environment, and individual coping mechanisms influence how students interact with digital devices and manage stress. The data indicate that excessive screen time, particularly before bedtime, contributes to sleep disruptions, which in turn heighten anxiety levels and affect students' daily functioning. The disruption of circadian rhythms, caused by suppressed melatonin production due to blue light exposure, leads to a repetitive cycle of poor sleep and increased stress, negatively impacting academic performance and overall well-being.

These findings emphasize the importance of addressing both sleep quality and anxiety through targeted interventions, particularly for LSIS and AUF students, who showed the highest risk in these areas. Schools should prioritize mental health initiatives and sleep education to mitigate these issues and improve student well-being.

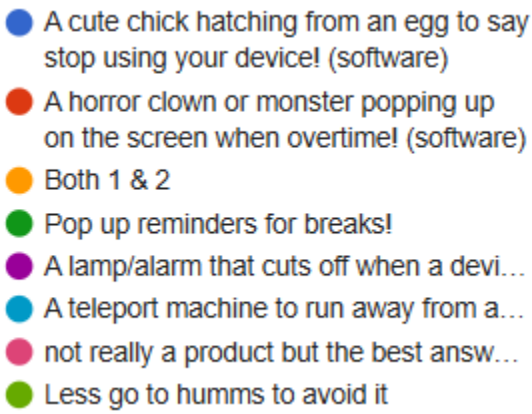
Recommendations

1. Mental Health Support Programs – Schools, particularly LSIS and AUF, should develop and implement programs focused on stress reduction, such as mindfulness sessions, counseling services, and peer support groups to help students manage anxiety effectively.

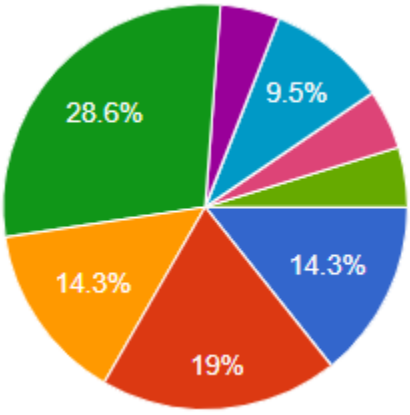
2. Sleep Hygiene Education – Institutions should educate students on the importance of sleep quality and provide strategies to improve sleep habits, such as reducing screen time before bed, establishing consistent sleep schedules, and limiting academic workload during late hours.
3. Institutional Policy Adjustments – Schools should review their academic structures to ensure a balanced workload that does not overly contribute to stress and sleep deprivation. Incorporating breaks and relaxation periods within the curriculum may help mitigate anxiety levels.
4. Further Research – Future studies should explore the specific institutional factors that contribute to the differences in anxiety and sleep quality across schools. Additionally, research should investigate individual coping mechanisms and resilience as moderators of stress and sleep disturbances.
5. Broader Study Sample – Expanding the sample to include more schools and diverse student populations would provide a more comprehensive understanding of the factors influencing mental health in different academic settings. A comparative study between STEM and non-STEM students could also reveal further insights into academic stress and coping strategies.

Here are the result recommendations decided by the participants for the researcher to possibly pursue:

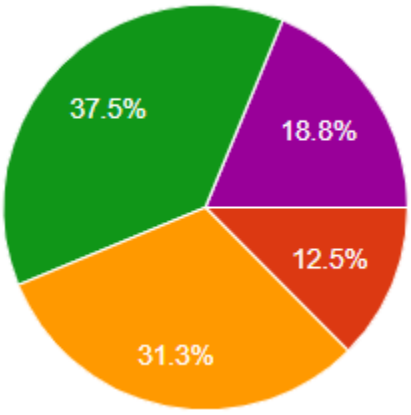
What product is best to stop using devices?



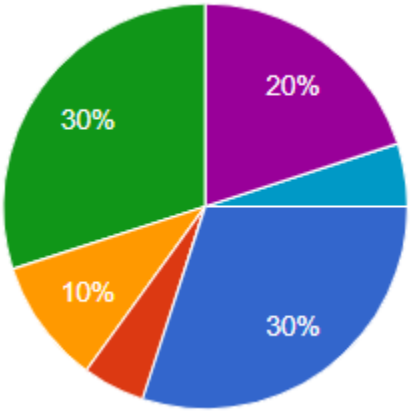
1. Livingstone International School



2. Angeles University Foundation



3. National University of Clark



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APPENDICES

Survey Questionnaire

Blue Light Survey (LSIS)

This is a Comparative Online Survey for Grade 11 students in LSIS ONLY. The aim is to aid in the study of "**Blue Light Exposure Comparison in Grade 11 Students from Livingstone International School, National University Clark, and Angeles University Foundation**". This study aims to compare adolescents' awareness, behaviors, and coping strategies regarding blue light exposure through comparative structured surveys. (This may look long...but it's a one sitting thing no worries <3)

Your email is just to confirm not a bot or repeats, thank you for understanding
Responses can be edited only until February 25th **Please be honest in answering**

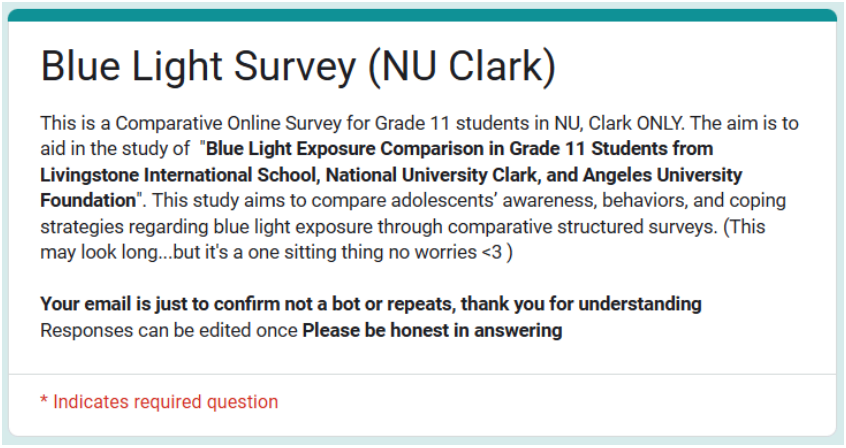
* Indicates required question

Blue Light Survey (AUF)

This is a Comparative Online Survey for Grade 11 students in AUF ONLY. The aim is to aid in the study of "**Blue Light Exposure Comparison in Grade 11 Students from Livingstone International School, National University Clark, and Angeles University Foundation**". This study aims to compare adolescents' awareness, behaviors, and coping strategies regarding blue light exposure through comparative structured surveys. (This may look long...but it's a one sitting thing no worries <3)

Your email is just to confirm not a bot or repeats, thank you for understanding
Responses can be edited only until February 25th **Please be honest in answering**

* Indicates required question



Blue Light Survey (NU Clark)

This is a Comparative Online Survey for Grade 11 students in NU, Clark ONLY. The aim is to aid in the study of "**Blue Light Exposure Comparison in Grade 11 Students from Livingstone International School, National University Clark, and Angeles University Foundation**". This study aims to compare adolescents' awareness, behaviors, and coping strategies regarding blue light exposure through comparative structured surveys. (This may look long...but it's a one sitting thing no worries <3)

Your email is just to confirm not a bot or repeats, thank you for understanding
Responses can be edited once **Please be honest in answering**

* Indicates required question

**From this point on, all surveys are alike

Significance of this study + Consent Form
PURPOSE OF THE STUDY

You are being invited to take part in a research study. Before you decide to participate in this study, you must understand why the research is being done and what your participation will involve. Please read the following information carefully and feel free to ask the researcher if there is anything that is not clear or if you need more information.

This quantitative research will focus on the blue light exposure emitted from digital devices on the sleep quality and mental health of grade 11 students. This study aims to compare adolescents' awareness, behaviors, and coping strategies regarding blue light exposure through comparative structured surveys.

This comparison is to be conducted between three different schools Livingstone International School, Angeles University Foundation, and the National University located in Clark grade 11 participants from the strands GAS, ABM, HUMSS, and STEM/STEAM.

Grade 11 students are among those most affected by extended screen time for both academic and recreational purposes. Blue light disrupts circadian rhythms by suppressing melatonin production, critical for initiating and maintaining sleep. This disruption leads to delayed sleep onset, reduced sleep duration, and poor sleep quality. Poor sleep, in turn, exacerbates mental health challenges, including heightened risks of anxiety, depression, and impaired cognitive performance.

Through this study the goal is to address the issue between students using devices in school and finding a solution for their sleep and mental health, by providing a statistical analysis to the effects of blue light, and how the students use it. To be later on creating a solution for this in Mixed Method research in Grade 12, Thank you.

STUDY PROCEDURES

The questions you will be asked will be as said and relate to Blue Light Effects **Please Do Not Research Prior to the next Sections** as this is the purpose for this study:

Awareness of Blue Light

Behavioral Patterns

Coping Strategies

And Average scores during 1st Term and 2nd Term (or 1st Quarter, 2nd Quarter depending on Institution) [This is optional portion]

DURATION

The duration is estimated approximately to take 15 - 20 minutes, exceeding depending on each student.

However, the availability of the form to stay open lasts till **February 25th**.

VOLUNTARY PARTICIPATION

Your participation in this study is voluntary. If you decide to participate, you will be asked to sign this consent form.

After you sign this consent form, you are still free to withdraw at any time and without giving a reason.

Withdrawing from this study will not affect the relationship you have, if any, with the researcher.

If you withdraw from the study before data collection is completed, your data will be deleted appropriately to protect your details.

Your participation in this study is entirely voluntary. If at any point you would like to review your remarks from your form, you have the right to do so. You may request to erase any part or all of the notes taken during.

(But pls consider my grds po tyyy)

RISKS

Entrusting to the researcher your identification alongside your responses. Personal or sensitive topics for mental health and personal privacy. To prevent these the researcher will fulfill their role in your responses to protect your confidentiality to the subject and identification from identification leaks and understanding towards your responses having it be your own words and will.

You may decline to answer any or all questions and you may withdraw your participation at any time if you choose. (Your name will not be included within the research, simply as Statistical Data)

BENEFITS

It is hoped that through this study you learn more of yourself and the topics involved. That being aimed towards the last 2 years of high school before college the goals and ideals for your future are closer to being visualized and preparing you for your future.

Offering insights into modern-day and future aspirations within the evolution of technology. May this study help benefit you by strengthening your many capabilities.

CONFIDENTIALITY

Your responses in this research will be anonymous. Every effort will be made by the researcher to preserve your confidentiality, including the following:

1. *Assigning codes/pseudonyms for participants that will be used on all research notes and documents*
2. *Keeping notes, interview transcriptions, and any other personal identifiers in the forms and not sharing with others not involved or required to know the information of the study.*
3. *Respect any requests the individuals taking the tests have, including removing or editing said comments or responses by the interviewee.*

The researcher will maintain the confidentiality of the data concerning both information of the participant and the participant's shares.

CONTACT INFORMATION

This study was approved by the Research Ethics Review Committee of LSIS. If you have any questions at any time about this study, or if you experience any non-normative sensations as a result of participation, you may contact the researcher whose contact information is on the first page. If you have any questions regarding your rights as a research participant, or if problems arise that you do not feel you can discuss with the researchers, please feel free to contact Smanuel180507@livingstone.ph.education. Thank you.

I have agreed to the terms and conditions *

- ◯ Yes, I am consenting to participate
- ◯ No, I am NOT consenting to participate

Next Clear form

Identity

In this section we are to gather basic information needed to be used in organizing the following responses. Please answer truthfully thank you.

Last Name (Will not be included in study results) *

Your answer

What strand are you pursuing? *

- ◯ STEM/STEAM (Science, Technology, Engineering, and Mathematics + Arts)
- ◯ ABM (Accounting, Business, and Management)
- ◯ HUMSS (Humanities and Social Sciences)
- ◯ GAS (General Academics Strand)

Gender *

- ◯ Female
- ◯ Male

Back Next Clear form

Blue Light

In this Section we are to analyze how well you understand what Blue Light is, Please do not research as in the next section context will be provided to those not knowing. Please be minded it is okay not to know, as such is the purpose for this study, thank you.

Do you Know what Blue Light means? * 0 points

- ◯ Yes
- ◯ No
- ◯ Yes...yet No (Uncertain)

Confidence level *

	1	2	3	4	5	
0 knowledge prior to this	◯	◯	◯	◯	◯	I know Blue Light well

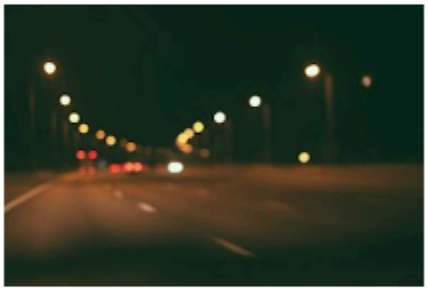
What do you think of when you hear Blue Light? *



Gadgets, Devices (Phones, Tablets, Laptops, Screens)



Blue Light (literally)



Ceiling Light, Lamp Lights, Street Lights



Radiation

Others

Answer this next portion if unaware, based on what you believe then

What do you think Blue Light effects **mainly**? *

- ◯ Our device screens
- ◯ Our Melatonin & Circadian Rhytms
- ◯ None or Poor Sleep
- ◯ Fatigue & Irritability
- ◯ Focus Issues (Cannot Concentrate)

Back Next Clear form

Awareness

This portion is to determine how well aware you are to the topic **Blue Light**

1 - Minimal Understanding
2 - Limited Understanding
3 - Basic Understanding
4 - Proficient Understanding
5 - Advance Understanding

What do you know about Blue Light and it's effects on sleep and Mental Health? *

	1	2	3	4	5	
Minimal Understanding	◯	◯	◯	◯	◯	Advance Understanding

Have you ever heard about how blue light influences melatonin levels or circadian rhythms? *

	1	2	3	4	5	
Minimal Understanding	◯	◯	◯	◯	◯	Advance Understanding

Back Next Clear form

Blue Light Definition

Summary:

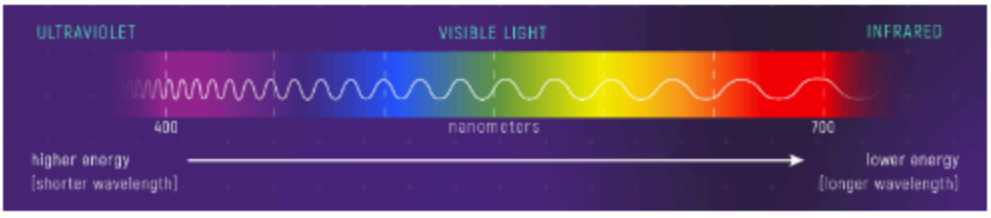
To put it simply Blue Light is a type of visible light displayed from our screens. Situationally, when you wake up in the morning, the sun helps, the reason to so is thanks to Blue Light, it is like a natural alarm clock. Now if you still have such an 'alarm' on your phone (If you use your devices at night), this hinders the amount of sleep you get by messing with your circadian rhythms (Sleep-Wake Cycle) and can lead to emotional distress.

Long Route (short RRL)

The literature review draws extensively on recent studies that explore the impact of blue light exposure on sleep and mental health. Key sources include the systematic review by Behar and Radrizzani (2022), which highlights how blue light suppresses melatonin and disrupts circadian rhythms, leading to delayed sleep and increased fatigue. Another crucial study by Prather and Hurst (2022) emphasizes the bidirectional relationship between sleep and circadian rhythms, detailing how blue light exposure exacerbates mental health issues like anxiety and depression. Wahl and Engelhardt (2021) contribute insights into the physiological mechanisms, showing how even short-wavelength light can misalign natural sleep-wake cycles. These findings collectively underscore the urgent need for interventions, such as using blue light filters and limiting screen time before bed, to mitigate the negative effects on adolescents. Together, the studies provide a strong foundation for understanding the pervasive risks associated with evening device use.

You may also now research online

Wavelength Chart - Harvard has their own quantitative research wherein they found blue light to be twice more damaging than green light.



Which part did you read? *

- ◯ Everything
- ◯ Nothing
- ◯ Just the Summary
- ◯ Just the RRL

Did you google or search online further explanations? *

- ◯ No
- ◯ Yes

Re-asking, How much do you know about Blue Light and it's effects on sleep and Mental Health? *

	1	2	3	4	5	
Minimal Understanding	◯	◯	◯	◯	◯	Advanced Understanding

Back Next Clear form

Behavioral Patterns, Perceived Effects, & Coping Strategies

Answer based on yourself this past month!!

How often do you use electronic devices at night? *

	1	2	3	4	5	
Not so Often	◯	◯	◯	◯	◯	Very Often

Do you take any breaks from screen use before going to bed? *

- ◯ Yes
- ◯ No

How long do you use your devices for before you sleep? *

	1	2	3	4	5	6	7	8	9	10	
8:00pm	◯	◯	◯	◯	◯	◯	◯	◯	◯	◯	6:00am

What is your Avg Screen Time? (all devices or phone) *

- ◯ Less than an hour
- ◯ 1 to 2 hours
- ◯ 3 to 4 hours
- ◯ 5 to 6 hours
- ◯ 7 to 8 hours
- ◯ 9 to 10 hours
- ◯ Over 11 hours

What types of activities (e.g., Social Media, Gaming, Studying) do you typically engage in on these devices? *

- ☐ SNS (Facebook, Youtube, Tiktok, Instagram, Discord, X, Threads, LINE, Snapchat, etc.)
- ☐ Gaming (RPG, FPS, TPS, MMO, MOBA, RTS, TBS, ARPG, BR, etc.)
- ☐ Studying (Google Classroom, Docs, Slides, etc.)
- ☐ Watching (Netflix, Youtube, Amazon, Disney+, Viu, Anilab, Tiktok, BiliBili, Dramabox, etc.)
- ☐ Reading (Wattpad, Webtoon, pirated sites, Manga, Manhwa, Manhua, Comics, Webnovels, etc.)
- ☐ Shopping (Lazada, Shoppee, Shein, Grab, FoodPanda, Temu, etc.)
- ☐ Music (Spotify, YT Music, Music player, etc.)
- ☐ Other

Do you feel that using devices at night affects your ability to fall asleep or stay asleep? *

- ◯ Yes
- ◯ No

Have you noticed any changes in your mood, stress levels, or energy after a night of disrupted sleep? *

- ◯ Yes
- ◯ No

What emotions do you associate with sleep difficulties? (Pick at most 3) *

- ☐ Frustration
- ☐ Anxiety
- ☐ Stress
- ☐ Fear
- ☐ Boredom
- ☐ Exhaustion
- ☐ Restlessness
- ☐ Irritability

Have you tried any methods to reduce your screen use at night? (Max 3) *

- ◯ None
- ◯ Keeping Devices out of reach (Charge outside of Bedroom, relying on a traditional alarm)
- ◯ Cutting off devices at a certain time in the day (Screen Curfew)
- ◯ Setting an Alarm or Reminder
- ◯ Locking certain apps at a certain time in the day (App Limits)
- ◯ Adjust Device Settings (Night Mode/Grayscale Model)
- ◯ Read a Physical Book
- ◯ Meditation & Relaxation
- ◯ Journaling
- ◯ Working out (Gym or Home)

Do you use blue light filters or apps to manage screen exposure in the evening? * (Warm light, Eye protection, or Dark mode)

- ◯ Yes, Always
- ◯ No, Never

Optional: What other strategies have you heard about or tried to improve your sleep or mental health in relation to screen use?

Your answer

Back Next Clear form

General Anxiety Disorder - 7 **GAD-7**

This portion is to compare the data of your anxiety levels

(Results will be provided to you upon request or you can solve it given the scoring guide)

0 = Not at all
1 = Several Days
2 = More than half the Days
3 = Nearly Every Day

Over the span of this month how often have you been bothered by the following problems?

Feeling nervous, anxious, or on edge *

	0	1	2	3	
Not at all	○	○	○	○	Nearly Everyday

Not being able to stop or control worrying *

	0	1	2	3	
Not at all	○	○	○	○	Nearly Everyday

Worrying too much about different things *

	0	1	2	3	
Not at all	○	○	○	○	Nearly Everyday

Trouble Relaxing *

	0	1	2	3	
Not at all	○	○	○	○	Nearly Everyday

Being so restless that it is hard to sit still *

	0	1	2	3	
Not at all	○	○	○	○	Nearly Everyday

Becoming easily annoyed or irritable *

	0	1	2	3	
Not at all	○	○	○	○	Nearly Everyday

Feeling afraid, as if something awful might happen *

	0	1	2	3	
Not at all	○	○	○	○	Nearly Everyday

If you checked any problems, how difficult have they made it for you to do your work, take care of things at home, or get along with other people?

- ◯ Not difficult at all
- ◯ Somewhat difficult
- ◯ Very Difficult
- ◯ Extremely difficult

Scoring GAD-7 Anxiety Severity
This is calculated by assigning scores of 0, 1, 2, and 3 to the response categories, respectively,
of "not at all," "several days," "more than half the days," and "nearly every day."

GAD-7 total score for the seven items ranges from 0 to 21.

0–4: minimal anxiety
5–9: mild anxiety
10–14: moderate anxiety
15–21: severe anxiety

Link instrument was taken from:
https://adaa.org/sites/default/files/GAD-7_Anxiety-updated_0.pdf

I would like: *

- ◯ To score my own total
- ◯ Request you send my total
- ◯ Not that interested to see my total scoring

Back Next Clear form

Pittsburgh Sleep Quality Index (**PSQI**)

This is the final portion for my research (Tysm for managing through this all)

This is an instrument used to measure the amount of sleep you get.

Instructions:

The following questions relate to your usual sleep habits during the past month only. Your answers
should indicate the most accurate reply for the majority of days and nights in the past month. Please answer
all questions.

During the past month, what time have you usually gone to bed at night? *

- ◯ 7:00pm to 8:00pm
- ◯ 9:00pm to 10:00pm
- ◯ 11:00pm to 12:00pm
- ◯ 1:00am to 2:00am
- ◯ 3:00am to 4:00am
- ◯ 5:00am to 6:00am
- ◯ Past 7:00am

During the past month, how long (in minutes) has it usually taken you to fall asleep each night?

Format: _##_ minutes ; If and only if longer in hours change minutes to hours (lowercase)

Your answer

During the past month, what time have you usually gotten up in the morning? (Consider most days in thee week or average) *

- ◯ 12:00am to 1:00am
- ◯ 2:00am to 3:00am
- ◯ 4:00am to 5:00am
- ◯ 6:00am to 7:00am
- ◯ 8:00am to 9:00am
- ◯ 10:00am to 11:00am
- ◯ I don't get up in the Morning

During the past month, how many hours of actual sleep did you get at night? (This may be different than the number of hours you spent in bed.) *

- ◯ 1 to 2 hours
- ◯ 3 to 4 hours
- ◯ 5 to 6 hours
- ◯ 7 to 8 hours
- ◯ 9 to 10 hours
- ◯ 11 to 12 hours
- ◯ Above 13 hours

Questions 5(a to j) till 7
0 = Not during the past month
1 = Less than once a week
2 = Once or twice a week
3 = Three or more times a week

5. During the past month, how often have you had trouble sleeping because you...

a. Cannot get to sleep within 30 minutes *

	0	1	2	3	
Not during the past month	○	○	○	○	Three or more times a week

b. Wake up in the middle of the night or early morning *

	0	1	2	3	
Not during the past month	○	○	○	○	Three or more times a week

c. Have to get up to use the bathroom *

	0	1	2	3	
Not during the past month	○	○	○	○	Three or more times a week

d. Cannot breathe comfortably *

	0	1	2	3	
Not during the past month	○	○	○	○	Three or more times a week

e. Cough or snore loudly *

	0	1	2	3	
Not during the past month	○	○	○	○	Three or more times a week

f. Feel to cold *

	0	1	2	3	
Not during the past month	○	○	○	○	Three or more times a week

g. Feel to hot *

	0	1	2	3	
Not during the past month	○	○	○	○	Three or more times a week

h. Have bad dreams *

	0	1	2	3	
Not during the past month	○	○	○	○	Three or more times a week

i. Have pain *

	0	1	2	3	
Not during the past month	○	○	○	○	Three or more times a week

j. Other reason(s), please describe: (and include if still 0 - 3); if none, type N/A *

Your answer

6. During the past month, how often have you taken medicine to help you sleep (prescribed or "over the counter")? *

	0	1	2	3	
Not during the past month	○	○	○	○	Three or more times a week

7. During the past month, how often have you had trouble staying awake while driving, eating meals, or engaging in social activity? *

	0	1	2	3	
Not during the past month	◯	◯	◯	◯	Three or more times a week

8. During the past month, how much of a problem has it been for you to keep up enough enthusiasm to get things done? *

- ◯ No problem at all
- ◯ Only a very slight problem
- ◯ Somewhat of a problem
- ◯ A very big problem

10. Do you have a bed partner or room mate? (Family, Peer, or SO) *

- ◯ No bed partner or room mate
- ◯ Partner/room mate in other room
- ◯ Partner in same room but not same bed
- ◯ Same bed

OPTIONAL: If you have a room mate or bed partner, ask him/her how often in the past month you have had:

a. Loud snoring
b. Long pauses between breaths while asleep
c. Legs twitching or jerking while you sleep
d. Episodes of disorientation or confusion during sleep
e. Other restlessness while you sleep, please describe

0 - Not during the past month
1 - Less than once a week
2 - Once or twice a week
3 - Three or more times a week

Your answer

Scoring for PSQI
Here is the instrument for the PSQI test, pages 3 & 4 show how to score this one (much more different than the GAS-7 and harder for me to copy paste)

Here is the link for instrument used:
https://www.goodmedicine.org.uk/files/assessment,%20pittsburgh%20psqi.pdf

I would like: *

- ◯ To score my own total
- ◯ Request you send my total
- ◯ Not that interested to see my total scoring

Back Next Clear form

Submit Form

From the researcher, Thank you so much for competing this form!!

I will be using this data as a foundation for next years product for my strand (STEM) speaking of such what way do you think will be most efficient? this is the last question and you are all freed

Optional: What product is best to stop using devices?

- ◯ A cute chick hatching from an egg to say stop using your device! (software)
- ◯ A horror clown or monster popping up on the screen when overtime! (software)
- ◯ Both 1 & 2
- ◯ Pop up reminders for breaks!
- ◯ A lamp/alarm that cuts off when a device is near it
- ◯ A teleport machine to run away from academics (joke pls my grds)
- ◯ Other: _____

Optional Remarks (cannot confirm I'll read these all)

Your answer

A copy of your responses will be emailed to smanuel180507@livingstone.ph.education.

Back Submit Clear form

Raw Data (Summary graphs) Those not included in the Results**

LSIS

Optional: What other strategies have you heard about or tried to improve your sleep or mental health in relation to screen use?

13 responses

Not using gadgets an hour before bed and doing some stretching.

the only thing that can make me sleep immediately is exhaustion, if I'm not exhausted enough, then I'll listen to music and then fall asleep after some minutes.

Reading manga or manhwa and playing guitar helped me a lot

I restrained myself by deleting apps (e.g., youtube), setting passwords, and blocking sites to improve my mental health from screen use, but not really for sleep. I guess it helped :D.

Brotha, you really think-

I sleep them off. Whenever I still feel tired, I don't use them at the moment in time and just fall asleep at any part of the day. It does not feel healthy, but it does help for me as I am an energetic person.

Screen time limits, and social media feeds

i use night mode for my phone and laptop all the time therefore i can less exposed from blue light,

spending time with loved ones

(Do I gotta do allat...) Whenever I socialize or go out, I do not use my phone entirely unless I have to, because using it while being out in the real world can really mess with your brain's dopamine levels and can worsen depression or anxiety if relied upon too much.

deep talk with my cousin, face to face. Also my mom's advice, Keep a light on to reduce eye strain if using screens at night or in the dark.

-

Go Outside

AUF

Optional: What other strategies have you heard about or tried to improve your sleep or mental health in relation to screen use?

3 responses

Swimming

Napping!

Drink warm milk

Optional: What other strategies have you heard about or tried to improve your sleep or mental health in relation to screen use?

6 responses

Less gadget/devices

Napping?

I tend to talk to my so a lot till one of us sleeps

I focus on doing my skincare

Nothing really

I tried to sleep early. But it didn't really work because I have this subconscious need to make up for all the time I spent in the day outside of school or doing other things by spending as much time as I can on my phone before my bed time. (Since it's the only time where I get to have my alone time)

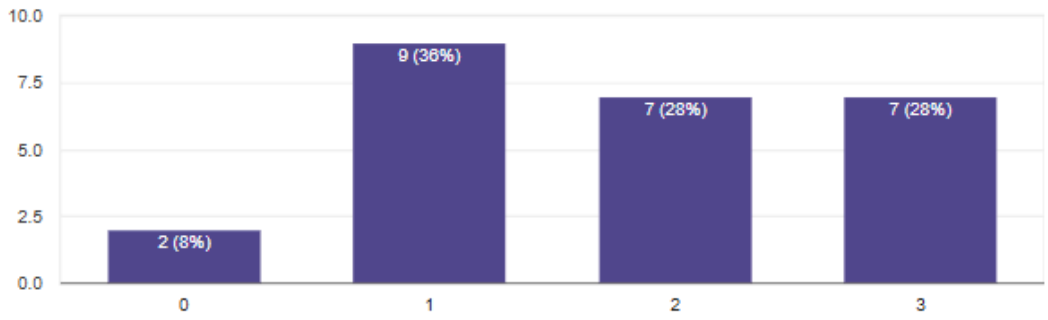
GAD results (Individually)
LSIS

General Anxiety Disorder - 7 **GAD-7**

Feeling nervous, anxious, or on edge

Copy chart

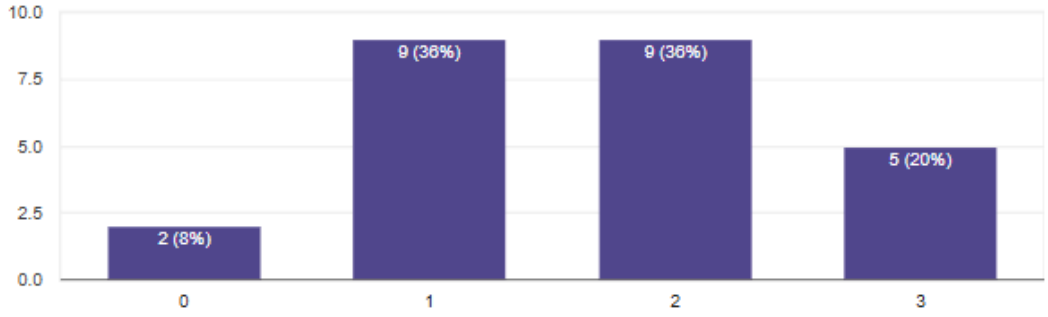
25 responses



Not being able to stop or control worrying

Copy chart

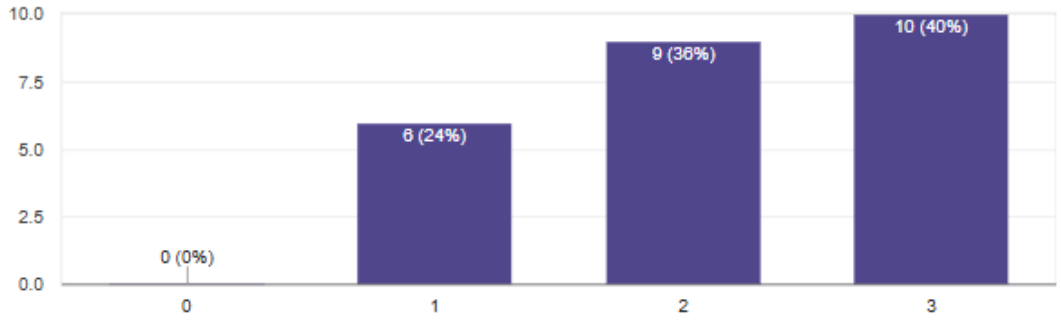
25 responses



Worrying too much about different things

Copy chart

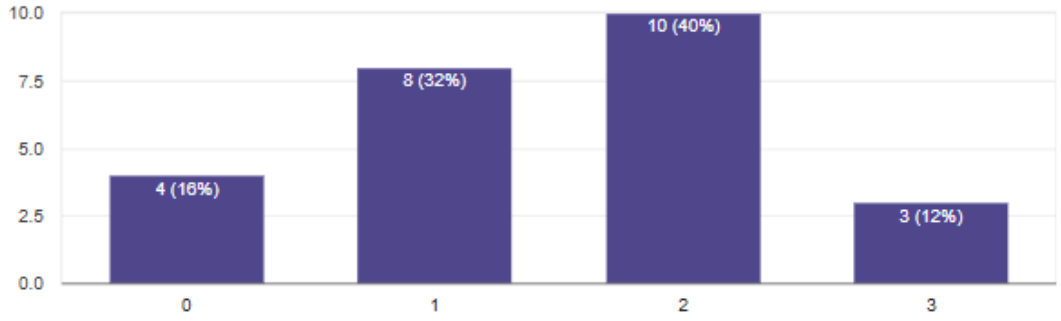
25 responses



Trouble Relaxing

Copy chart

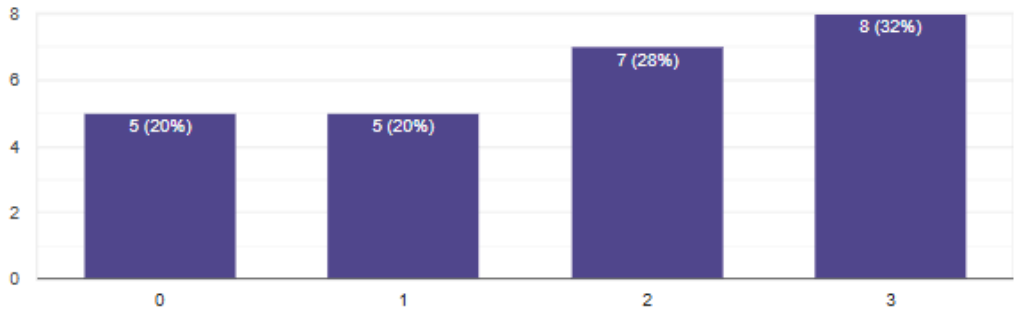
25 responses



Feeling afraid, as if something awful might happen

Copy chart

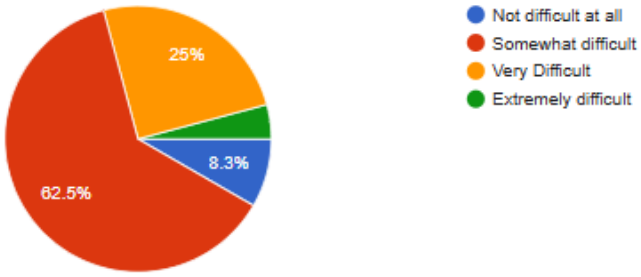
25 responses



If you checked any problems, how difficult have they made it for you to do your work, take care of things at home, or get along with other people?

Copy chart

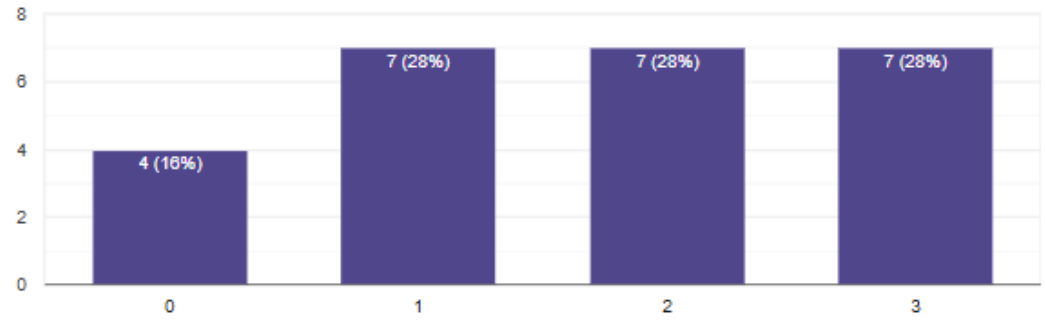
24 responses



Being so restless that it is hard to sit still

Copy chart

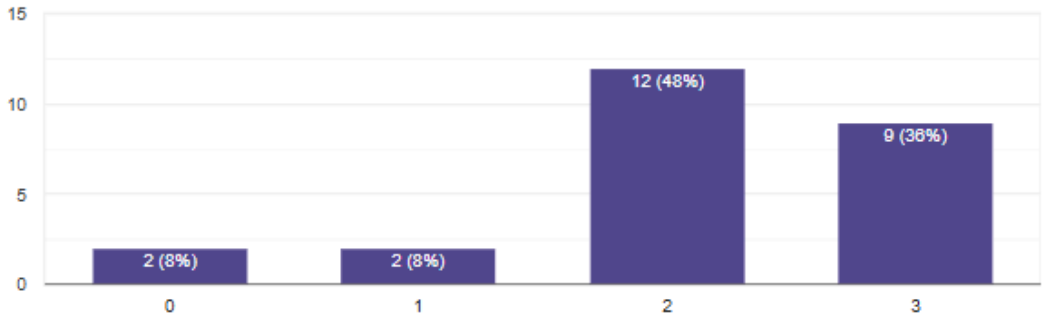
25 responses



Becoming easily annoyed or irritable

Copy chart

25 responses



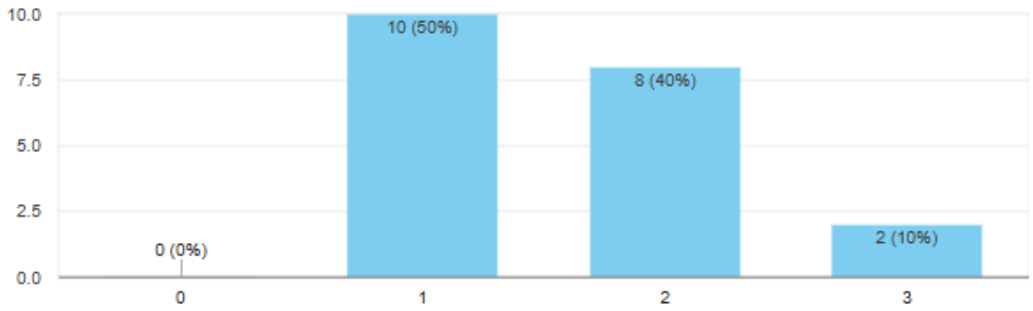
AUF:

General Anxiety Disorder - 7 **GAD-7**

Feeling nervous, anxious, or on edge

Copy chart

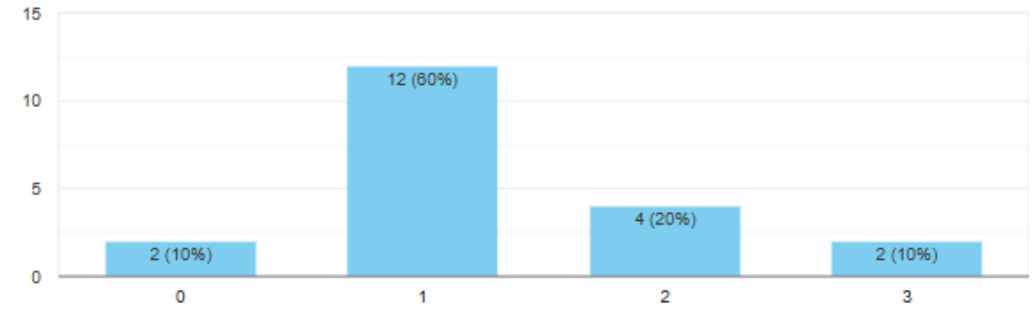
20 responses



Not being able to stop or control worrying

Copy chart

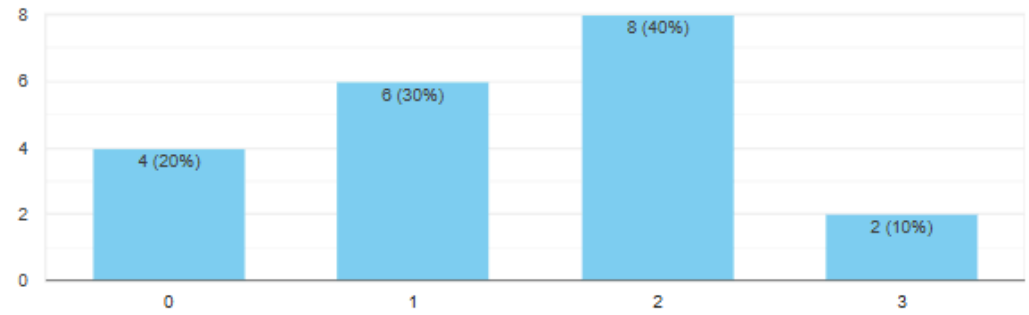
20 responses



Worrying too much about different things

Copy chart

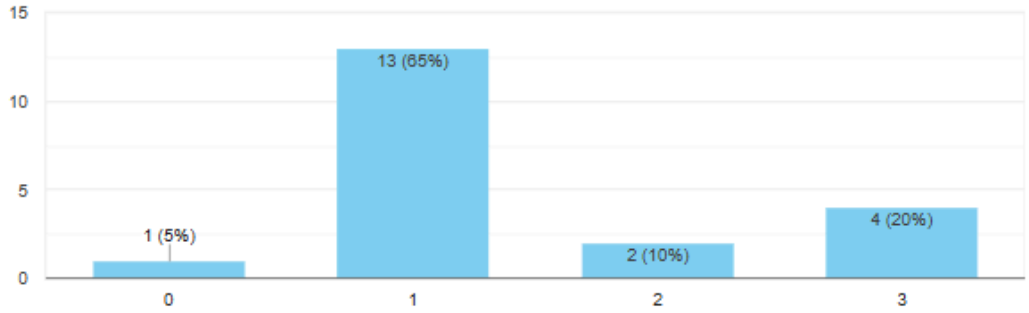
20 responses



Trouble Relaxing

Copy chart

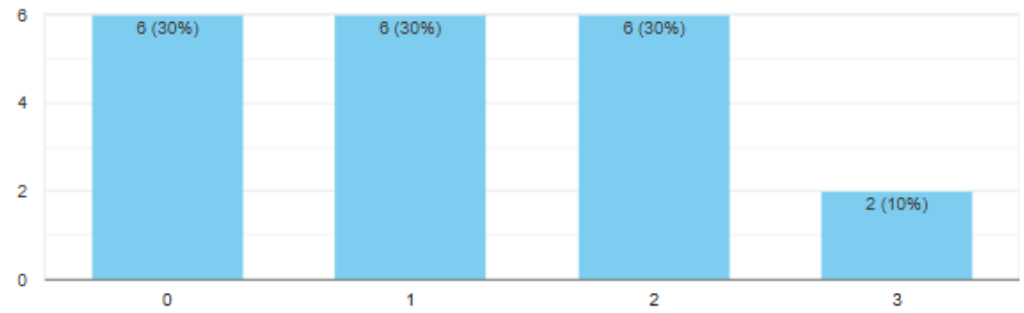
20 responses



Being so restless that it is hard to sit still

Copy chart

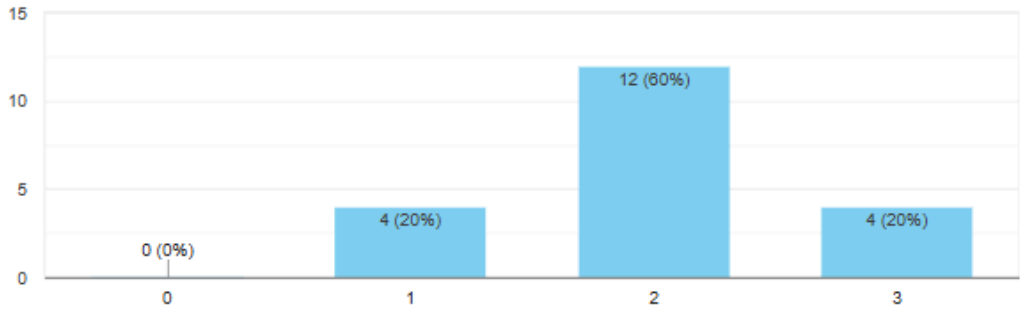
20 responses



Becoming easily annoyed or irritable

Copy chart

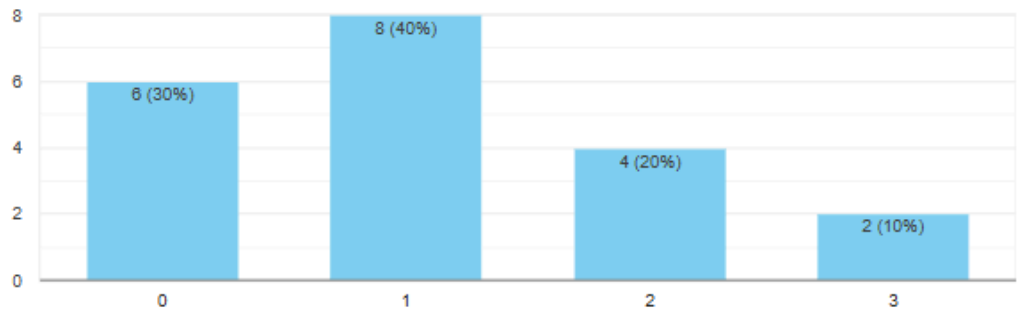
20 responses



Feeling afraid, as if something awful might happen

Copy chart

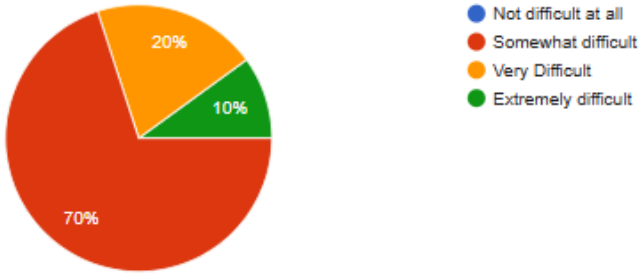
20 responses



If you checked any problems, how difficult have they made it for you to do your work, take care of things at home, or get along with other people?

Copy chart

20 responses



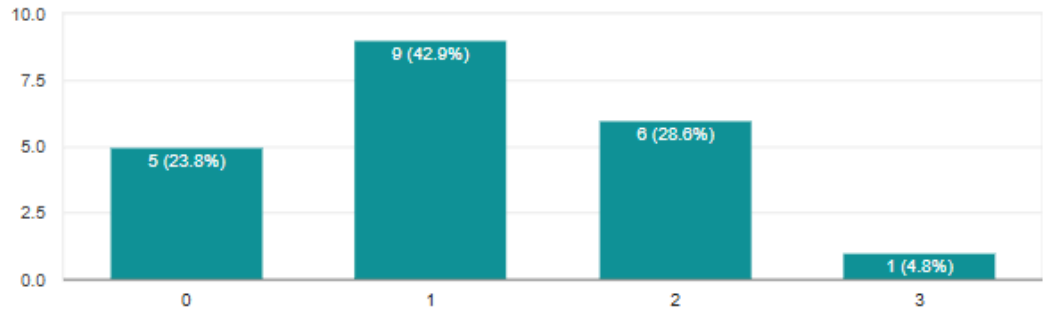
NU:

General Anxiety Disorder - 7 **GAD-7**

Feeling nervous, anxious, or on edge

Copy chart

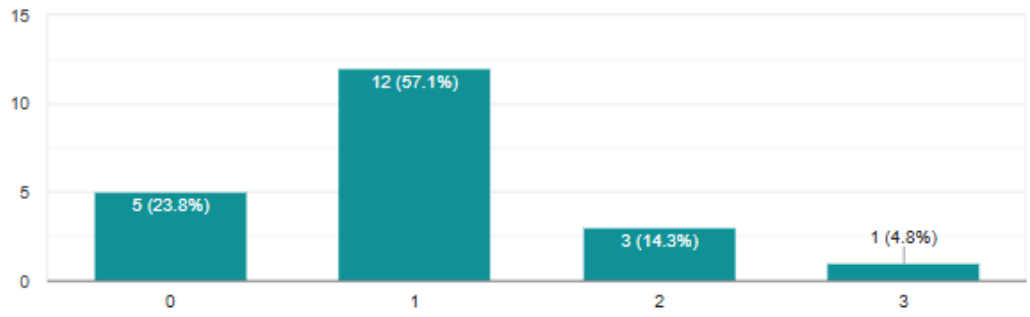
21 responses



Not being able to stop or control worrying

Copy chart

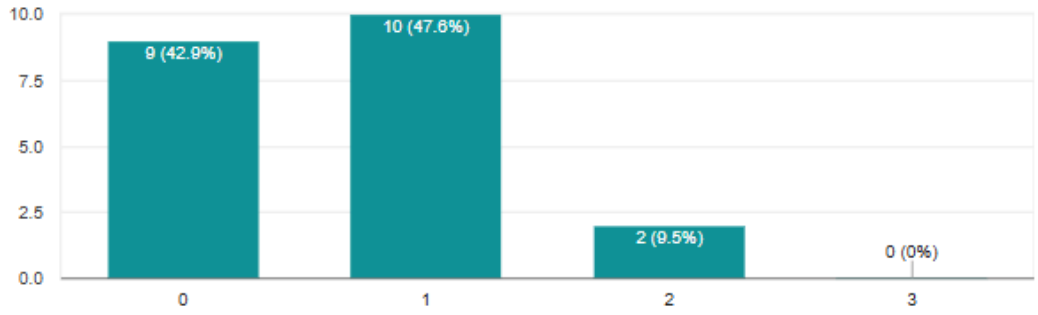
21 responses



Worrying too much about different things

Copy chart

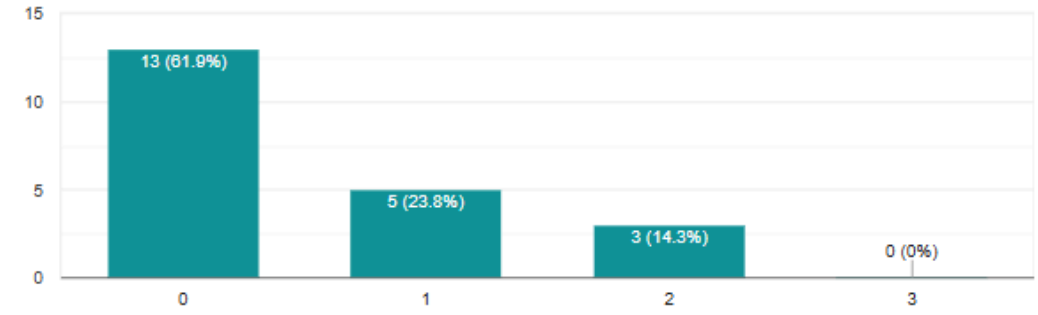
21 responses



Trouble Relaxing

Copy chart

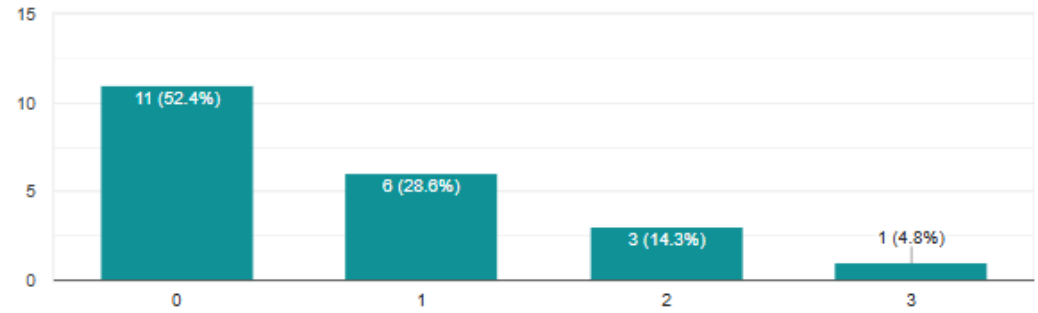
21 responses



Being so restless that it is hard to sit still

Copy chart

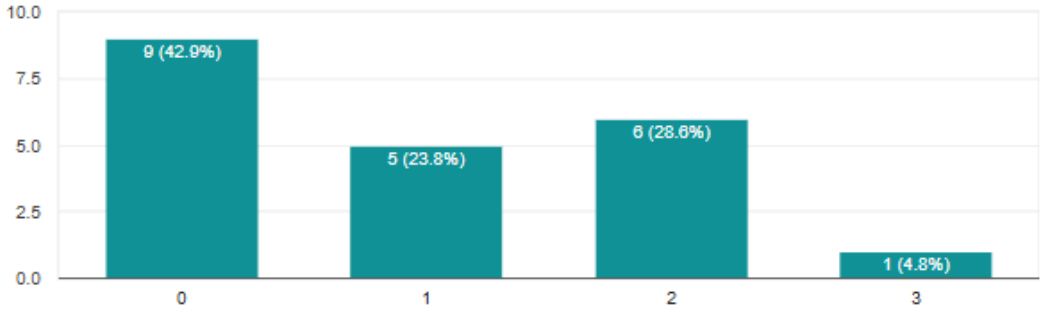
21 responses



Becoming easily annoyed or irritable

Copy chart

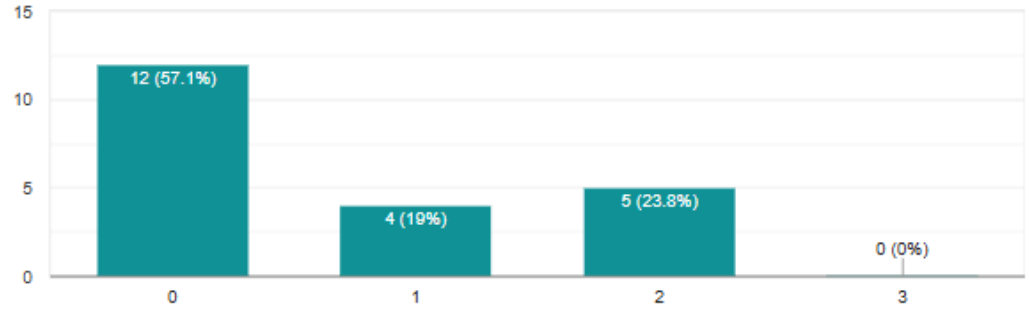
21 responses



Feeling afraid, as if something awful might happen

Copy chart

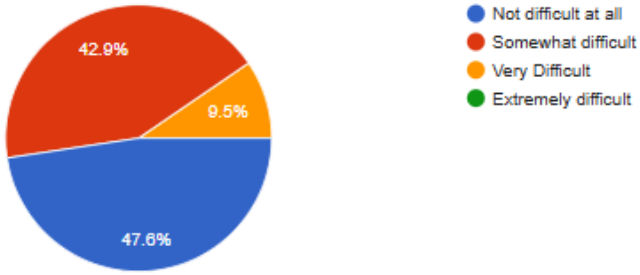
21 responses



If you checked any problems, how difficult have they made it for you to do your work, take care of things at home, or get along with other people?

Copy chart

21 responses

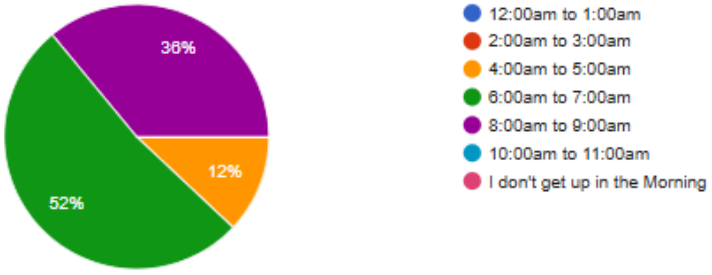


PSQI Results (Individually:
LSIS:

During the past month, what time have you usually gotten up in the morning? (Consider most days in thee week or average)

Copy chart

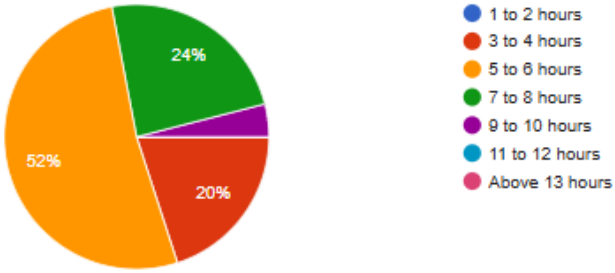
25 responses



During the past month, how many hours of actual sleep did you get at night? (This may be different than the number of hours you spent in bed.)

Copy chart

25 responses

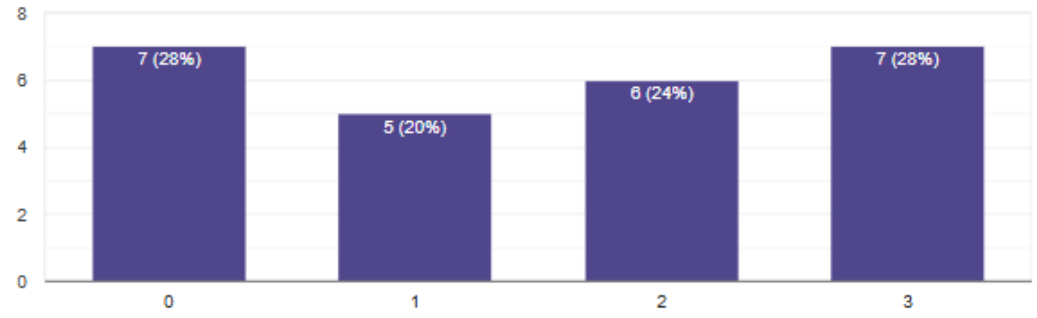


Questions 5(a to j) till 7

a. Cannot get to sleep within 30 minutes

Copy chart

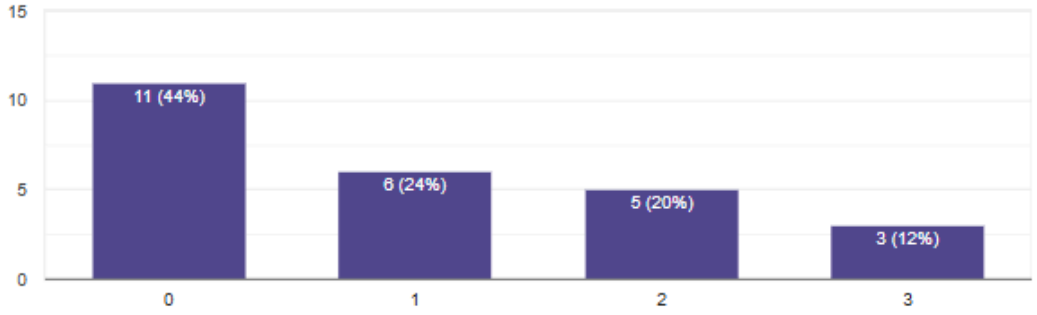
25 responses



b. Wake up in the middle of the night or early morning

Copy chart

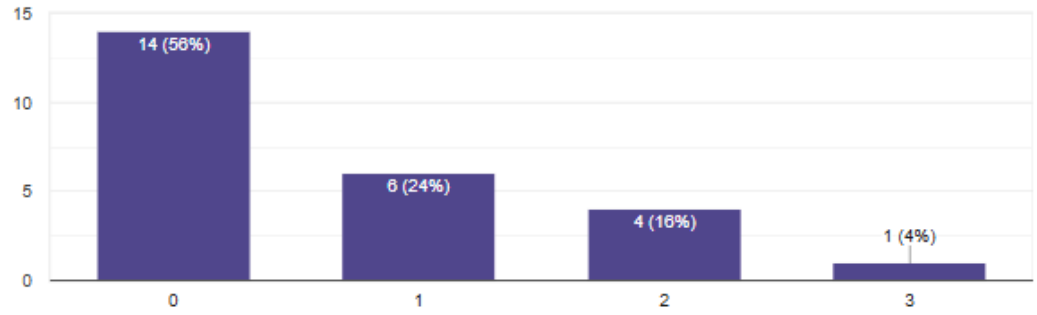
25 responses



c. Have to get up to use the bathroom

Copy chart

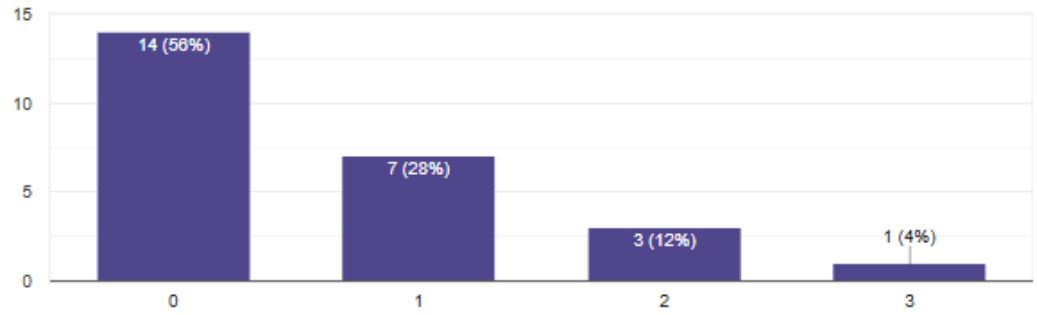
25 responses



d. Cannot breathe comfortably

Copy chart

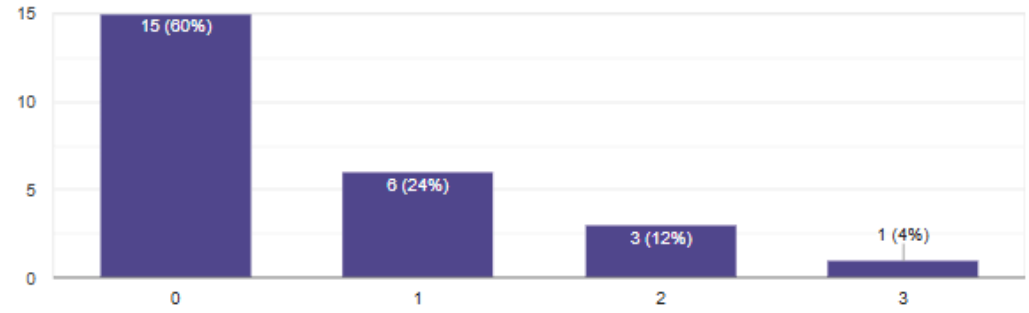
25 responses



e. Cough or snore loudly

Copy chart

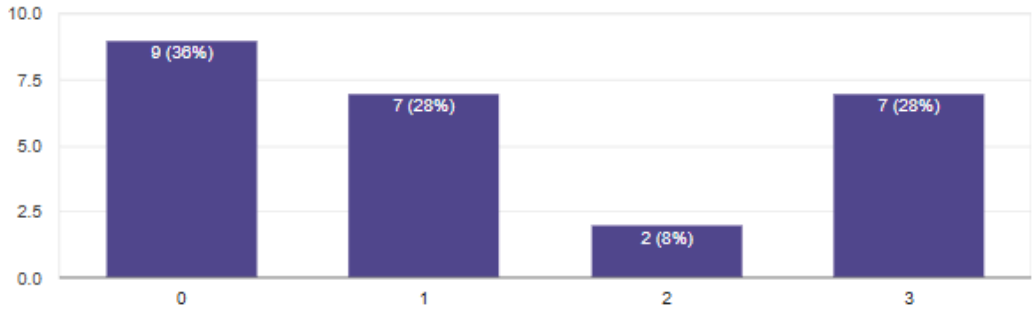
25 responses



f. Feel to cold

Copy chart

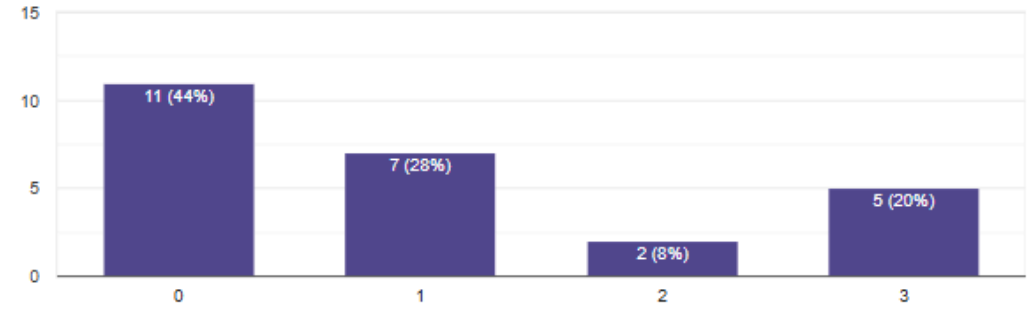
25 responses



g. Feel to hot

Copy chart

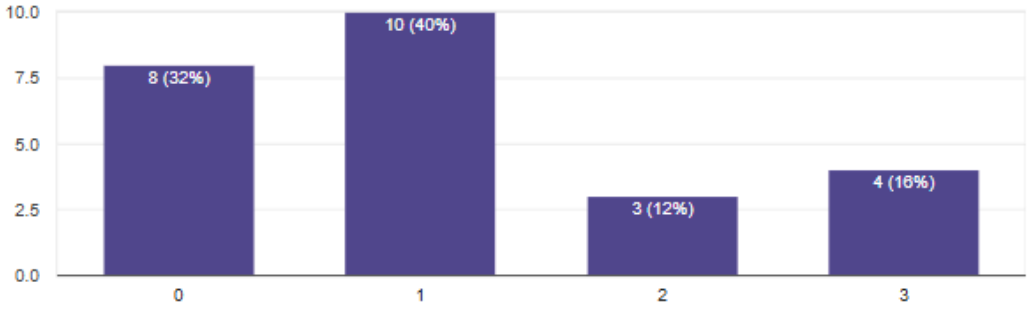
25 responses



h. Have bad dreams

Copy chart

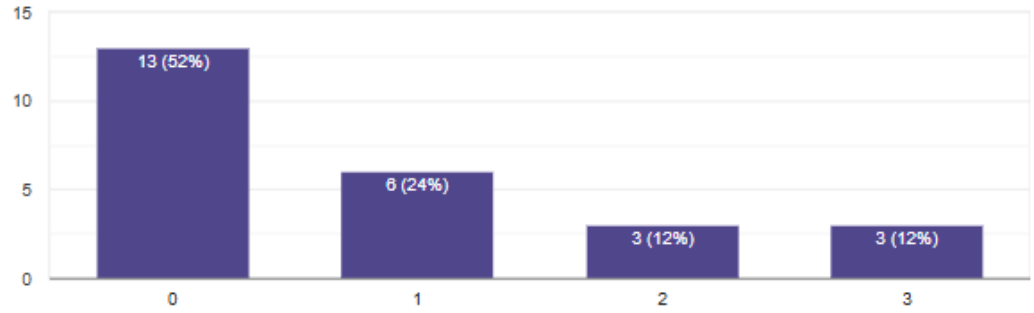
25 responses



i. Have pain

Copy chart

25 responses



j. Other reason(s), please describe: (and include if still 0 - 3); if none, type N/A

25 responses

N/A

na

Can't stop thinking (3)

Energy too high, 2

Background noise (jeepsor loud motor cycles)

.

n/a

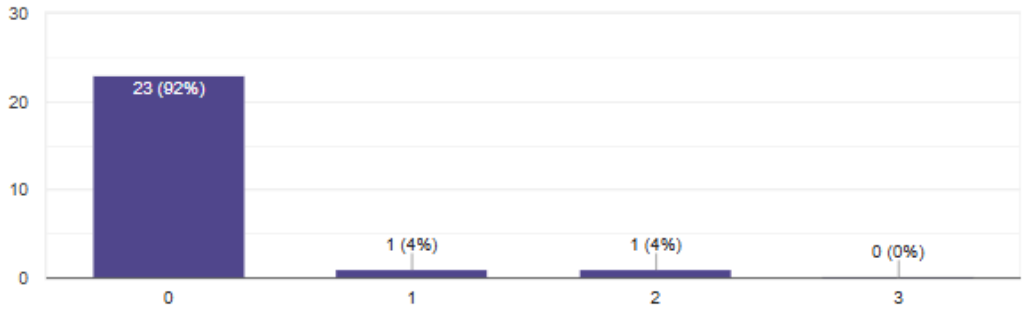
Randomly waking up without reason and being unable to fall back asleep for a few minutes

Empty

6. During the past month, how often have you taken medicine to help you sleep (prescribed or "over the counter")?

Copy chart

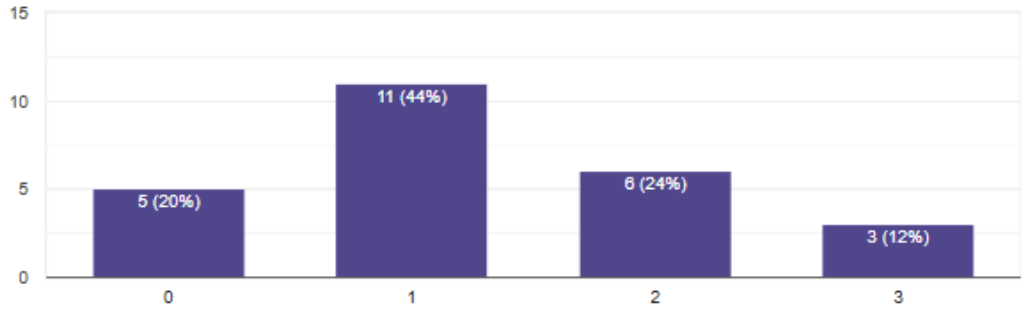
25 responses



7. During the past month, how often have you had trouble staying awake while driving, eating meals, or engaging in social activity?

Copy chart

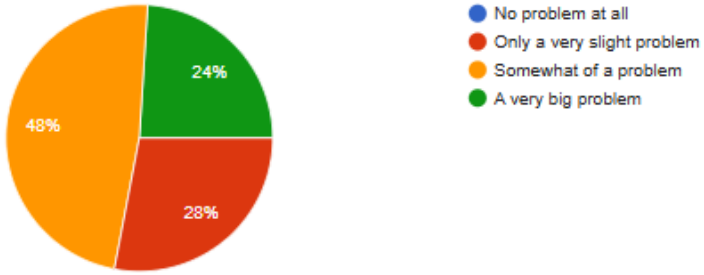
25 responses



8. During the past month, how much of a problem has it been for you to keep up enough enthusiasm to get things done?

Copy chart

25 responses



10. Do you have a bed partner or room mate? (Family, Peer, or SO)

Copy chart

25 responses



OPTIONAL: If you have a room mate or bed partner, ask him/her how often in the past month you have had:

a. Loud snoring
b. Long pauses between breaths while asleep
c. Legs twitching or jerking while you sleep
d. Episodes of disorientation or confusion during sleep
e. Other restlessness while you sleep, please describe

0 - Not during the past month
1 - Less than once a week
2 - Once or twice a week
3 - Three or more times a week

8 responses

0

He doesn't have any complaints about any of the following except for when I talk in my sleep but it's a rare occurrence.

a.3 b. 0 c. 3 d. 0 e. 3

0
0
3
0
0

a. 1 b. 1 c. 0 d. 1 e. none

C. 3

3, 0, 2, 0, 0

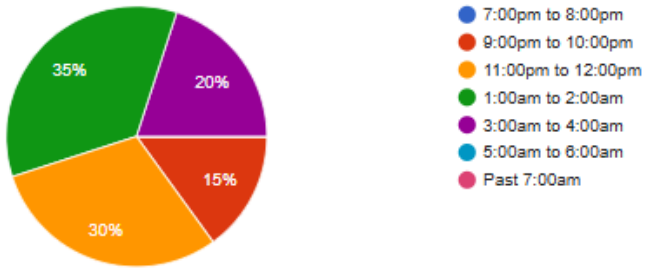
AUF:

Pittsburgh Sleep Quality Index (**PSQI**)

During the past month, what time have you usually gone to bed at night?

Copy chart

20 responses

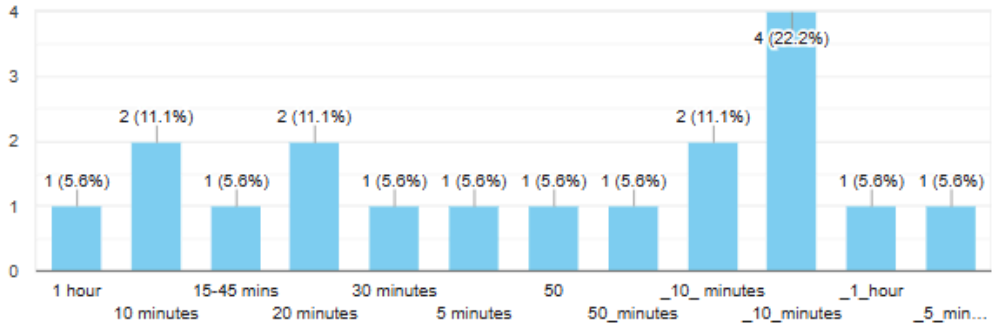


During the past month, how long (in minutes) has it usually taken you to fall asleep each night?

Copy chart

Format: _##_ minutes ; If and only if longer in hours change minutes to hours (lowercase)

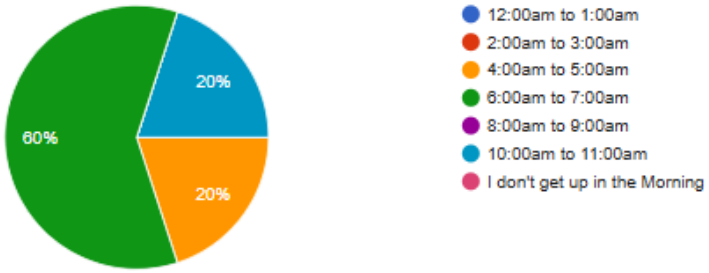
18 responses



During the past month, what time have you usually gotten up in the morning? (Consider most days in thee week or average)

Copy chart

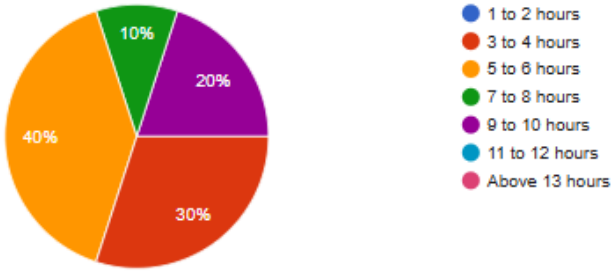
20 responses



During the past month, how many hours of actual sleep did you get at night? (This may be different than the number of hours you spent in bed.)

Copy chart

20 responses

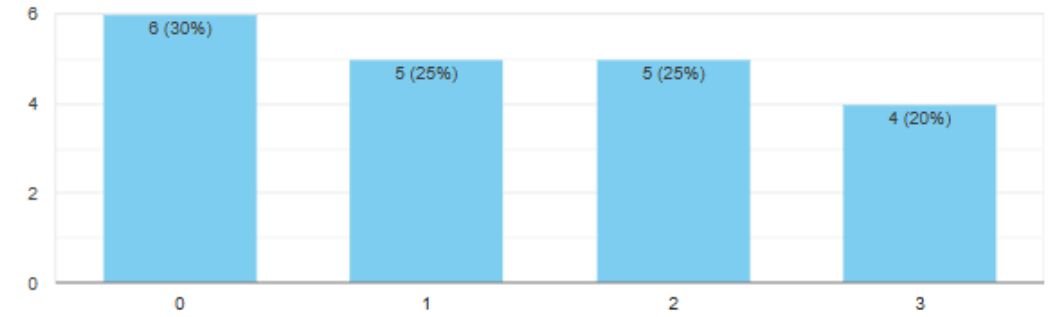


Questions 5(a to j) till 7

a. Cannot get to sleep within 30 minutes

Copy chart

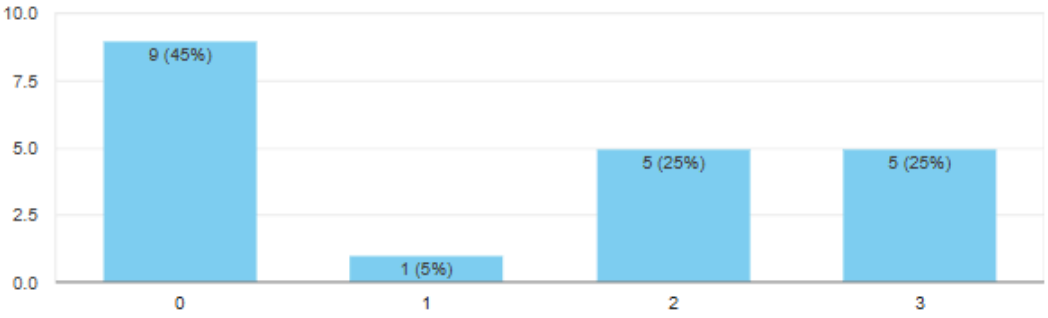
20 responses



b. Wake up in the middle of the night or early morning

Copy chart

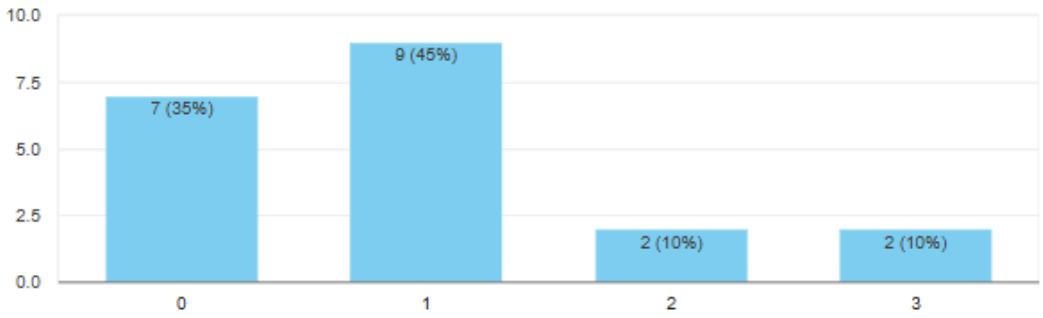
20 responses



c. Have to get up to use the bathroom

Copy chart

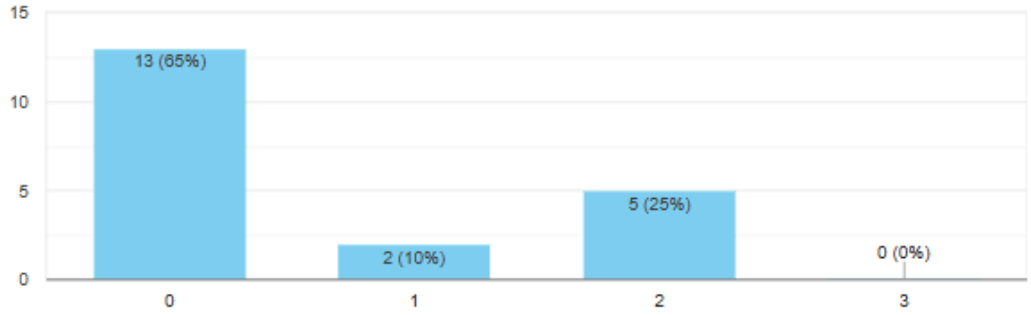
20 responses



d. Cannot breathe comfortably

Copy chart

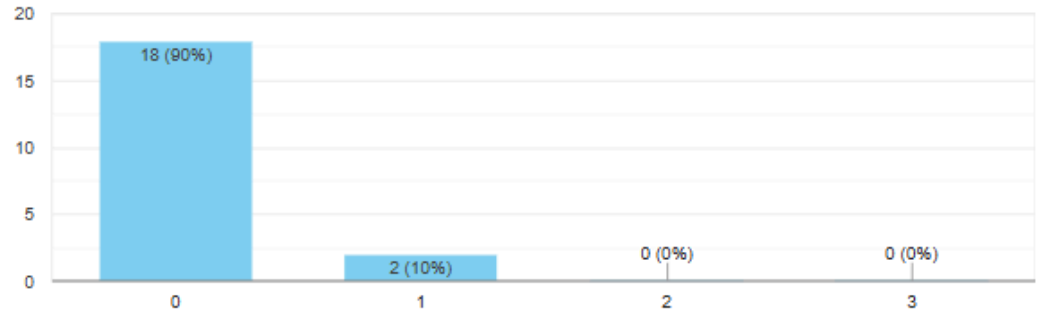
20 responses



e. Cough or snore loudly

Copy chart

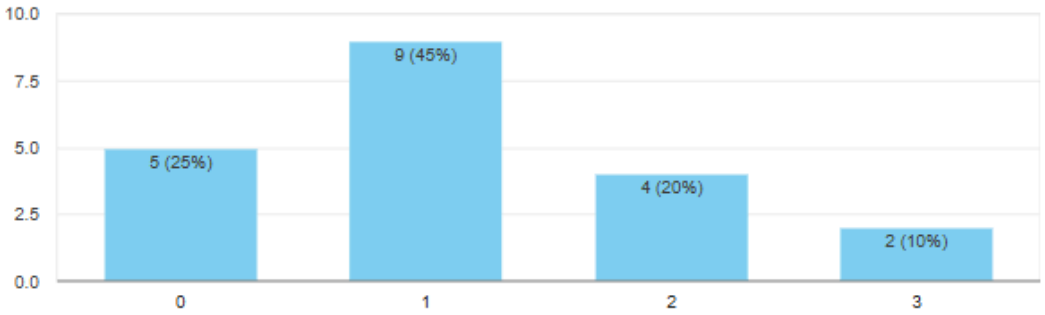
20 responses



f. Feel to cold

Copy chart

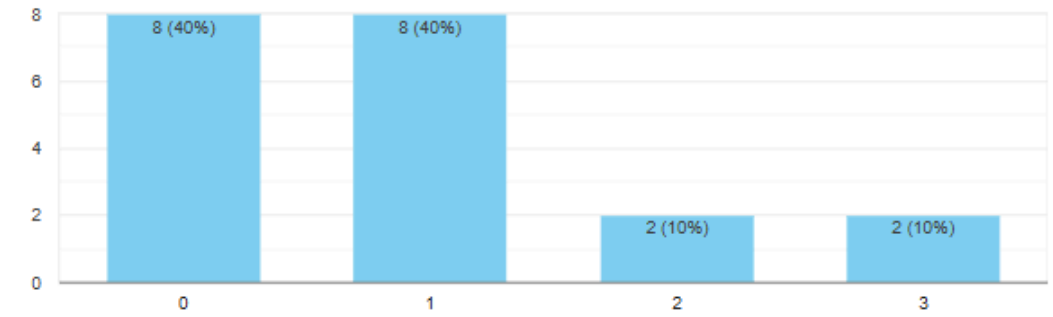
20 responses



g. Feel to hot

Copy chart

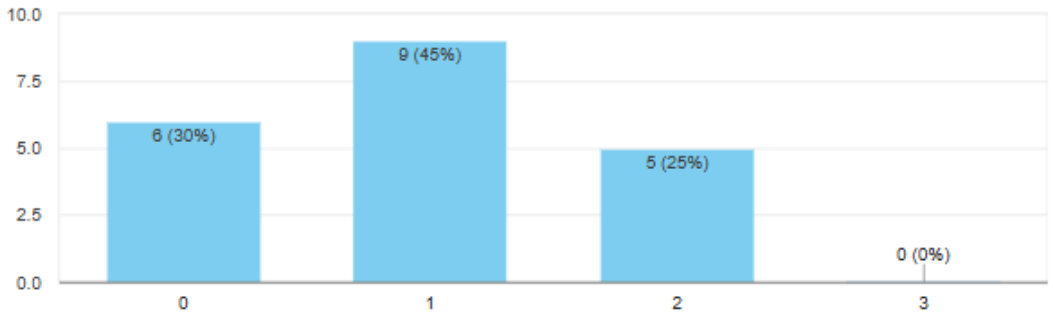
20 responses



h. Have bad dreams

Copy chart

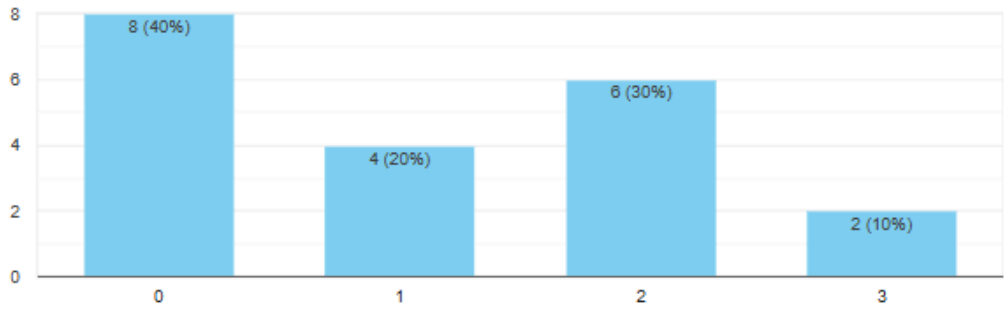
20 responses



i. Have pain

Copy chart

20 responses



j. Other reason(s), please describe: (and include if still 0 - 3); if none, type N/A

20 responses

N/A

My asthma; 2

Cystic fibrosis- 3

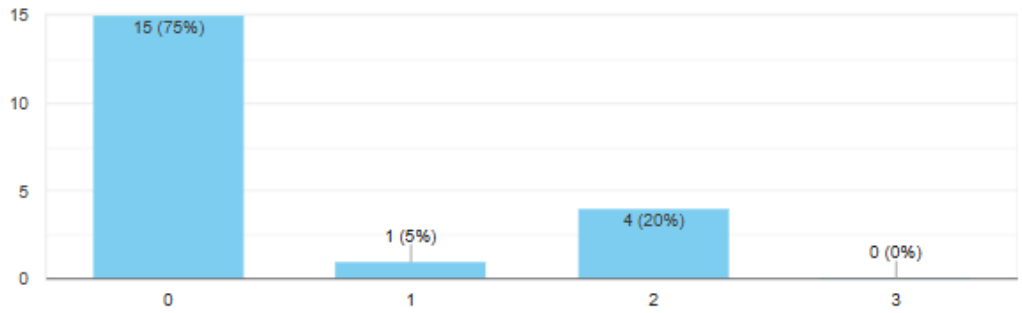
N/a

Family?

6. During the past month, how often have you taken medicine to help you sleep (prescribed or "over the counter")?

Copy chart

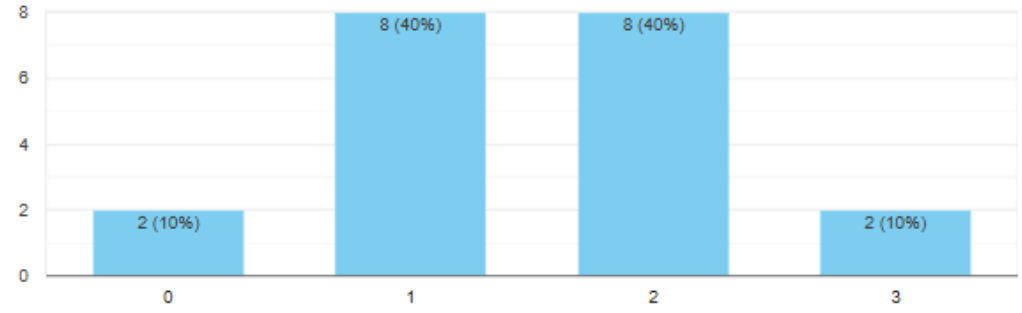
20 responses



7. During the past month, how often have you had trouble staying awake while driving, eating meals, or engaging in social activity?

Copy chart

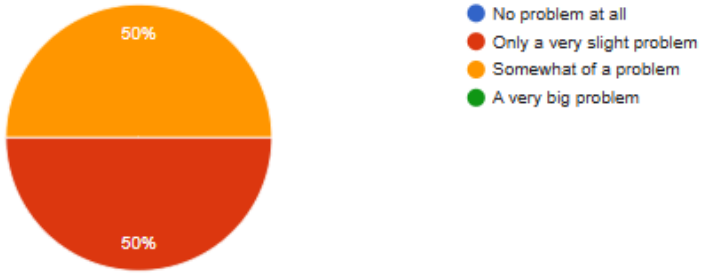
20 responses



8. During the past month, how much of a problem has it been for you to keep up enough enthusiasm to get things done?

Copy chart

20 responses



10. Do you have a bed partner or room mate? (Family, Peer, or SO)

Copy chart

20 responses

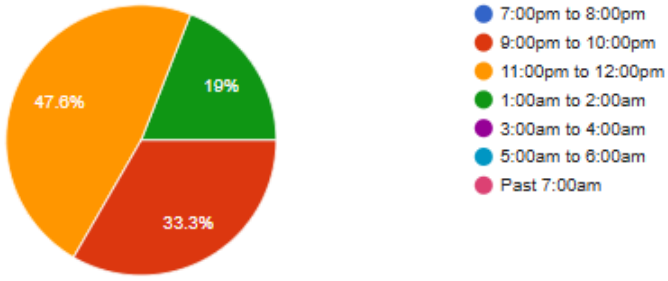


NU:

Pittsburgh Sleep Quality Index (**PSQI**)

During the past month, what time have you usually gone to bed at night? Copy chart

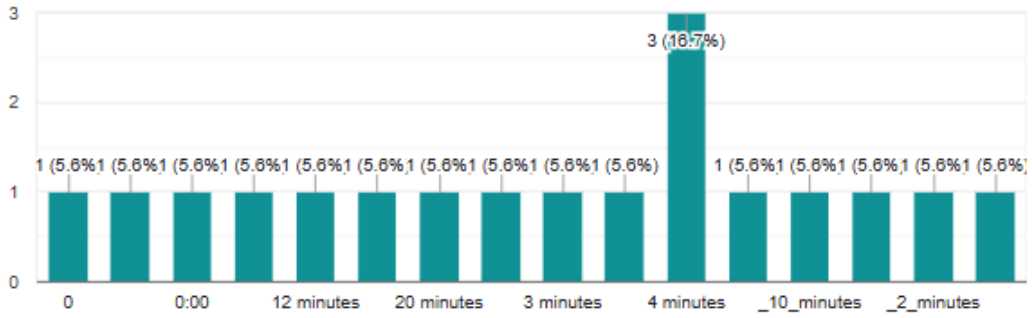
21 responses



During the past month, how long (in minutes) has it usually taken you to fall asleep each night? Copy chart

Format: _##_ minutes ; If and only if longer in hours change minutes to hours (lowercase)

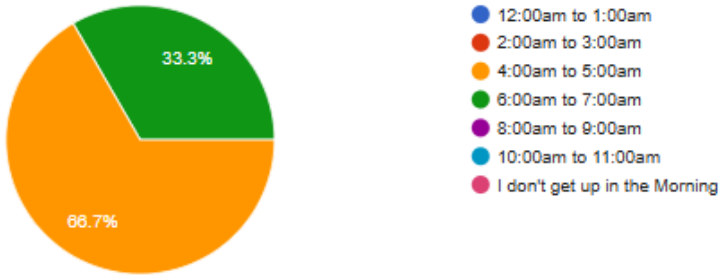
18 responses



During the past month, what time have you usually gotten up in the morning? (Consider most days in thee week or average)

Copy chart

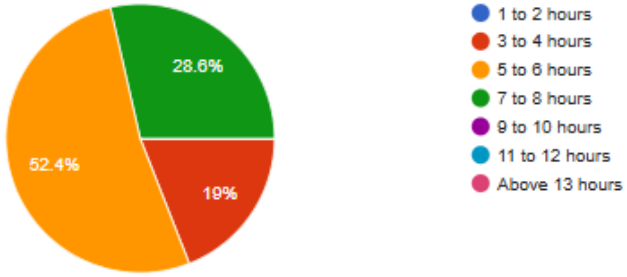
21 responses



During the past month, how many hours of actual sleep did you get at night? (This may be different than the number of hours you spent in bed.)

Copy chart

21 responses

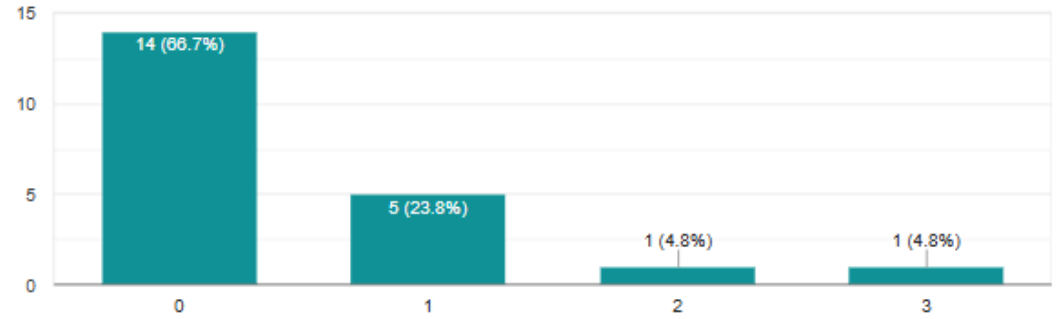


Questions 5(a to j) till 7

a. Cannot get to sleep within 30 minutes

Copy chart

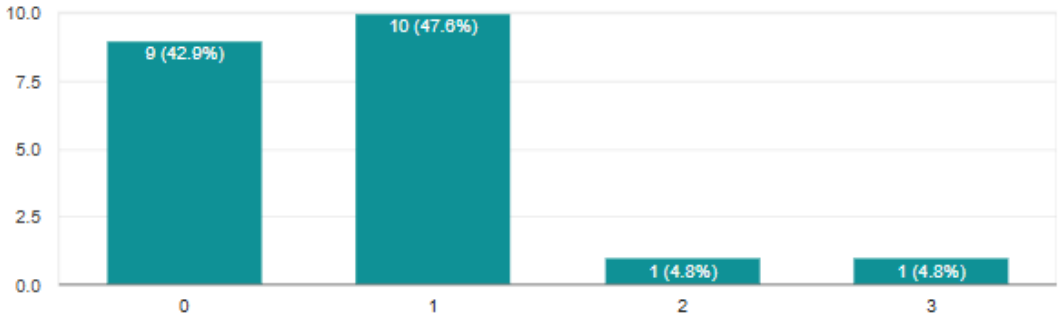
21 responses



b. Wake up in the middle of the night or early morning

Copy chart

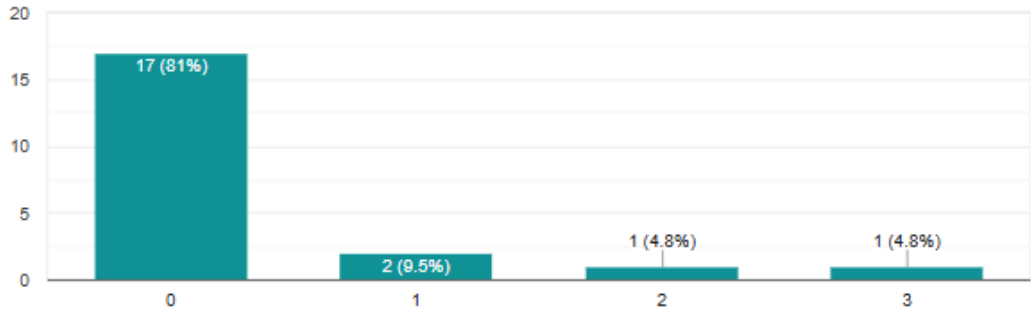
21 responses



c. Have to get up to use the bathroom

Copy chart

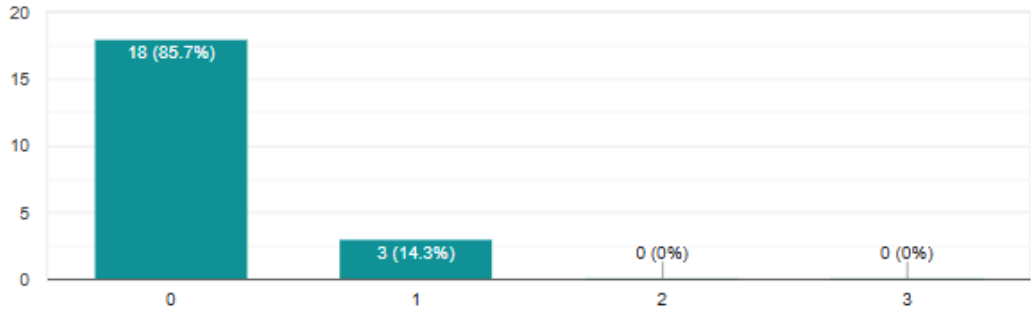
21 responses



d. Cannot breathe comfortably

Copy chart

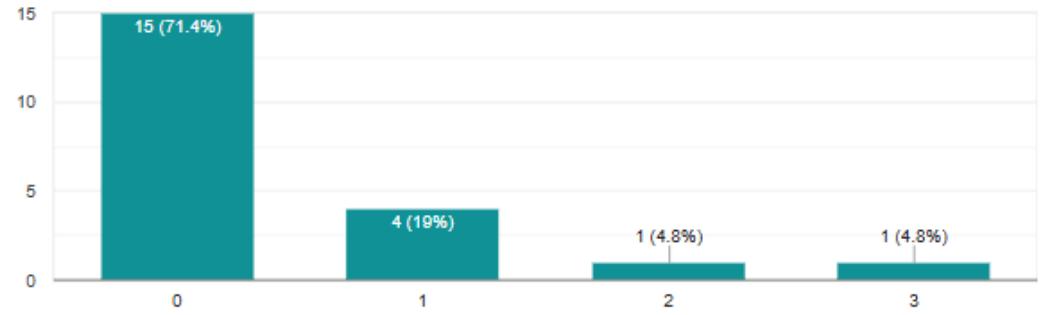
21 responses



e. Cough or snore loudly

Copy chart

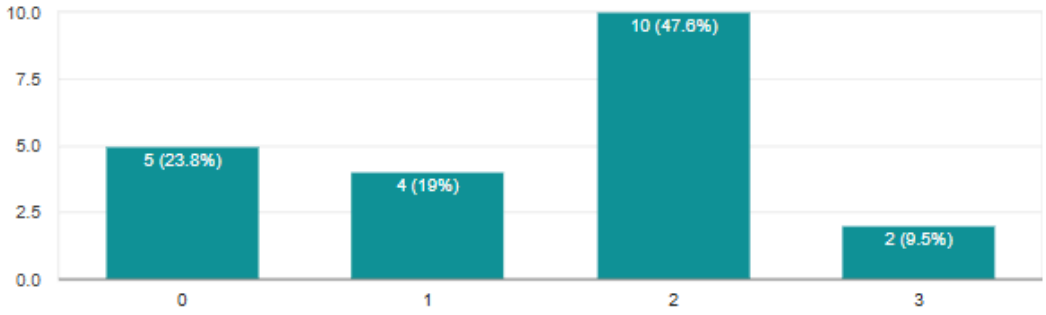
21 responses



f. Feel to cold

Copy chart

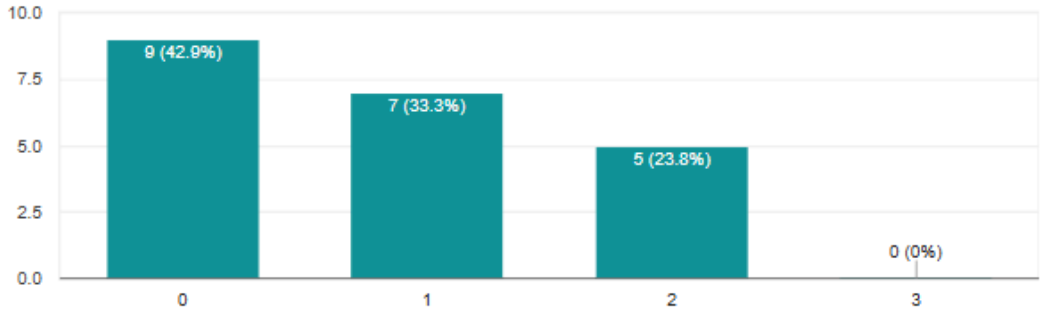
21 responses



g. Feel to hot

Copy chart

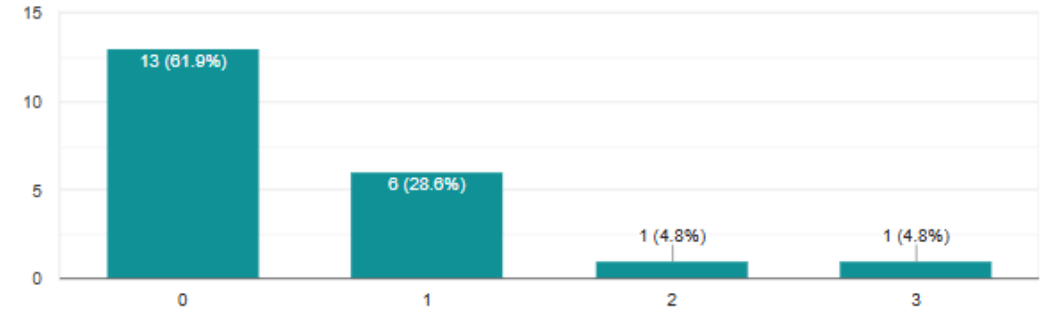
21 responses



h. Have bad dreams

Copy chart

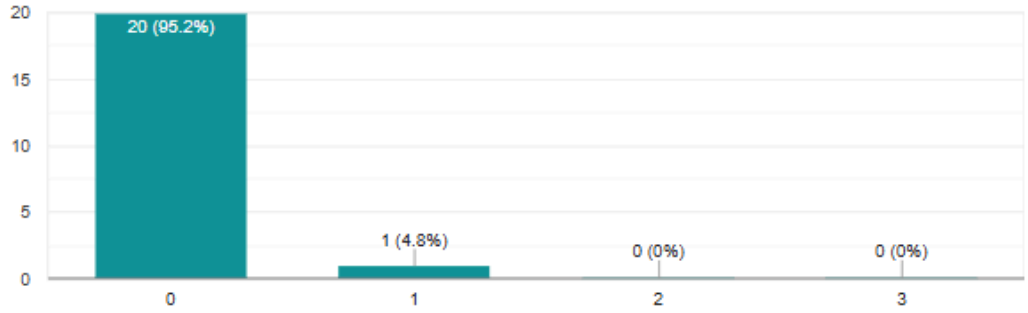
21 responses



i. Have pain

Copy chart

21 responses



j. Other reason(s), please describe: (and include if still 0 - 3); if none, type N/A

21 responses

N/A

N/a

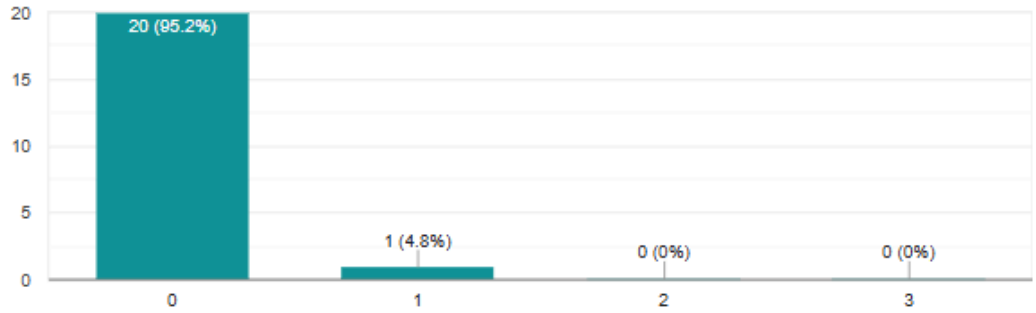
n/a

N/A

6. During the past month, how often have you taken medicine to help you sleep (prescribed or "over the counter")?

Copy chart

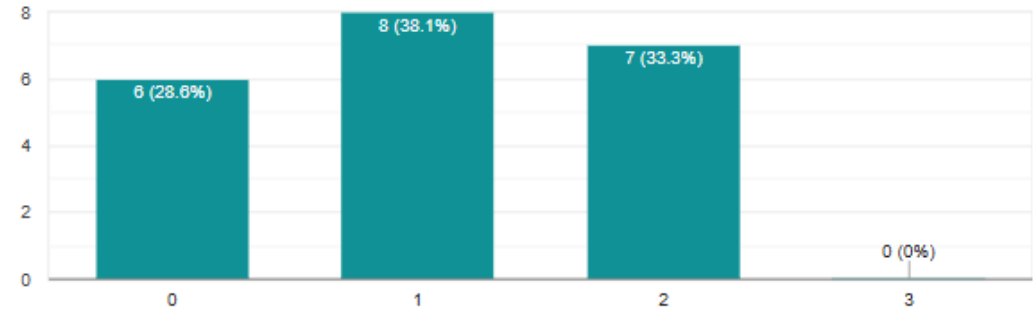
21 responses



7. During the past month, how often have you had trouble staying awake while driving, eating meals, or engaging in social activity?

Copy chart

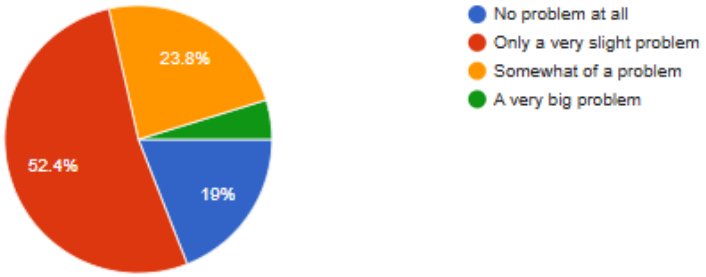
21 responses



8. During the past month, how much of a problem has it been for you to keep up enough enthusiasm to get things done?

Copy chart

21 responses



10. Do you have a bed partner or room mate? (Family, Peer, or SO)

Copy chart

21 responses



GAD-7 Anxiety

Over the last two weeks, how often have you been bothered by the following problems?	Not at all	Several days	More than half the days	Nearly every day
1. Feeling nervous, anxious, or on edge	0	1	2	3
2. Not being able to stop or control worrying	0	1	2	3
3. Worrying too much about different things	0	1	2	3
4. Trouble relaxing	0	1	2	3
5. Being so restless that it is hard to sit still	0	1	2	3
6. Becoming easily annoyed or irritable	0	1	2	3
7. Feeling afraid, as if something awful might happen	0	1	2	3

Column totals _____ + _____ + _____ + _____ =

Total score _____

If you checked any problems, how difficult have they made it for you to do your work, take care of things at home, or get along with other people?

Not difficult at all	Somewhat difficult	Very difficult	Extremely difficult
☐	☐	☐	☐

Source: Primary Care Evaluation of Mental Disorders Patient Health Questionnaire (PRIME-MD-PHQ). The PHQ was developed by Drs. Robert L. Spitzer, Janet B.W. Williams, Kurt Kroenke, and colleagues. For research information, contact Dr. Spitzer at ris8@columbia.edu. PRIME-MD® is a trademark of Pfizer Inc. Copyright© 1999 Pfizer Inc. All rights reserved. Reproduced with permission

Name:_____ Date: _____

Pittsburgh Sleep Quality Index (PSQI)

Instructions: The following questions relate to your usual sleep habits during the past month only. Your answers should indicate the most accurate reply for the majority of days and nights in the past month. **Please answer all questions.**

1. During the past month, what time have you usually gone to bed at night? _____
2. During the past month, how long (in minutes) has it usually taken you to fall asleep each night? _____
3. During the past month, what time have you usually gotten up in the morning? _____
4. During the past month, how many hours of actual sleep did you get at night? (This may be different than the number of hours you spent in bed.) _____

5. During the past month, how often have you had trouble sleeping because you…	Not during the past month	Less than once a week	Once or twice a week	Three or more times a week
a. Cannot get to sleep within 30 minutes				
b. Wake up in the middle of the night or early morning				
c. Have to get up to use the bathroom				
d. Cannot breathe comfortably				
e. Cough or snore loudly				
f. Feel too cold				
g. Feel too hot				
h. Have bad dreams				
i. Have pain				
j. Other reason(s), please describe:				
6. During the past month, how often have you taken medicine to help you sleep (prescribed or "over the counter")?				
7. During the past month, how often have you had trouble staying awake while driving, eating meals, or engaging in social activity?				
	No problem at all	Only a very slight problem	Somewhat of a problem	A very big problem
8. During the past month, how much of a problem has it been for you to keep up enough enthusiasm to get things done?				
	Very good	Fairly good	Fairly bad	Very bad
9. During the past month, how would you rate your sleep quality overall?				

	No bed partner or room mate	Partner/room mate in other room	Partner in same room but not same bed	Partner in same bed
10. Do you have a bed partner or room mate?				
	Not during the past month	Less than once a week	Once or twice a week	Three or more times a week
If you have a room mate or bed partner, ask him/her how often in the past month you have had:				
a. Loud snoring				
b. Long pauses between breaths while asleep				
c. Legs twitching or jerking while you sleep				
d. Episodes of disorientation or confusion during sleep				
e. Other restlessness while you sleep, please describe:				

Left GAD; Right PSQI

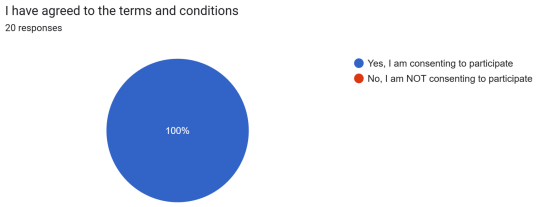
	A	B	C	D	E	F	G	H
1	7	15	12		A NU F	3	7	7
2	8	13	8		B LSIS G	6	6	13
3	8	13	11		C AUF H	11	6	12
4	7	11	10			4	8	9
5	7	21	20			6	13	8
6	10	14	9			7	10	5
7	5	5	9			8	3	8
8	0	4	7			6	5	8
9	0	10	8			5	12	5
10	4	17	8			4	14	10
11	13	18	4			5	11	6
12	5	15	9			4	5	8
13	4	12	11			5	9	2
14	5	11	14			3	3	5
15	3	10	21			1	5	11
16	10	20	7			4	7	5
17	4	8	9			3	9	6
18	3	14	7			4	11	6
19	7	10	7			4	8	12
20	1	16	11			6	10	11
21	8	12				3	5	
22		7					8	
23		17					9	
24		6					6	
25		15					5	
26								
27								

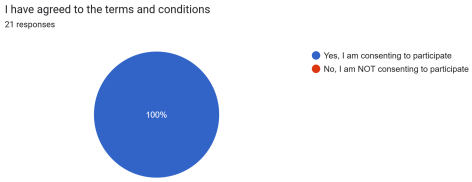
	J	K	L	M
1	Awareness	7	10	5
2		8	6	4
3		7	6	8
4		7	6	9
5		7	5	10
6		8	5	4
7		4	2	6
8		4	8	10
9		4	8	5
10		6	5	4
11		7	5	7
12		7	4	10
13		7	4	4
14		10	6	8
15		7	4	10
16		7	5	5
17		5	7	4
18		4	8	5
19		6	6	7
20		8	6	5
21		5	5	
22			6	
23			5	
24			2	
25			10	
26				
27				
28				
29				
30				

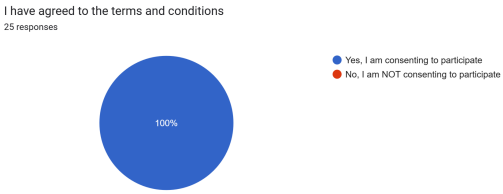
Ethical Documents (Some did signatures)



Was later changed to yes or no for people to easily answer**







Operation Definitions of Variables

Scoring GAD-7 Anxiety Severity

This is calculated by assigning scores of 0, 1, 2, and 3 to the response categories, respectively, of "not at all," "several days," "more than half the days," and "nearly every day."
GAD-7 total score for the seven items ranges from 0 to 21.

0–4: minimal anxiety

5–9: mild anxiety

10–14: moderate anxiety

15–21: severe anxiety

Scoring the PSQI

The order of the PSQI items has been modified from the original order in order to fit the first 9 items (which are the only items that contribute to the total score) on a single page. Item 10, which is the second page of the scale, does not contribute to the PSQI score.

In scoring the PSQI, seven component scores are derived, each scored 0 (no difficulty) to 3 (severe difficulty). The component scores are summed to produce a global score (range 0 to 21). Higher scores indicate worse sleep quality.

Component 1: Subjective sleep quality—question 9

Response to Q9	Component 1 score
Very good	0
Fairly good	1
Fairly bad	2
Very bad	3

Component 1 score:_____

Component 2: Sleep latency—questions 2 and 5a

Response to Q2	Component 2/Q2 subscore
≤ 15 minutes	0
16-30 minutes	1
31-60 minutes	2
> 60 minutes	3

Response to Q5a	Component 2/Q5a subscore
Not during past month	0
Less than once a week	1
Once or twice a week	2
Three or more times a week	3

Sum of Q2 and Q5a subscores	Component 2 score
0	0
1-2	1
3-4	2
5-6	3

Component 2 score:_____

Component 3: Sleep duration—question 4

Response to Q4	Component 3 score
> 7 hours	0
6-7 hours	1
5-6 hours	2
< 5 hours	3

Component 3 score:_____

Component 4: Sleep efficiency—questions 1, 3, and 4

Sleep efficiency = (# hours slept/# hours in bed) X 100%

hours slept—question 4

hours in bed—calculated from responses to questions 1 and 3

Sleep efficiency	Component 4 score
> 85%	0
75-84%	1
65-74%	2
< 65%	3

Component 4 score:_____

Component 5: Sleep disturbance—questions 5b-5j

Questions 5b to 5j should be scored as follows:

Not during past month	0
Less than once a week	1
Once or twice a week	2
Three or more times a week	3

Sum of 5b to 5j scores	Component 5 score
0	0
1-9	1
10-18	2
19-27	3

Component 5 score:_____

Component 6: Use of sleep medication—question 6

Response to Q6	Component 6 score
Not during past month	0
Less than once a week	1
Once or twice a week	2
Three or more times a week	3

Component 6 score:_____

Component 7: Daytime dysfunction—questions 7 and 8

Response to Q7	Component 7/Q7 subscore
Not during past month	0
Less than once a week	1
Once or twice a week	2
Three or more times a week	3

Response to Q8	Component 7/Q8 subscore
No problem at all	0
Only a very slight problem	1
Somewhat of a problem	2
A very big problem	3

Sum of Q7 and Q8 subscores	Component 7 score
0	0
1-2	1
3-4	2
5-6	3

Component 7 score:_____

Global PSQI Score: Sum of seven component scores:_____

Copyright notice: The Pittsburgh Sleep Quality Index (PSQI) is copyrighted by Daniel J. Buysse, M.D. Permission has been granted to reproduce the scale on this website for clinicians to use in their practice and for researchers to use in non-industry studies. For other uses of the scale, the owner of the copyright should be contacted.

Citation: Buysse, DJ, Reynolds CF, Monk TH, Berman SR, Kupfer DJ: The Pittsburgh Sleep Quality Index (PSQI): A new instrument for psychiatric research and practice. Psychiatry Research 28:193-213, 1989

PowerPoint Presentation (Appendix)



Comparative study on Blue Light Exposure in Grade 11 students from LSIS, NU, and AUF

Manuel Seraphim 11 - STEM

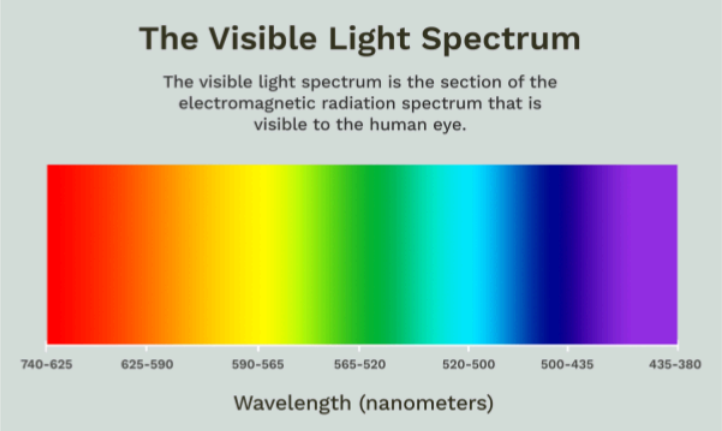


Background

- Increasing reliance on electronic devices among adolescents
- Blue light affects melatonin and Circadian rhythms

Background

Living Stone International School



Problem Statement

Living Stone International School

- Pervasive evening use disrupts sleep & mental health
- Limited research on tailored interventions for adolescents

Problem Statement

Living Stone International School

"Is there a significant gap difference between the three schools' students in Awareness, behavioral Patterns, and Coping Strategies?"

Hypotheses

Living Stone International School

Null hypothesis: *There is no significant difference in awareness, behavioral patterns, and coping strategies between students from the three schools.*

Alternative Hypothesis: *There is significant difference in awareness, behavioral patterns, and coping strategies between students from the three schools.*

RRL: Overview

Living Stone International School

- Key Findings: Blue Light suppresses melatonin, disrupts circadian rhythms
- Increased risks of anxiety, depression, and fatigue

Objectives

Living Stone International School

- Explore adolescents' awareness of Blue Light exposure
- Investigate evening use behaviors through the students PSQI and GAD-7
- Identify coping strategies

Supporting Evidence

Living Stone International School

- Evidence linking screen time to poor sleep quality & mental health issues
- Effective interventions include Blue Light filters and Behavior modifications

Participants

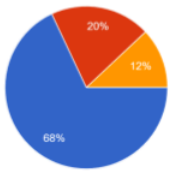
Living Stone International School

- Grade 11 students (66 Respondents)
- Within:
  - Livingstone International School
  - Angeles University Foundation
  - National University of Clark
- Selected via purposive sampling

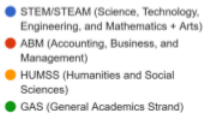
Participants

Living Stone International School

What strand are you pursuing?
25 responses

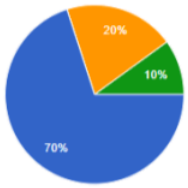


LSIS



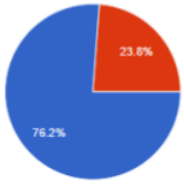
20 responses

AUF



21 responses

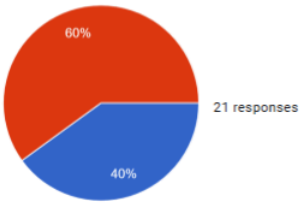
NU



Participants

Living Stone International School

Gender
25 responses

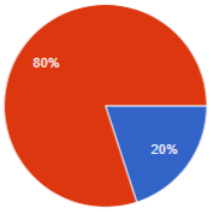


LSIS

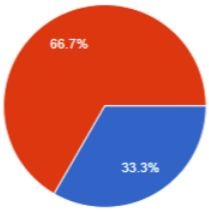
21 responses

20 responses

AUF



NU



Data Collection

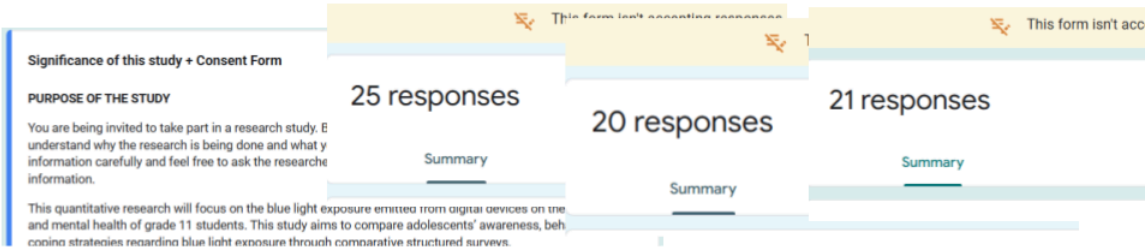
Living Stone International School

- February 16 till March 3, 2025
- Surveys: Pittsburgh Sleep Quality Index(PSQI), General Anxiety Disorder-7(GAD-7)
- Structured Survey containing 45 questions

Living Stone International School

Ethical Considerations

Informed consent, confidentiality, and voluntary participation



Living Stone International School

Limitations & Delimitations

Limitations

- Lack of generalizability
- Small sample size

Delimitations

- GAD-7 & PSQI measurements
- Three Schools

Awareness

Living Stone International School







LSIS	AUF	NU
5.76	6.5	6.42

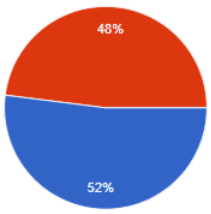
Behavioral Patterns

Living Stone International School

Do you take any breaks from screen use before going to bed?

LSIS

25 responses

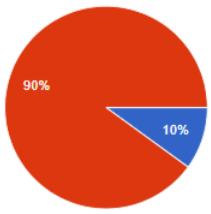


21 responses

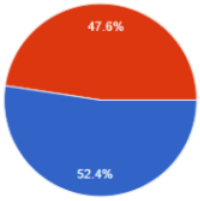
Yes
No

AUF

20 responses



NU



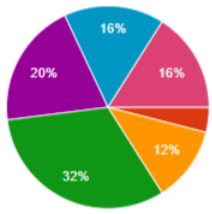
Behavioral Patterns

Living Stone International School

What is your Avg Screen Time? (all devices or phone)

LSIS

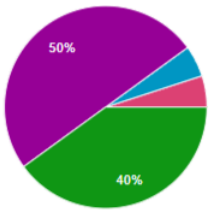
25 responses



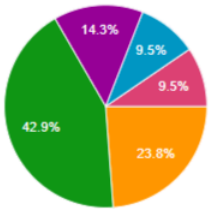


AUF

20 responses



NU



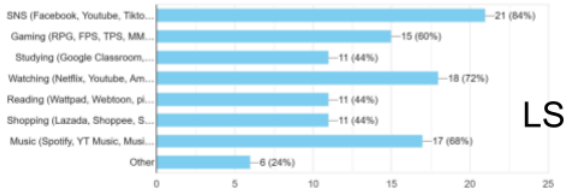
Behavioral Patterns

Living Stone International School

What types of activities (e.g., Social Media, Gaming, Studying) do you typically engage in on these devices?

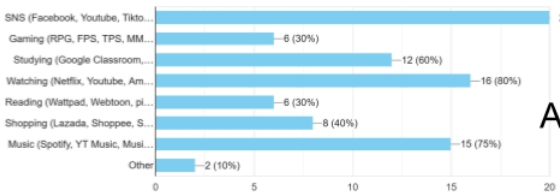
LSIS

25 responses



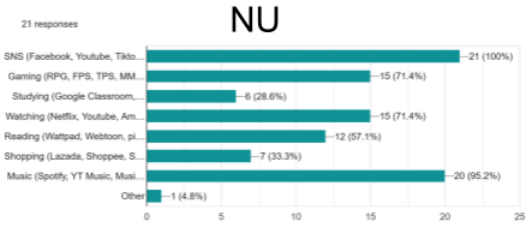
AUF

20 responses



NU

21 responses



Perceived Effects

Living Stone International School

Do you feel that using devices at night affects your ability to fall asleep or stay asleep?

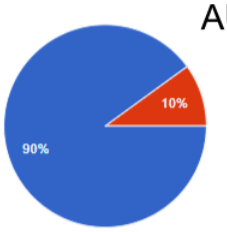
LSIS

25 responses



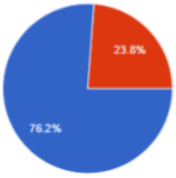
AUF

20 responses



NU

21 responses



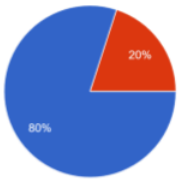
Perceived Effects

Living Stone International School

Have you noticed any changes in your mood, stress levels, or energy after a night of disrupted sleep?

LSIS

25 responses



AUF

20 responses



NU

21 responses



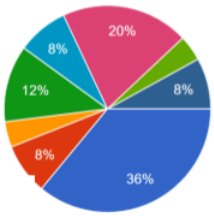
Coping Strategies

Living Stone International School

Have you tried any methods to reduce your screen use at night? (Max 3)

25 responses

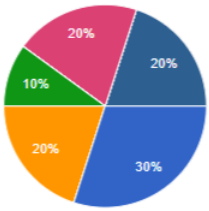
LSIS



20 responses

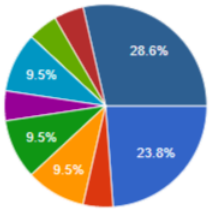
- Journaling
- Working out (Gym or Home)
- None
- Keeping Devices out of reach (Charge…
- Cutting off devices at a certain time in…
- Setting an Alarm or Reminder
- Locking certain apps at a certain time…
- Adjust Device Settings (Night Mode/G…
- Read a Physical Book
- Meditation & Relaxation

AUF



21 responses

NU



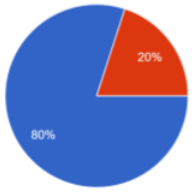
Coping Strategies

Living Stone International School

Do you use blue light filters or apps to manage screen exposure in the evening? (Warm light, Eye protection, or Dark mode)

25 responses

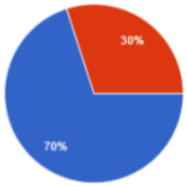
LSIS



21 responses

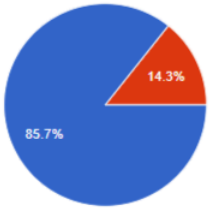
- Yes, Always
- No, Never

AUF



20 responses

NU



One-Way ANOVA

Living Stone International School

One-Way ANOVA (Welch's)

	F	df1	df2	p
GAD-7	18.7	2	41.2	< .001
PSQI	10.5	2	40.7	< .001

Group Descriptives

	Schools	N	Mean	SD	SE
GAD-7	AUF	20	10.10	4.18	0.934
	NU	21	5.67	3.35	0.731
	LSIS	25	12.56	4.47	0.895
PSQI	AUF	20	7.85	2.94	0.658
	NU	21	4.86	2.13	0.464
	LSIS	25	7.80	2.99	0.597

Assumption Checks

Homogeneity of Variances Test (Levene's)

	F	df1	df2	p
GAD-7	0.777	2	63	0.464
PSQI	2.087	2	63	0.133

[3]

One-Way ANOVA

Indicates significant differences

Post Hoc Tests

Living Stone International School

Games-Howell Post-Hoc Test – GAD-7

		AUF	NU	LSIS
AUF	Mean difference	—	4.43	−2.46
	p-value	—	0.002	0.151
NU	Mean difference		—	−6.89
	p-value		—	< .001
LSIS	Mean difference			—
	p-value			—

Games-Howell Post-Hoc Test – PSQI

		AUF	NU	LSIS
AUF	Mean difference	—	2.99	0.0500
	p-value	—	0.002	0.998
NU	Mean difference		—	−2.9429
	p-value		—	< .001
LSIS	Mean difference			—
	p-value			—

Post Hoc Test

Identified groups difference

GAD: AUF>NU, AUF=LSIS, LSIS>NU

PSQI: AUF>NU, AUF=LSIS, LSIS>NU

One-Way ANOVA (Non-parametric)

Kruskal-Wallis

	χ^2	df	p
GAD-7	24.4	2	< .001
PSQI	15.4	2	< .001

Kruskal-Wallis

Assumes normality,

Significant difference for both found

Analysis

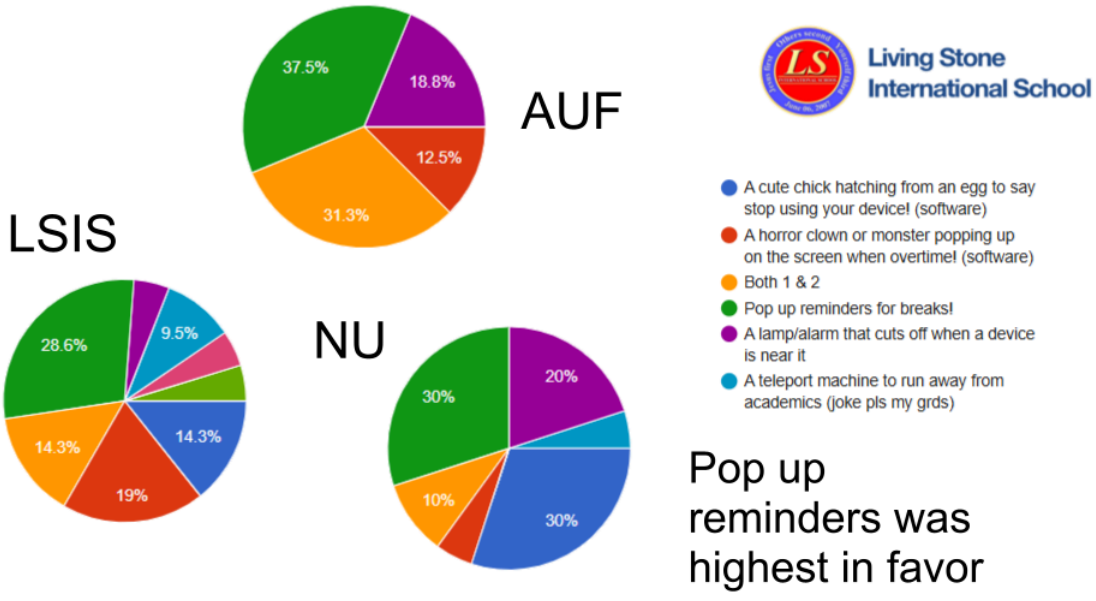
1. ANOVA results show difference between the three schools (p<.001) for both GAD-7 and PSQI
2. Post Hoc Test shows NU significantly differs from the other two schools in anxiety and sleep quality
3. Kruskal-Wallis test confirms these differences in case normality was a concern

Conclusion

Reject Null Hypothesis because there is significant differences between schools in terms of anxiety(GAD-7) and Sleep Quality (PSQI) along with the former questions for Perceived Effects, Awareness and Coping Strategies

Recommendations

- Education on Blue Light Effects
- Consistent use of Blue-Light reduction tools



Pop up reminders was highest in favor